

**RELAP5/MOD3.3 CODE MANUAL
VOLUME II:
USER'S GUIDE AND INPUT REQUIREMENTS**

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1 INTRODUCTION

The purpose of this volume is to help educate the code user by documenting the modeling experience accumulated from developmental assessment and application of the RELAP5 code. This information includes a blend of the model developers' recommendations with respect to how the model is intended to be applied and the application experience that indicates what has been found to work or not to work. Where possible, approaches known to work are definitely recommended, and approaches known not to work are pointed out as pitfalls to avoid.

1.1 General

The objective of the user's guide is to reduce the uncertainty associated with user simulation of light water reactor (LWR) systems. However, we do not imply that uncertainty can be eliminated or even quantified in all cases, since the range of possible system configurations and transients that could occur is large and constantly evolving. Hence, the effects of nodalization, time step selection, and modeling approach are not completely quantified. As the assessment proceeds, there will be a continual need to update the user guidelines document to reflect the current state of simulation knowledge.

1.2 Areas of Application

RELAP5 is a generic transient analysis code for thermal-hydraulic systems using a fluid that may be a mixture of steam, water, noncondensables, and a nonvolatile solute.

The fluid and energy flow paths are approximated by one-dimensional stream tube and conduction models. The code contains system component models applicable to LWRs. In particular, a point neutronics model, pumps, turbines, generator, valves, separator, and controls are included. The code also contains a jet pump component and an ecc mixer component.

The LWR applications for which the code is intended include accidents initiated from small break loss-of-coolant accidents, operational transients such as anticipated transients without SCRAM, loss of feed, loss-of-offsite power, and loss of flow transients. The reactor coolant system (RCS) behavior can be simulated up to and slightly beyond the point of fuel damage.

1.3 Modeling Philosophy

RELAP5 is designed for use in analyzing system component interactions; it does not offer detailed simulations of fluid flow within components. As such, it contains limited ability to model multidimensional effects, either for fluid flow, heat transfer, or reactor kinetics. Exceptions are the modeling of crossflow effects in a pressurized water reactor (PWR) core and the reflood modeling that uses a two-dimensional conduction solution in the vicinity of a quench front. To further enhance the overall system modeling capability, a control system model is included. This model provides a way to perform basic mathematical operations, such as addition, multiplication, integration, and control components such as proportional-integral, lag, and lead-lag controllers, for use with the basic fluid,

thermal, and component variables calculated by the remainder of the code. This capability can be used to construct models of system controls or components that can be described by algebraic and differential equations. The code numerical solution includes the evaluation and numerical time advancement of the control system coupled to the fluid and thermal system.

The hydrodynamic model and the associated numerical scheme are based on the use of fluid control volumes and junctions to represent the spatial character of the flow. The control volumes can be viewed as stream tubes having inlet and outlet junctions. The control volume has a direction associated with it that is positive from the inlet to the outlet. Velocities are located at the junctions and are associated with mass and energy flow between control volumes. Control volumes are connected in series, using junctions to represent a flow path. All internal flow paths, such as recirculation flows, must be explicitly modeled in this way since only single liquid and vapor velocities are represented at a junction. (In other words, a countercurrent liquid-liquid flow cannot be represented by a single-junction.) For flows in pipes, there is little confusion with respect to nodalization. However, in a steam generator having a separator and recirculation flow paths, some experience is needed to select a nodalization that will give correct results under all conditions of interest. Nodalization of branches or tees also requires more guidance.

Heat flow paths are also modeled in a one-dimensional sense, using a finite difference mesh to calculate temperatures and heat flux vectors. The heat conductors can be connected to hydrodynamic volumes to simulate a heat flow path normal to the fluid flow path. The heat conductor or heat structure is thermally connected to the hydrodynamic volume through a heat flux that is calculated using heat transfer correlations. Electrical or nuclear heating of the heat structure can also be modeled as either a surface heat flux or as a volumetric heat source. The heat structures are used to simulate pipe walls, heater elements, nuclear fuel pins, and heat exchanger surfaces.

A special, two-dimensional, heat conduction solution method with an automatic fine mesh rezoning is used for low-pressure reflood. Both axial and radial conduction are modeled, and the axial mesh spacing is refined as needed to resolve the axial thermal gradient. The hydrodynamic volume associated with the heat structure is not rezoned, and a spatial boiling curve is constructed and used to establish the convection heat transfer boundary condition. At present, this capability is specialized to the LWR core reflood process, but the plan is to generalize this model to higher pressure situations so that it can be used to track a quench front anywhere in the system.

The point reactor kinetics model is advanced in a serial and implicit manner after the heat conduction-transfer and hydrodynamic advancements but before the control system advancement. The kinetics model consists of a system of ordinary differential equations integrated using a modified Runge-Kutta technique. The integration time step is regulated by a truncation error control and may be less than the hydrodynamic time step; however, the thermal and fluid boundary conditions are held fixed over each hydrodynamic time interval. The reactivity feedback effects of fuel temperature, moderator temperature, moderator density, and boron concentration in the moderator are evaluated, using averages over the hydrodynamic control volumes and associated heat structures that represent the core. The averages are weighted averages established a priori such that they represent the effect on total core power. Certain nonlinear or multidimensional effects caused by spatial variations of the feedback parameters cannot be

accounted for with such a model. Thus, the user must judge whether or not the model is a reasonable approximation of the physical situation being modeled.

The control system model provides a way for simulating any lumped process, such as controls or instrumentation, in which the process can be defined in terms of system variables through logical, algebraic, differentiating, or integrating operations. These models do not have a spatial variable and are integrated with respect to time. The control system is coupled to the thermal and hydrodynamic components serially and implicitly. The control system advancement occurs after the heat conduction transfer, hydrodynamic, and reactor kinetics advancements and uses the same time step as the hydrodynamics so that new time thermal and hydrodynamic information is used in the control model advancement. However, the control variables are fed back to the thermal and hydrodynamic model in the succeeding time step, i.e., they are explicitly coupled.

A system code such as RELAP5 contains numerous approximations to the behavior of a real, continuous system. These approximations are necessitated by the finite storage capability of computers, by the need to obtain a calculated result in a reasonable amount of computer time, and in many cases because of limited knowledge about the physical behavior of the components and processes modeled. For example, knowledge is limited for components such as pumps and separators, processes such as two-phase flow, and heat transfer. Examples of approximations required because of limited computer resources are limited spatial nodalization for hydrodynamics, heat transfer, and kinetics; and density of thermodynamic and property tables. In general, the accuracy effect of each of these factors is of the same order; thus, improving one approximation without a corresponding increase in the others will not necessarily lead to a corresponding increase in physical accuracy. At the present time, very little quantitative information is available regarding the relative accuracies and their interactions. What is known has been established through applications and comparison of simulation results to experimental data. Progress is being made in this area as the code is used; but there is, and will be for some time, a need to continue the effort to quantify the system simulation capabilities.

2 HYDRODYNAMICS

The hydrodynamics simulation is based on a one-dimensional model of the transient flow for a steam-water noncondensable mixture. The numerical solution scheme used results in a system representation using control volumes connected by junctions. A physical system consisting of flow paths, volumes, areas, etc., is simulated by constructing a network of control volumes connected by junctions. The transformation of the physical system to a system of volumes and junctions is an inexact process, and there is no substitute for experience. General guidelines have evolved through application work using RELAP5. The purpose here is to summarize these guidelines.

In selecting a nodalization for hydrodynamics, the following general rules should be followed:

1. The length of volumes should be such that all have similar material Courant limits, i.e., flow length divided by velocity about the same. (Expected velocities during the transient must be considered.)
2. The volumes should have $\frac{L}{D} \geq 1$, except for special cases such as the bottom of a pressurizer where a smaller $\frac{L}{D}$ is desired to sharpen the emptying characteristic.
3. The total system cannot exceed the computer resources. RELAP5 dynamically allocates memory based on the requirements of each problem, and most models require memory based on factors such as the number of volumes, junctions, number of heat structures and the number of meshes, and the number and length of various user-input tables. The number of hydrodynamic volumes is a reasonable measure of problem size, and typical LWR systems with over 600 volumes have been run on workstations with 32 Mbytes of memory. The memory should be sufficiently large to avoid paging during transient advancement.
4. If possible, a nodalization sensitivity study should be made in order to estimate the uncertainty owing to nodalization. Volume V provides guidance and examples of appropriate nodalizations for reactor systems.
5. Avoid nodalizations where a sharp density gradient coincides with a junction (a liquid interface, for example) at steady-state or during most of the transient. This type of situation can result in time-step reduction and increased computer cost.
6. Eliminate minor flow paths that do not play a role in system behavior or are insignificant compared to the accuracy of the system representation. This can not usually be done until some preliminary trial calculations have been made that include all the flow paths. Care must be used here because in certain situations flow through minor flow paths can have a significant effect on system behavior. An example is the effect of hot-to-cold-leg leakage on core level depression in a PWR under small break loss-of-coolant accident conditions.

7. Establish the flow and pressure boundaries of the system beyond which modeling is not required and specify appropriate boundary conditions at these locations.

2.1 Basic Flow Model

The RELAP5 flow model is a nonhomogeneous, nonequilibrium two-phase flow model. See Section 3 of Volume I for a detailed description of the model and the governing equations. Options exist for homogeneous, equilibrium, or frictionless models if desired. These options are included to facilitate comparisons with other homogeneous and/or equilibrium codes. Generally, the code will not run faster if these options are selected.

The RELAP5 flow model is a one-dimensional, stream-tube formulation in which the bulk flow properties are assumed to be uniform over the fluid passage cross-section. The control volumes are finite increments of the flow passage and may have a junction at the inlet or outlet (normal junctions) or at the side of a volume (crossflow junctions). The stream-wise variation of the fluid passage is specified through the volume cross-sectional area, the junction areas, and through use of the smooth or abrupt area change options at the junctions. The smooth or abrupt area change option affects the way in which the flow is modeled, both through the calculation of loss factors at the junction and through the method used to calculate the volume average velocity. (Volume average velocity enters into momentum flux, boiling heat transfer, and wall friction calculations.) The abrupt area change model should be used to model the effect of sudden area changes such as reducers, orifices, or any obstruction in which the flow area variation with length is great enough to cause turbulence and flow separation. Only flow passages having a low wall angle (< 10 degrees, including angle) should be considered smooth. An exception to this rule is the case where the user specifies the kinetic loss factor at a junction and uses the smooth option. This type of modeling should only be attempted for cases where the actual flow area change is modest (less than a factor of two).

The hydrodynamic boundaries of a system are modeled using time-dependent volumes and junctions. For example, a reservoir condition would normally be modeled as a constant pressure source of mass and energy (a sink in the case of an outflow boundary). The reservoir is connected to the system through a normal junction, and the inflow velocity is determined from the momentum equation solution. For this type of boundary, some caution is required, since the energy boundary condition is in terms of the thermal energy rather than total energy. Thus, as the velocity increases, the total energy inflow increases owing to the increase in kinetic energy. This effect can be minimized for simulation of a reservoir by making the cross-sectional area of the time-dependent volume very large compared to the inlet junction area. This policy should be followed for outflow boundaries as well, or else flow reversals may occur.

A second way of specifying a flow boundary is using the time-dependent junction in addition to a time-dependent volume. This type of boundary condition is analogous to a positive displacement pump where the inflow rate is independent of the system pressure. In this case, the cross-sectional area of the time-dependent volume is not used because the velocity is fixed and the time-dependent volume is only used to specify the properties of the inflow. Thus, the total energy of the inflow is specified. When only time-dependent junctions are used as boundary conditions, the system pressure entirely depends on the

system mass, and, in the case of all liquid systems, a very stiff system results. An additional fact that should be considered when using a time-dependent junction as a boundary is that pump work is required for system inflow if the system pressure is greater than the time-dependent volume pressure. In particular, any energy dissipation associated with a real pumping process is not simulated. The flow work done against the system pressure is approximated by work terms in the thermal energy equation.

In RELAP5, any volume that does not have a connecting junction at an inlet or outlet is treated as a closed end. Thus, no special boundary conditions are required to simulate a closed end.

The fluid properties at an outflow boundary are not used unless flow reversal occurs. In this respect, some caution is necessary and is best illustrated by an example. In the modeling of a subatmospheric pressure containment, saturated steam is often specified for the containment volume condition. This will result in the outflow volume containing pure steam at low pressure and temperature. If in the course of calculation a flow reversal occurs, even a very minute one (possibly caused by numerical noise), a cascading result occurs. The low-pressure or low-temperature steam can rush into a volume at higher pressure and rapidly condense. The rapid condensation leads to depressurization of the volume and increased flow. Such a result can be avoided by using air or superheated steam in the containment volume.

A general guide to modeling hydrodynamic boundary conditions is to simulate the actual process as closely as possible. This guideline should be followed unless initial calculations result in unphysical results because of unanticipated numerical idiosyncrasies.

Only the algebraic sign is needed in the one-dimensional hydrodynamic components to indicate the direction of vector quantities, i.e., the volume and junction velocities. Both the volumes and the junctions have coordinate directions that are specified through input. Each hydrodynamic volume has three coordinate directions, named x, y, and z, and each coordinate direction has an associated inlet and outlet face. The coordinate direction is positive from the inlet to the outlet. The normal, one-dimensional flow is along the x-coordinate. Normal volume connections are to the inlet and outlet faces associated with the x-coordinate. Crossflow connections are to the inlet and outlet faces associated with coordinates orthogonal to the x-coordinate, that is, the y- and z-coordinates.

Which faces of a volume are the inlet or outlet faces depend upon the specifications of the volume orientation. For a positive vertical elevation change, the inlet is at the lowest elevation, whereas for a negative vertical elevation change, the inlet is at the highest elevation of the volume. For a horizontal volume, whether the inlet is at the left or right depends upon the azimuthal angle. (A zero value implies an orientation with the inlet at the left.) This orientation of a horizontal volume is not important as far as hydrodynamic calculations are concerned but is important if one tries to construct a three-dimensional picture of the flow path. Several possible volume orientations, depending upon the input values for the azimuthal and inclination angles, are illustrated in **Figure 2.1-1**.

The junction coordinate direction is established through input of the junction connection code (e.g., Words W1 and W2 of Cards CCC0101 through CCC0109, Section A-7.4 of Appendix A for a single-junction component). The junction connection codes designate a *from* and a *to* component, and the velocity

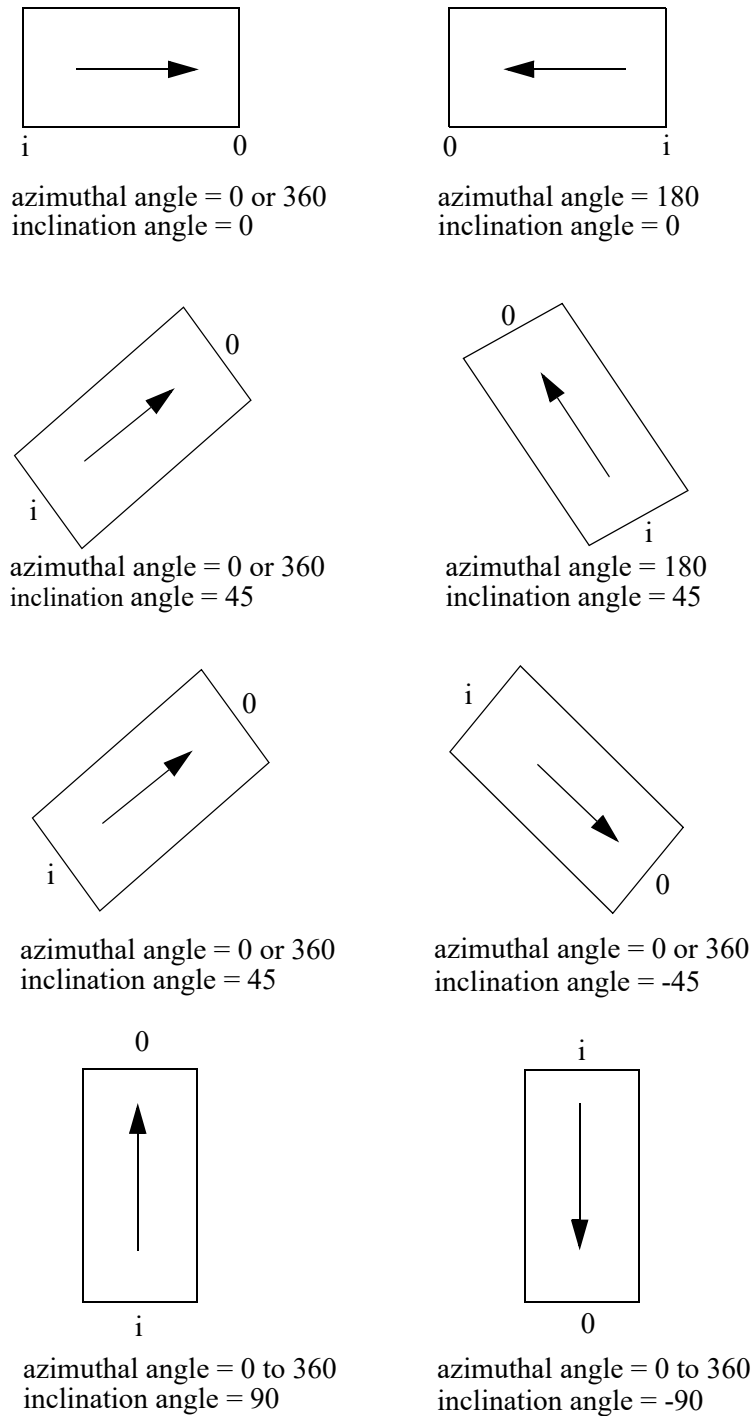


Figure 2.1-1 Possible Volume Orientation Specifications

is positive in the direction from the *from* component to the *to* component. The connection codes can be entered in an old or an expanded format. The expanded format is recommended, but the old format is still valid.

A connection code has the format CCCVV000N, where CCC is the component number, VV is the volume number, and N is the face number, where zero indicates the old format and nonzero indicates the expanded format. The old format (N = 0) can only specify connections to the faces associated with normal flow, that is, flow along the x-coordinate. In the old format, VV is not a volume number but, instead, VV = 00 specifies the inlet face of the component, and VV = 01 specifies the outlet face of the component. The volume number is only implied. For components specifying single-volumes (currently only a pipe specifies multiple volumes), normal flow (as opposed to crossflow) to either the inlet or outlet face can be specified. For a pipe, however, the old format allows specification of normal flow only to the inlet of the first pipe volume or to the outlet of the last pipe volume. Crossflow meaning connections to faces associated with y- or z-faces cannot be specified with the old format.

The expanded connection code assumes that a volume has six faces, i.e., an inlet and outlet for each of three coordinate directions (see **Figure 2.1-2** and **Figure 2.1-3**). The expanded connection code indicates the volume being connected and through which face it is being connected. In the new format (N nonzero), N is the face number and VV is the volume number. For components specifying single-volumes, VV is 01; but for pipes, VV can vary from 01 for the first pipe volume to the last pipe volume number. The quantity N is 1 and 2 for the inlet and outlet faces, respectively, for the volume's normal or x-coordinate direction. The quantity N is 3 and 4 to indicate inlet and outlet faces, respectively, for the volume's y-coordinate direction, and N is 5 and 6 to indicate inlet and outlet faces, respectively, for the volume's z-coordinate direction. Entering N as 1 or 2 specifies normal connections to a volume; entering N as 3 through 6 specifies a crossflow connection to a volume. In **Figure 2.1-2** and **Figure 2.1-3**, the world (inertial) coordinates are indicated by x_0 , y_0 , and z_0 , whereas the local coordinates for the volume are indicated by x, y, and z. **Figure 2.1-2** is for a horizontal volume. **Figure 2.1-3** is for a vertical volume, and is obtained from the horizontal volume (**Figure 2.1-2**) by a 90° counter-clockwise rotation about the local y-axis.

Average volume velocities are computed along each coordinate direction that is active. The x-coordinate is assumed active, and a warning message is issued during input processing if no junctions attach to normal faces. A y- or z-coordinate is active only if a junction attaches to one of the associated faces. The average volume velocity for each coordinate direction involves only junction velocities at the faces associated with that coordinate direction. Thus, a crossflow entering a y-face does not contribute to the computation of the volume velocity in the x-direction. But that crossflow does contribute to the average velocity in the y-direction.

Users of previous versions of RELAP5 will note that the crossflow discussed above is different from older versions. The crossflow capability has been improved, but unfortunately the differing meanings for the term *crossflow* may lead to misunderstanding. The previous use of crossflow implied the following: Flow entered a face orthogonal to the normal flow; crossflows never contributed to any average volume

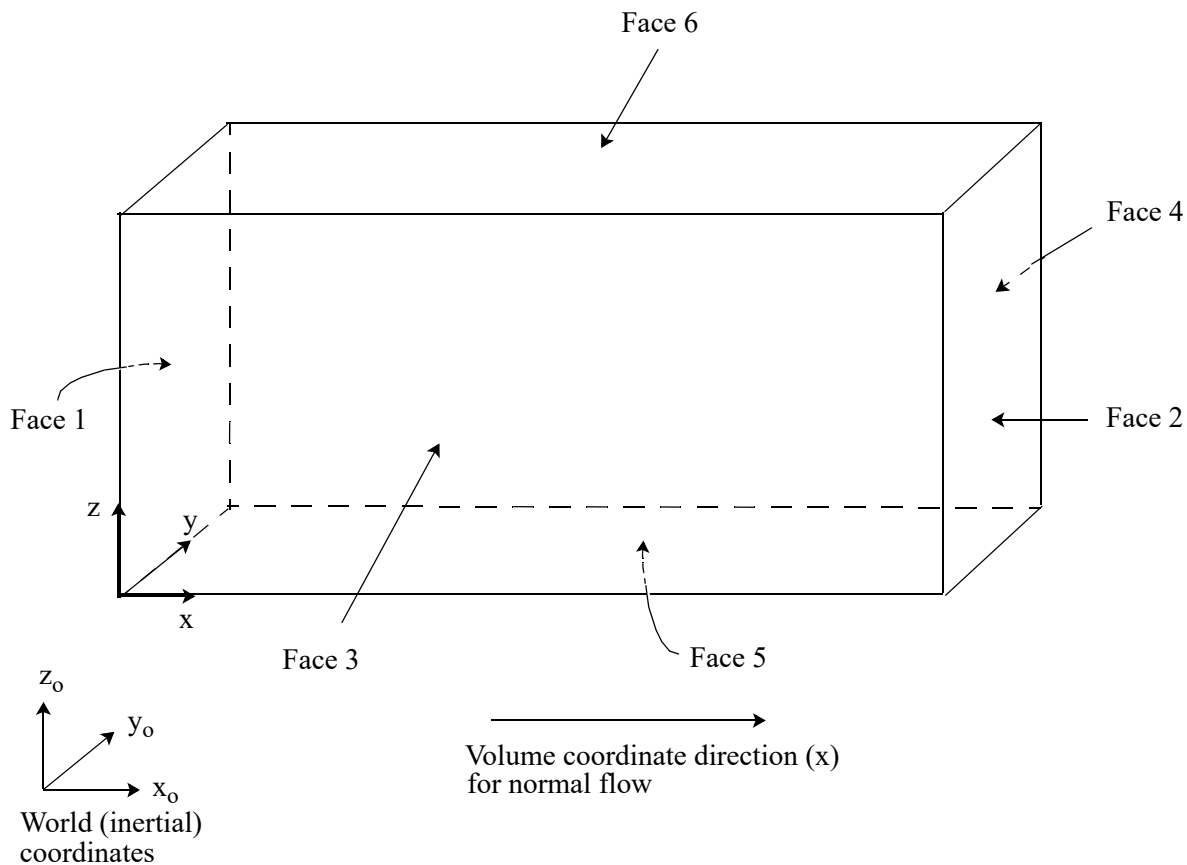


Figure 2.1-2 Horizontal Volume Schematic Showing Face Numbers

velocity; a limited form of the momentum equation was used; and face numbers 3 through 6 and/or junction flags could specify a crossflow connection. The limited momentum equation ignored momentum flux, wall friction, and gravity terms. Now, crossflow means only that the connection is to a face other than one of the normal faces. Note especially that crossflow does not imply a modified form of the momentum equation. The same momentum equation options are available to both normal flows and crossflows. The standard one-dimensional momentum equations can be applied to both normal and crossflows. Optionally, and only through the use of the momentum flux junctions flags, the momentum flux contribution in the *from* or *to* volume can be ignored for normal and crossflow connections.

There is no difference in the application of the conservation equations to the normal and crossflow types of connections. The only difference is that the term normal is applied to the flow that would occur in a strictly one-dimensional volume; crossflow is an approximation to multidimensional effects consisting of applying the one-dimensional momentum equation to each of the coordinate directions in use. To give some perspective to the approximation, the three-dimensional momentum equation contains nine terms for momentum flux; the momentum in each of the three directions being convected by velocities in the three directions. In the crossflow model, only three momentum flux terms are used--the momentum in each direction convected by velocity in the same direction.

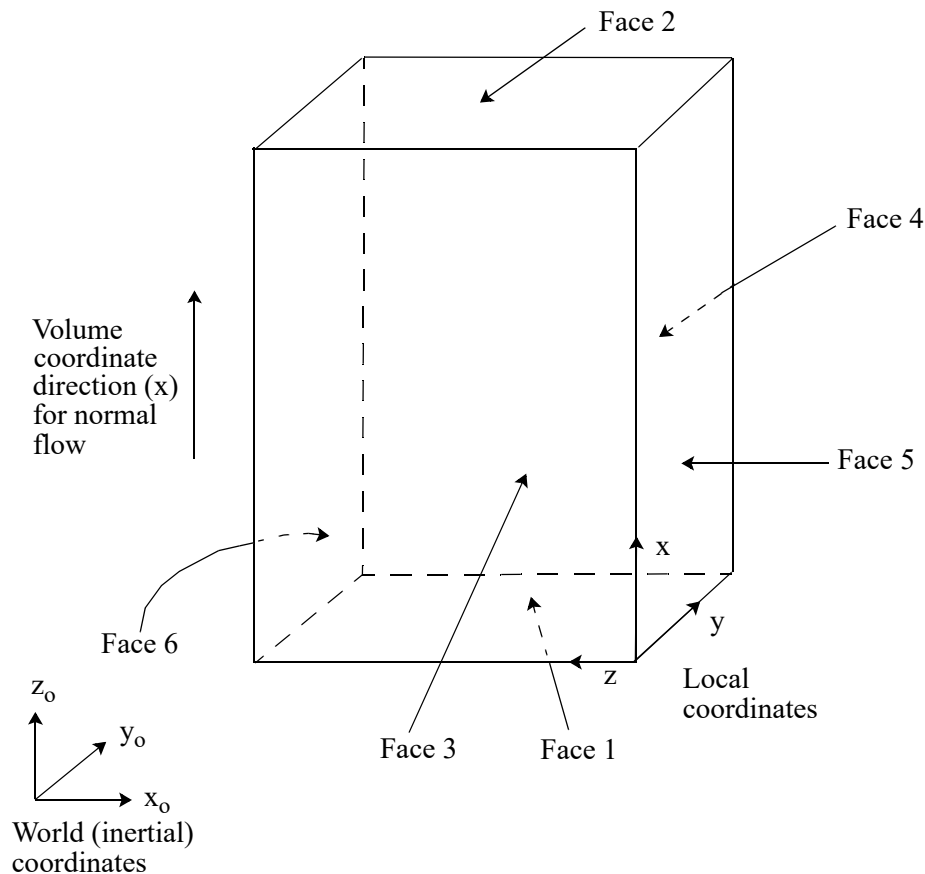


Figure 2.1-3 Vertical Volume Schematic Showing Face Numbers

The code input provides junction flags to ignore momentum flux effects in either the *from* volume, the *to* volume, both volumes, or to include momentum effects in both volumes (the default). Intuitively, including momentum effects is more accurate modeling, and momentum effects should be included in junctions attached to the normal faces. In previous versions of the code, a restricted form of the momentum equation was used that omitted momentum flux, wall friction, and gravity terms. One reason was that the geometric information necessary for computing these terms was not available and average volume velocity terms in the crossflow directions were not computed. The earliest motive for the crossflow model was to treat recirculation flows in the reactor core, and these restrictions were acceptable since velocities were low and there were no elevation changes. The crossflow model was subsequently used for tees since the crossflow model, even with the restrictions, was a better model than previous approaches for tees. The current recommendation is to include the momentum flux terms for crossflows but remove them if computational difficulties involving crossflow junctions are encountered. The crossflow model is currently under developmental assessment. A more definite recommendation is not to have multiple junctions with differing momentum flux options attached to the same coordinate direction. Even though the momentum flux is ignored in one junction, its velocity contributes to the average velocity in that coordinate direction and thus other junctions using momentum flux terms use that average volume velocity.

The current crossflow model requires input information for the y- and z-coordinates similar to that entered for the x-coordinate. Default data for the y and z-coordinates are obtained from the x-coordinate data by assuming the volume is a section of a right circular pipe. Optional input data may be entered when this assumption is not valid.

In major edits and similar input edits, the junction connection code is edited in the new format. Note that the new logic allows branching and merging flow (i.e., multiple junctions at a face) at any volume, including interior pipe volumes. The primary reason for this change is to permit crossflow to all volumes in a pipe. Now it is possible to use pipe volumes to represent axial levels in a vessel and to use multiple pipe components to represent radial or azimuthal dependence. Single-junctions can crosslink any of the pipe volumes at the same axial level.

A simpler method to crosslink volumes is to use the multiple junction component. This component describes one or more junctions, with the limitation that all volumes connected by the junctions must be part of the same hydrodynamic system. Although this component can be considered a collection of single-junctions, its common use is to crosslink adjacent volumes of parallel pipes. Because the junctions linking pipe volumes tend to be similar, N junctions crosslinking N volumes per pipe can be entered with the amount of input comparable to one junction.

A sketch showing a series of three horizontal volumes connected by two junctions is shown in **Figure 2.1-4** to illustrate some of the possible coordinate orientations that result from combinations of the connection codes and the volume orientation data. In **Figure 2.1-5**, two possible combinations are illustrated for the connection of two vertical volumes. **Figure 2.1-5a** shows the two volumes unconnected; **Figure 2.1-5b** shows the result when the outlet of Volume 1 is joined to the inlet of Volume 2; and **Figure 2.1-5c** shows the result when the inlet of Volume 1 is connected to the inlet of Volume 2. In particular, note that the geometry can be modified from a straight passage to a manometer configuration by simply reversing the inlet/outlet designator in the junction connection code.

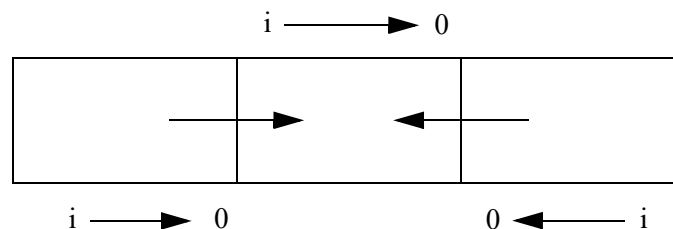


Figure 2.1-4 Sketch of Possible Coordinate Orientation for Three Volumes and Two Junctions

When systems of volumes or components are connected in a closed loop, the summation of the volume elevations must close when they are summed according to the junction connection codes and sequence, or an unbalanced gravitational force will result. RELAP5 has an input processing feature that finds all loops or closed systems (which are defined by the input) and checks for elevation closure around each loop. The error criterion is 10^{-4} m. If closure is not obtained, the fail flag is set, and no transient or

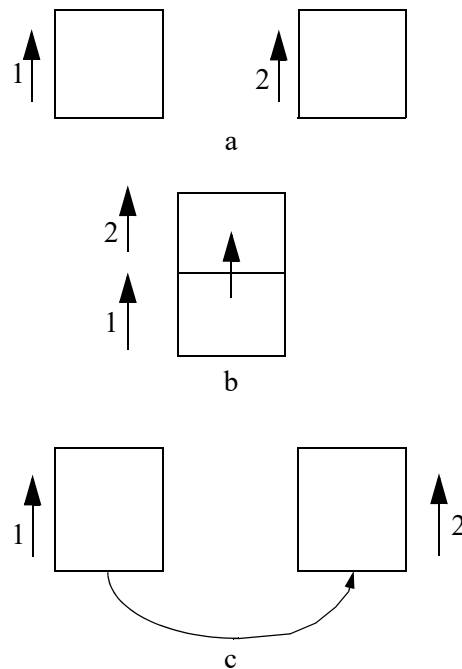


Figure 2.1-5 Sketch of Possible Vertical Volume Connections

steady-state calculations will be made. The elevation checker will print out that elevation closure does not occur at a particular junction that formed a closed loop during input processing. The junction at which closure of the loop occurs is somewhat arbitrary and depends on the input order of the components.

The elevation checking with crossflows differs from earlier versions of RELAP5. The elevation checking starts from the center of a volume with the initial volume and its elevation obtained from input data or defaulted. Using *to* and *from* junction information and elevation change information from the connected volumes, the elevation to the common face of the volumes is computed; then, the elevation of the center of the connected volume is computed. This computing of the elevations by tracking the junctions continues until all junctions have been used. Whenever a volume is reentered, the newly obtained elevation is compared to the previously computed elevation, and an error occurs if they do not match. With the previous crossflow model, the elevation from the center to a face was zero for a crossflow connection. This meant that the same elevation would be obtained regardless of which face the crossflow connection used. The face number is now important, both for elevation checking and in computing elevation effects, momentum flux effects, and friction. We recommend that decks prepared for previous versions of the code have all crossflow connections reviewed for use with the newer crossflow model.

The junctions are printed out in the major edits in the hydrodynamic junction information sections (Section 8.3.2.9 and Section 8.3.2.10). The *from* and *to* volumes are listed for each junction. In addition, the flow regimes for the volumes (flore_g) and the junctions (florg_j) are also listed using three letters. It is

also possible to list the flow regime for the volumes and the junctions in the minor edits and plots, where a number is used. **Table 2.1-1** shows the three-letter code and number used for each flow regime.

Table 2.1-1 Flow Regime Letters And Numbers

Flow regime	Three-letter code (major edits)	Number (minor edits/plots)
None	X	0
High mixing bubbly	CTB	1
High mixing bubbly/mist transition	CTT	2
High mixing mist	CTM	3
Bubbly	BBY	4
Slug	SLG	5
Annular mist	ANM	6
Mist pre-CHF	MPR	7
Inverted annular	IAN	8
Inverted slug	ISL	9
Mist	MST	10
Mist post-CHF	MPO	11
Horizontal stratified	HST	12
Vertical stratified	VST	13
Level tracking	LEV	14
Jet junction	JET	15
ECC mixer wavy	MWY	16
ECC mixer wavy/annular mist	MWA	17
ECC mixer annular mist	MAM	18
ECC mixer mist	MMS	19
ECC mixer wavy/slug transition	MWS	20
ECC mixer wavy-plug-slug transition	MWP	21
ECC mixer plug	MPL	22
ECC mixer plug-slug transition	MPS	23
ECC mixer slug	MSL	24
ECC mixer plug-bubbly transition	MPB	25

Table 2.1-1 Flow Regime Letters And Numbers (Continued)

Flow regime	Three-letter code (major edits)	Number (minor edits/plots)
ECC mixer bubbly	MBB	26
Supercritical pressure “liquid” side of Terit	SPF	27
Supercritical pressure “vapor” side of Terit	SPV	28

Table 2.1-2 Bubbly/Slug Flow Regime Numbers for Vertical Junctions

Geometry and flow conditions	Correlations used	Numbers (minor edits/plots)
Rod bundles	EPRI	2
High up/down flows in small pipes	EPRI	3
Low up/down countercurrent flows in small pipes	Zuber-Findlay slug	4
Transition regions between 3 and 4	EPRI & Zuber-Findlay slug	5
High up/down flows in intermediate pipes	EPRI	9
Low up/down countercurrent flows in intermediate pipes	Churn-turbulent bubbly	10
Transition regions between 10 and 12	Churn-turbulent bubbly & Kataoka-Ishii	11
Low up/down countercurrent flows in intermediate pipes	Kataoka-Ishii	12
Transition between regions 9 and 10-11-12	EPRI & Churn-turbulent bubbly/Kataoka-Ishii	13
Large pipes	Churn-turbulent bubbly	14
Transition regions between 14 and 16	Churn-turbulent bubbly & Kataoka-Ishii	15
Large pipes	Kataoka-Ishii	16

In the bubbly and slug flow regimes for vertical junctions, it is possible to list an additional flow regime number (iregj) in the minor edits and plots that is associated with a particular geometry/flow and correlation that is used in the interphase drag. If not in bubbly or slug flow and not a vertical function, the

number will be zero. **Table 2.1-2** shows the number used for each regime. In the transition regions (11 and 15), a fraction is added to the number (between 0 and 1) that indicates how far the junction conditions are between churn-turbulent bubbly and Kataoka-Ishii, based on the dimensionless vapor superficial velocity (j_g^+) .

The interphase friction model for bundles (i.e., core and steam generator) can be activated with a volume control flag (b). The model is based on a correlation from EPRI, as discussed in Volume I of this manual. When in bubbly or slug flow, the flow regime number is 2, as indicated in **Table 2.1-2**; otherwise it is 0.

The user should be aware that all plant or experimental facility geometries that are not circular should have an input junction hydraulic diameter to specify the necessary information required for the code calculated interphase friction. For bundles and steam generators, the junction hydraulic diameter should match the volume hydraulic diameter (including grid spacers, which should use the volume hydraulic diameter at the junction). In addition for grid spacers, the volume flow area should be used at the junction and the user-input loss should be multiplied by ratio of squared areas of the volume and the grid spacer. For area changes, the donor diameter for the normal flow direction is recommended. For orifices, the actual diameter is recommended.

2.2 State Relationships

The steam table subroutine has upper and lower limits on pressure and temperature. A pressure-temperature diagram is shown in **Figure 2.2-1**.

If the calculation predicts a pressure and temperature in the shaded region, a water property error will result, and the code will cut the time step. If the calculation continues to predict pressure and temperature in the shaded region down to the minimum time step, the calculation will be terminated.

2.3 Process Models

In RELAP5, process models are used for simulation of processes that involve large spatial gradients or which are sufficiently complex that empirical models are required. The flow processes for an abrupt area change, a choked flow, a branch, reflood, noncondensables, water packer, CCFL, level tracking, and thermal stratification are all simulated using specialized modeling. These particular processes are not peculiar to a component and will be discussed as a group. Some components, such as pumps and separators, also involve special process models; these models will be discussed with the component models. The use of the process models is specified through input, and proper application is the responsibility of the user. Under certain circumstances, we recommend that the user not mix process models; e.g., we recommend the user not use the choking model at a junction connected to either the side of a volume where the abrupt area change is activated for the junction and more than one junction is connected. The purpose of this section is to advise the user regarding proper application of the process models.

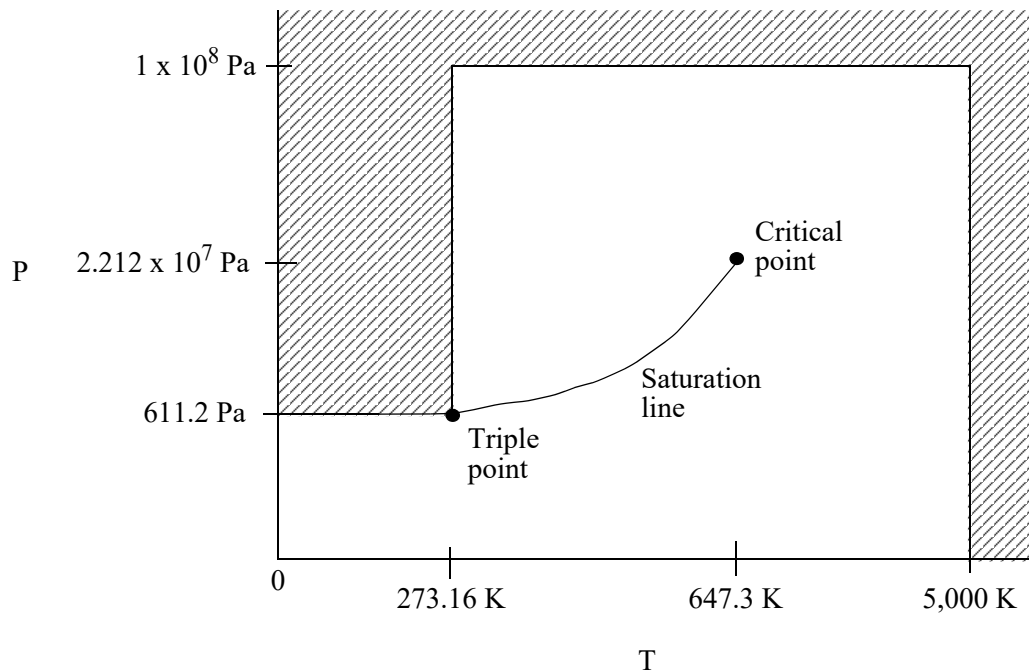


Figure 2.2-1 Pressure-Temperature Diagram

2.3.1 Abrupt Area Change

The abrupt area change option should generally be used in the following situations:

1. Sharp edged area changes.
2. Manifolds and plena connecting parallel flow passages.
3. At break locations.

For the abrupt area model, the junction area (upon which the velocities are based) is the minimum area of the two connecting volumes. The abrupt area change model is discussed in more detail in Section 2.4.1.

In addition to the computed form loss from the abrupt area change model, users have the option of input form loss factors to achieve the desired pressure drop. See Section 2.3.3.3 for discussion for modeling of minor flow paths. It is recommended that the abrupt area change model not be used in situations where the area change ratio is greater than 10. For this situation, the smooth area change option and an appropriate loss coefficient is recommended.

The pressure drop calculated by using form losses is a function of junction velocity.

2.3.2 Choked Flow

The choked flow option is specified in the junction flags on the junction geometry card. In general, the choked flow model should be used at all exit junctions of a system. We recommend that the choked flow model be usually used at the choke plane and that the user not model anything past this plane. (Therefore, just use a time-dependent volume downstream of the choke plane.) Internal choking is allowed but may not be desirable under certain conditions. Some applications of RELAP5 require that volumes downstream of the choke plane be modeled with non-time-dependent volumes. For this case, the user should monitor the mass error in the downstream volumes to ensure that the total mass error is not governed by these volumes. This is done by examining the hydrodynamic time step control information in the major edits (see Section 8.3.2); one of columns labeled LRGST.MASS ERR gives the number of times a volume had the largest mass error. If the mass error in these volumes is large (i.e., the number of times a volume had the largest mass error is high), the user should consider adjusting the size of the volumes. This would involve reducing the size of these volumes, since the mass error is given by the density error times the volume.

The recommended input junction flags when the choking model is on ($c = 0$) are smooth area ($a = 0$) and nonhomogeneous ($h = 0$). Since the smooth area change option is recommended where the choking flag is active, the user will need to specify loss factors appropriate for the geometry of the choking plane (i.e., orifice, nozzle, etc.). The smooth and abrupt area options are discussed in Section 2.4.1. With regard to the nonhomogeneous ($h = 0$) option, it is generally recommended that $h = 0$ be used. There may be rare situations where the combined interphase drag is too low, resulting in too much slip and too low mass flow. For this situation, the homogeneous option ($h = 1$ or 2) is recommended. Note, the user should monitor the calculated results for nonphysical choking. If this occurs, the user should turn choking off ($c = 1$) at junctions where this occurs.

Guidelines for the discharge coefficients (subcooled and two-phase) are as follows. For a break nozzle/venturi geometry, a discharge coefficient of nearly 1.0 should be used. For an orifice geometry, the discharge coefficient depends on the break configuration and may be somewhat less than 1.0.

The throat $\frac{dA}{dx}$ used in subcooled choking, which is denoted by $\left(\frac{dA}{dx}\right)_t$ in Volume I of this manual, is calculated differently for the normal junction abrupt area option and the normal junction smooth area option.

For the recommended abrupt area change option, the following formula is used:

$$\left(\frac{dA}{dx}\right)_{t, \text{abrupt}} = \frac{A_K - A_t}{10.0 D_K} \quad (2.3-1)$$

where

A_K = the upstream volume flow area in the coordinate direction of the junction

A_t = the throat or junction area (minimum physical area)

D_K = the upstream volume diameter in the coordinate direction of the junction.

It is recommended the user input the actual physical values for A_K , A_t , and D_K . This formula is empirical, and the data base is limited. It was developed primarily to obtain the proper subcooled discharge at the break for the LOFT-Wyle Blowdown Test WSB03R,^{2.3-1} which is one of the developmental assessment separate-effects test problems. In addition, it has been used successfully in many Semiscale test comparisons for the break flow.^{2.3-2}

If the user selects the smooth area change option, the code uses the following formula:

$$\left(\frac{dA}{dx}\right)_{t, \text{smooth}} = \frac{A_K - A_t}{0.5\Delta x_K} \quad (2.3-2)$$

where

A_K = the upstream volume flow area in the coordinate direction of the junction

A_t = the throat or junction area (minimum physical area)

Δx_K = is the upstream volume length in the coordinate direction of the junction.

The smooth area option is intended to be used for smoothly varying geometries. The length $0.5 \Delta x_K$ would be the actual length of the upstream volume (A_K) to the throat (A_t). Since the smooth area change option is not recommended, this formula has had little assessment.

Sometimes, it is observed that the choking junction oscillates in time between the inlet and outlet junctions of a control volume. This may induce flow oscillations and should be avoided. The situation most often occurs in modeling a break nozzle. The choking plane is normally located in the neighborhood of the throat. The break can be adequately modeled by putting the break junction at the throat and including only the upstream portion of the nozzle. If the entire nozzle is modeled, the choked flow option should be applied only to the junction at the throat.

The internal choking option must be removed when supersonic flows are anticipated or when its application causes unphysical flow oscillations. Typical cases are propagation of shock waves downstream from a choked junction. Sometimes, it is necessary to remove the choking option at junctions near a known internal choked junction in order to avoid oscillations.

2.3.3 Branching

A fundamental and vital model needed for simulation of fluid networks is the branched flow path. Two types of branches are common, the tee and the plenum. The tee involves a modest change in flow area from branch to branch and a large change in flow direction, while the plenum may involve a very large change in flow area from branch to branch and little or no change in flow direction. In PWR simulations, a tee model would be used at pressurizer surge line connections, hot leg vessel connections, and cold leg connections to the vessel inlet annulus. A plenum model would be used for modeling upper and lower reactor vessel plenums, steam generator models, and low-angle wyes.

Two special modeling options are available for modeling branched flow paths. These are a crossflow junction model and a flow stratification model, in which the smaller pipe at a tee or plenum may be specified as connected to the top, center, or bottom of a larger connecting pipe. When stratified flow is predicted to exist at such a branch, vapor pullthrough and/or liquid entrainment models are used to predict the void fraction of the branched flow. The use of these models for simulating tees, plenums, and leak paths are discussed in greater detail below.

2.3.3.1 Tees

The simplest tee is the 90-degree tee, in which all branches have the same or comparable diameters. The recommended nodalization for this flow process is illustrated in **Figure 2.3-1**. The small volume at the intersection of the side branch with the main flow path should have a length equal to the pipe diameters. Generally, this length will be shorter than most other hydraulic volumes and will have a relatively small material Courant limit. The code, however, has a time step scheme that permits violation of the material Courant for an isolated volume for the semi-implicit scheme. Thus, this modeling practice may not result in a time step restriction. User experience has shown that if the code runs too slowly and is Courant-limited in the small volume, it is possible to increase the length of the volume to allow faster running without adversely affecting the results.

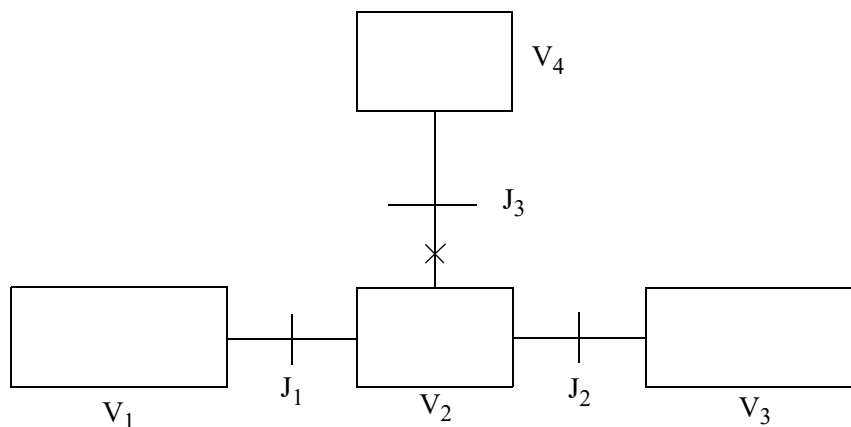


Figure 2.3-1 A 90-Degree Tee Model Using a Crossflow Junction

The Junction J3 is specified as a half normal junction and half crossflow junction. The half of Junction J3 associated with Volume V4 is a normal junction, whereas the half associated with Volume V2 is a crossflow junction. The junction specification is made using the junction flag *efvcahs*, which (for a single-junction) is Word W6(I) of Cards ccc0101 through ccc0109. As noted in previous crossflow discussions, the same momentum equation options are used available in both normal and crossflow. Both flow types allow ignoring of momentum flux and wall friction terms through the use of volume and junction flags. User experience shows that temperature oscillations may develop in Volume V2. It may be necessary to increase the length of Volume V2 to remove the oscillations. In general, a user loss coefficient will be needed at Junction J3. This coefficient should be determined to obtain the proper pressure drop.

A tee can also be modeled using the branch component, as illustrated in **Figure 2.3-2**. This approach has the advantage that fewer volumes are used. Disadvantages are that the calculated result may be altered, depending on whether Junction J₂ is connected to Volume V₁ or V₂, and that the flow division has less resolution at the tee in the presence of sharp density gradients. In cases where the Volumes V₁ and V₃ are nearly parallel, the model illustrated in **Figure 2.3-2** may be a more accurate representation of the physical process (such as for a wye).

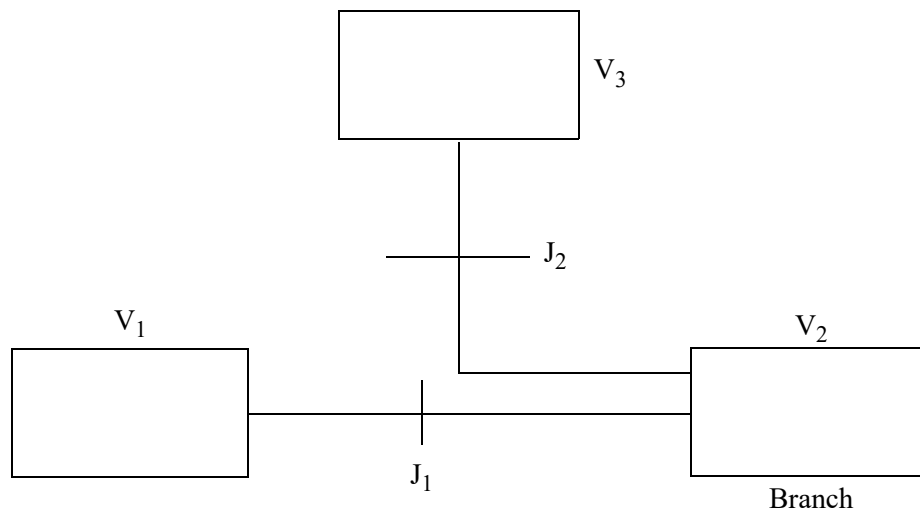


Figure 2.3-2 Tee Model Using a Branch Component

2.3.3.2 Branch

The branch model approximates the flow process that occurs at merging or dividing flows, such as at wyes and plenums. This model does not include momentum transfer caused by mixing and thus is not suited for high-velocity merging flows. A special component, the JETMIXER, is provided for modeling the mixing of high-velocity, parallel streams. Application of this model is discussed in Section 2.4.9.

A branch component consists of one system volume and zero to nine junctions. The limit of nine junctions is due to a card numbering constraint. Junctions from other components, such as single-junctions, pumps, other branches, or even time-dependent junction components, may be connected to the branch component. The results are identical whether junctions are attached to the branch volume as part of the branch component or as part of other components. Use of junctions connected to the branch but defined in other components is required in the case of pump and valve components. Any of these may also be used to attach more than the maximum of nine junctions that can be described in the branch component input.

A typical one-dimensional branch is illustrated in **Figure 2.3-3**. The figure is only one example and implies merging flow. Additional junctions could be attached to both ends, and any of the volume and junction coordinate directions could be changed. The actual flows may be in any direction; thus, flow out of Volume V_3 through Junction J_1 and into Volume V_3 through Junction J_2 is permitted.

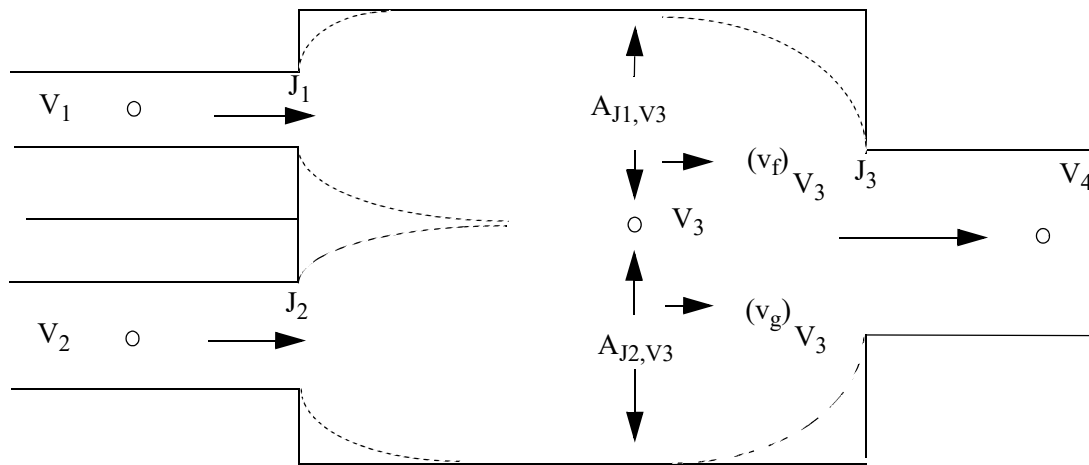


Figure 2.3-3 Typical Branching Junctions

The volume velocities are calculated by a method that averages the phasic mass flows over the volume cell inlets and outlets. The volume velocities of Volume V_3 are used to evaluate the momentum

flux terms for all junctions connected to Volume V_3 . The losses associated with these junctions are calculated using a stream tube formulation based on the assumption that the fraction of volume flow area associated with a junction stream tube is the same as the volumetric flow fraction for the junction within the respective volume. Also, using the junction flow area, the adjacent volume flow areas, and the branch volume stream tube flow area, the stream tube formulation of the momentum equation is applied at each junction. However, if the smooth area change is specified, large changes in flow can lead to nonphysical results. Therefore, it is normally recommended that the abrupt area change option be used at branches.

Plenums are modeled using the branch component. Typical LWR applications of a plenum are the upper and lower reactor vessel regions, steam generator plenums, and steam domes. The use of a branch to

model a plenum having four parallel connections is illustrated in **Figure 2.3-4**. The flows in such a configuration can be either inflows or outflows. The junctions connecting the separate flow paths to the plenum are ordinary junctions with the abrupt area change option recommended. It is possible to use crossflow junctions at a branch for some or all of the connections.

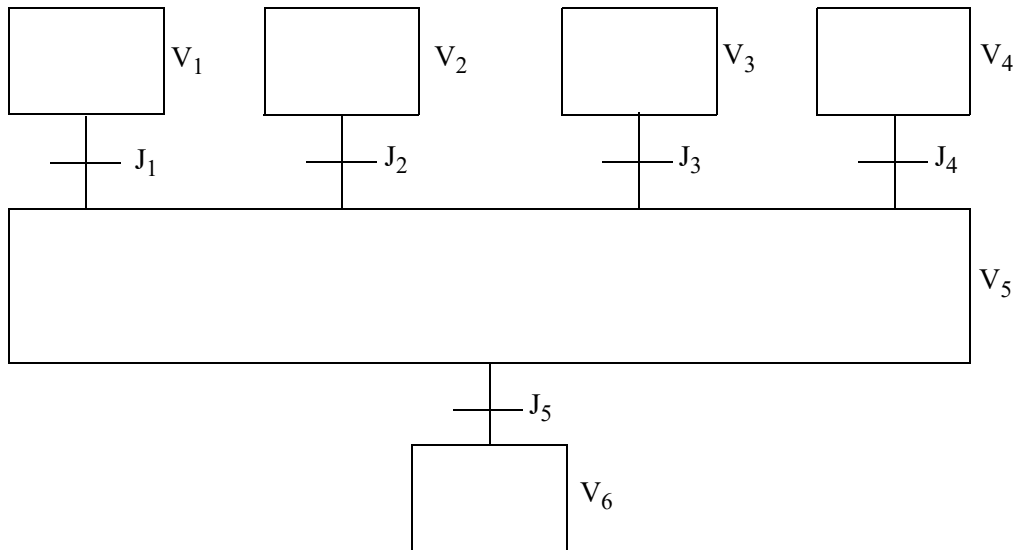


Figure 2.3-4 Plenum Model Using a Branch

A wye is modeled, as illustrated in **Figure 2.3-3**, using the branch component. The flow can either merge or divide. Either the smooth or the abrupt area change option may be used. If the smooth area change is specified, large changes in flow can lead to nonphysical results. Therefore, it is normally recommended that the abrupt area change option be used at wyes.

2.3.3.3 Leak Paths

An application, that may or may not involve branching but which is frequently a source of problems, is the modeling of small leak paths. These may be high-resistance paths or may involve extreme variations in flow area. The approximation of the momentum flux terms for such flow paths is highly uncertain and can lead to large forces, resulting in numerical oscillations. Modeling of small leak paths was one of the primary motivations for developing the crossflow connections. As needed, the momentum flux and wall friction can be omitted, and the flow resistance could instead be computed from a user-specified kinetic loss factor.

In applying the crossflow junction to leak path models, the actual area of the leak path is used as the junction area. A kinetic loss factor is input, based on the fluid junction area velocity for the forward and reverse loss factors. The forward and reverse loss factors should be equal unless there is a physical reason why they should be different. In particular, a very large forward and small reverse loss factor should not be

used to simulate a check valve. This approach can cause code failure. A typical leak path model between vertical volumes is illustrated in **Figure 2.3-5**.

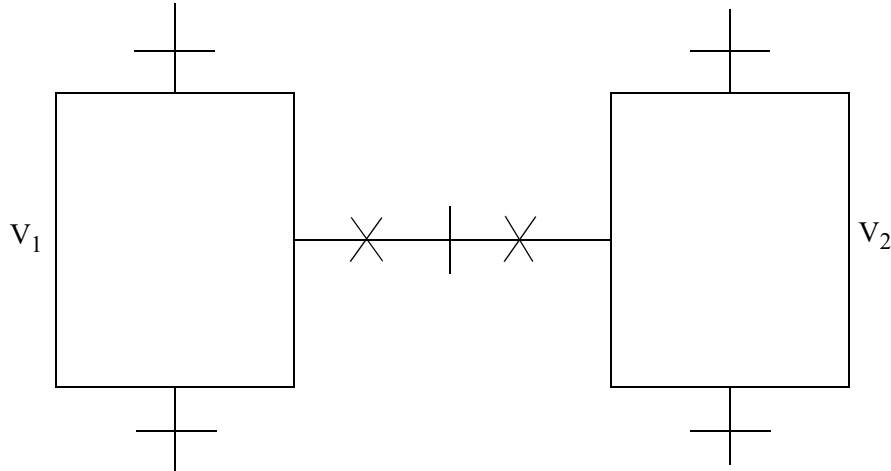


Figure 2.3-5 Leak Path Model Using the Crossflow Junction

Minor flow paths having extreme area variations or flow splits, in which the minor flow is a small fraction of the main flow (< 0.1), can also be modeled using the standard junction by the following special procedures. The smooth area change option is used for the junction (the *efvcahs* flag with $a = 0$), and the junction area is allowed to default (the minimum area of the adjoining volume areas). It may be necessary for the user to input a more reasonable flow area if the default area is too large. With this specification, it is necessary to enter user-input form loss coefficients normalized to the default area in order to give the proper flow rate and pressure drop relationship. The loss factor to be input can be estimated using the following equation:

$$K = 2\Delta P A^2 \frac{\rho}{\dot{m}^2} \quad (2.3-3)$$

where

K = loss factor

ΔP = nominal pressure drop (Pa)

A = junction area (m^2)

ρ = fluid density (kg/m^3)

$\dot{m}$ = nominal mass flow rate (kg/s)

The value computed for K in this way may be very large because the default area is much larger than the actual flow area. Also, critical flow would not be detected with this approach. Both the forward and reverse loss coefficients should be equal unless there is a reason why they are physically different. In this case, Equation (2.3-3) should be used to calculate the effective loss factor for both the forward and reverse flow conditions (i.e., assume ΔP and $\dot{m}$ also correspond to the reverse flow case). The geometric relationship between the actual situation and the model is illustrated schematically in **Figure 2.3-6**.

In the case of minor flow paths that connect at branches having large main flows, a similar approach can be used. In this case, let the junction area default to the minimum of the adjoining volumes (presumably the area of the minor flow path) and use the smooth option (*efvcahs* with $a = 0$). The determination of the loss factor may require some experimentation because of the possible large momentum flux effect, which is ignored in the derivation of Equation (2.3-3). If one of the volumes is quite large compared to the other, a modified Bernoulli equation can be used in which the overall loss factor defined by Equation (2.3-3) can be replaced by $K + 1$. [In other words, the user-input loss factor is computed by substituting $K + 1$ for K in Equation (2.3-3).]

All of the development herein assumes that known pressure drop flow relations exist for the single-phase case and that compressibility effects are small. If such is not the case, then the effective loss factor values must be determined experimentally by running the code for a series of cases. Some experimentation may be required, since the actual momentum flux calculation is complicated by several factors and may differ slightly from the simple Bernoulli form.

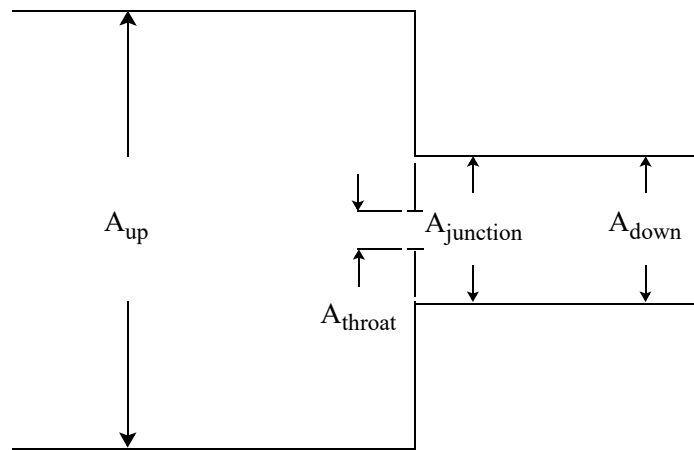
Another problem relative to a minor leak path can occur when an incorrect flow rate through an orifice for a given ΔP and loss coefficient K is calculated. As noted previously, this problem can be avoided if the user inputs reasonable values for the flow area and the loss coefficient K, rather than allowing the flow area to default and using a very large K.

2.3.4 Reflood Model

The reflood model is designed so that it can be activated at low pressure (less than 1.2×10^6 Pa) with nearly empty conditions (average void fraction in connected stack of hydraulic volumes > 0.9) or dryout beginning (average void fraction > 0.1) or by user command through a trip. The model considers a heat-structure geometry composed of 1 to 99 heat structures as a reflood unit. As there is no input specification for the length of a structure (except for the heat structure surface), such length is inferred from the length of the boundary volume connected to the heat structure. It is the user's responsibility to make certain that the length of a heat structure corresponds to the length of its connected volume for reflood calculations.

Additional suggestions concerning the use of the reflood model are listed below:

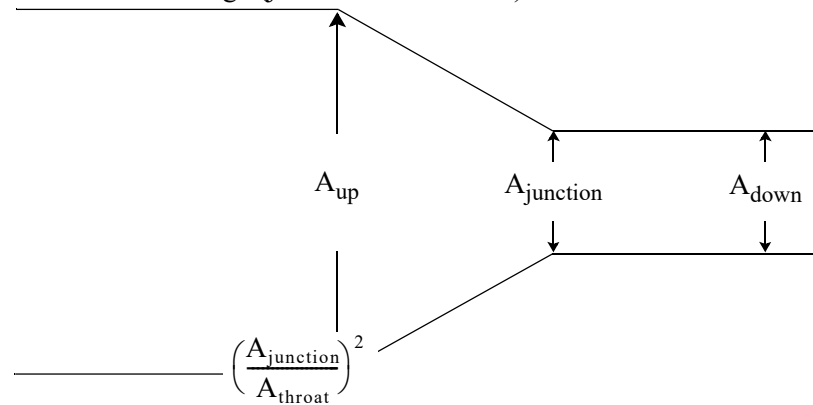
Abrupt area change (jefvcahs = 00000100)



$$\text{Throat ratio} = \frac{A_{\text{throat}}}{A_{\text{junction}}}$$

Physical situation with a loss factor K_{actual}

Smooth area change (jefvcahs = 00000000)



$$K_{\text{effective}} = K_{\text{actual}}$$

Equivalent model with effective loss factor for the same pressure drop-flow relation

Figure 2.3-6 High-Resistance Flow Path Model

1. The appropriate user-specified maximum number of axial fine mesh intervals is 8 to 32. No significant differences have been found in using 16 to 128 axial nodes for 0.18-m (0.6 ft) long heat structures.
2. The appropriate length of hydrodynamic volumes is 0.15 to 0.61 m (0.5 to 2.0 ft).

3. The maximum user-specified time step size is 0.01 to 0.05 seconds.
4. Each reflood unit should have its own flow channel and parallel flow channels should be connected by crossflow junctions.

The number of heat-structure geometries that can be specified for a reflood calculation is limited only by computer storage capacity. Once the reflood model is activated for a particular heat-structure geometry, only the structures where the critical heat flux are located will have a value in the critical heat flux column of the output. The heat transfer modes that appear in the mode column of the major edit are the same as those that appear when reflood is not activated except that 40 is added (see Section 3.2).

2.3.5 Noncondensables

The noncondensable model has the ability to be applied at every hydrodynamic volume in a system model. While in operation, the model affects interface mass and heat transfer, wall heat transfer, and the output of several variables that may cause discontinuities in plotted output. The purpose of this discussion is to clarify the operation of the noncondensable model and its affect on the calculated results and to give guidance for its use in system calculations.

In order to properly understand the operation of the noncondensable model, the fundamental assumptions used in the model need to be discussed. First, the steam/noncondensable mixture is assumed to be in thermal equilibrium. Second, the total pressure is the sum of the partial pressures of the steam and the noncondensable. Next, the specific vapor internal energy is the mass weighted sum of the steam specific internal energy and the noncondensable specific internal energy. Last, the saturation properties of the liquid and steam are assumed to be a function of the partial pressure of the steam.

One of the effects of these assumptions is to force the phasic temperatures and the saturation temperature to the same value. This causes a reduced driving potential for the interface mass and heat transfer models. The interface heat transfer coefficients are reduced in the presence of noncondensables. Consequently, low interfacial heat transfer regimes, such as the vertical stratification flow regime, may give heat transfer coefficients that are too low for stable calculations (as evidenced by oscillatory behavior). When this occurs, the vertical stratification model should be turned off on a volume basis. The highest probability for this occurrence is under very low flow conditions.

A second problem may occur when noncondensables first appear in a system volume. At times, again depending on the convection of noncondensable into the volume, the noncondensable iteration may fail or water property errors occur at the minimum time step. This problem can usually be overcome by reducing the size of the minimum time step. If this procedure fails, the convection rate or the concentration of the noncondensable convection must be changed. This may be accomplished by modifying the boundary conditions or by renodalizing the problem areas with acceptable thermodynamic conditions.

Last, the output from the code may contain discontinuities as noncondensables appear or disappear. The variables that will have these discontinuities are the partial pressure of steam, phasic temperatures,

saturation temperature, vapor specific internal energy, and noncondensable quality. The partial pressure of steam is set to the system pressure if noncondensables are not present, to 1.0 Pa if the volume state is pure noncondensable, or to the calculated value otherwise. When a pure noncondensable is present in a volume, all temperatures are set to equal values and are a function of the gas energy. As a fluid is injected into the volume, the temperatures, the partial pressure of steam, and possibly the vapor energy will abruptly change to new values based on the calculated thermodynamic conditions. Additionally, the liquid and saturation temperature may appear at the fluid triple point value if the partial pressure of the steam is calculated to be lower than the minimum fluid property table value.

As an example, a checkout problem used for development^{2,3-3} consisted of 322 K liquid water being injected into 436 K helium. The liquid temperature and saturation temperature both changed from 436 to 273 K in one time step as the volume changed from a pure noncondensable state to a steam/noncondensable mixture state. As more water was injected, the liquid transitioned to the correct value.

Selecting noncondensable input consists of specifying type and concentration of species on Cards 110 and 115 and by selecting options 4, 5, and 6 on the volume initial condition cards. Option 4, which consists of pressure, temperature, and static quality at equilibrium conditions, is the easiest to use. A restriction on the temperature is that it has to be less than the saturation temperature as a function of pressure. Little experience has been obtained in using option 5 (equilibrium), and it has not been checked out. Option 6 (nonequilibrium or equilibrium) is generally used to renode system models from Pygmalion decks.^{2,3-4} See Volume I of this manual for the equations and variables used. If option 6 is selected, but noncondensable quality is 0.0, the coding uses option 0. The static quality used in option 4 (equilibrium) is given by $\frac{M_g}{M_g + M_f}$, where $M_g = M_s + M_n$.

The capability for initializing and performing transients with pure noncondensables has been implemented. Input options 4 and 6 have provisions for initializing a system volume to a pure noncondensable state. This is accomplished in option 4 by using the equilibrium quality variable as a flag. By setting this quantity to 0, the logic specifies an ideal noncondensable gas equation of state. The variable is reset to 1.0 for transient calculations. To use option 6 for initializing a pure noncondensable, both void fraction and noncondensable quality must be set to 1.0 and the vapor energy must be set to a value that gives the desired gas temperature.

The code will not allow a noncondensable to exist as just liquid water and no steam. The code will add a little bit of steam (keeps steam quality $X_g = \frac{M_g}{M_g + M_n} \geq 10^{-8}$). Thus, we recommend that users input some steam when noncondensables and liquid water are present.

At the present time, option 4 is recommended. Only a saturated noncondensable state (100% humidity) is obtained by this option. Improvement of input conveniences for initial noncondensable states is under consideration. Users have reported some success with option 6 (nonequilibrium or equilibrium). The users employ the following method. For a given total pressure (P), gas temperature (T_g), liquid

temperature (T_f), void fraction (α_g), and noncondensable gas quality $\left(X_n = \frac{M_n}{M_n + M_g}\right)$, first calculate the steam quality ($X_s = 1 - X_n$). The partial pressure of steam (P_s) is determined from the relation $P_s = X_s \cdot P$, which is true only if both steam and noncondensable are ideal gases, obey Dalton's mixture law, and have similar molecular weight. Using steam tables to get the steam energy (U_s) from the known values of P_s and T_g ; then, calculate the noncondensable energy (U_n) from Equations in Volume I of the code manual. The mixture gas energy (U_g) is then calculated from equations in Volume I of the code manual. Finally, the liquid energy (U_f) is determined using steam tables from the known values of P and T_f .

Option 4, with static quality = 1.0, is recommended for containment volumes.

2.3.6 Water Packing

The volume control flag p is used to activate the water packing mitigation scheme. The scheme is invoked if the detection criteria are met.

The number of partial time step repeats is shown in the hydrodynamic volume statistics block (time step control information) in the major edit. Both the number of repeats since the last major edit and for the whole calculation are shown.

2.3.7 Countercurrent Flow Limitation Model

The countercurrent flow limitation (CCFL) model (discussed in detail in Section 3 of Volume I) is controlled by the junction control flag. The CCFL flag ($\underline{f}$) can be used with a single-junction, pipe, annulus, branch, valve, pump, and multiple junction. It cannot be used with a time-dependent junction, separator, jet mixer, ECC mixer, turbine, or accumulator. Setting $\underline{f} = 1$ will activate the CCFL model if all other conditions are met, and setting $\underline{f} = 0$ will not activate the model. The other conditions are as follows:

1. The orientation of both the connecting volumes cannot be horizontal (i.e., the elevation angle must be greater than or equal to 45 degrees).
2. Both gas and liquid phases must be present.
3. Countercurrent flow must exist, with liquid flowing down and gas flowing up.

As with the choking model, we recommend that if a junction is designated CCFL ($\underline{f} = 1$), then an adjacent junction should not be designated CCFL ($\underline{f} = 0$). It is anticipated that this flag will find use in activating the CCFL model in such internal structures as the upper core tie plate, downcomer annulus, steam generator tube support plates, and entrance to the tube sheet in the steam generator inlet plenum.

Junction data cards can be used to input four quantities (junction hydraulic diameter, correlation form, gas intercept, and slope). For these CCFL junction data cards, all four quantities must be entered (must have five quantities for pipe and multiple junction). If no card is entered but the CCFL flag $\underline{f}$ is set to 1, then default values of the four quantities will be used. Presently, the default values are

$$D_j = 2 \left(\frac{A_j}{\pi} \right)^{1/2}$$

$$\beta = 0$$

$$c = 1$$

$$m = 1$$

This corresponds to a Wallis CCFL correlation with a gas intercept of 1 and a slope of 1, which, according to Wallis,^{2.3-5} is the case for turbulent flow ($m = 1$) and when end effects are minimized ($c = 1$).

The input was made general so that the user can input CCFL correlations for the particular geometry of interest. Wallis,^{2.3-5} Bankoff et. al.,^{2.3-6} and Tien et al.^{2.3-7} discuss numerous examples, and these, along with other references, should be consulted in order to justify the use of a particular correlation for a given geometry. Wallis suggests $m = 1$ for a turbulent flow, $c = 0.725$ for tubes with sharp-edged flanges, and $c = 0.88$ to 1.0 for tubes when end effects are minimized. Bankoff suggests $\beta = \tanh(\gamma k_c D_j)$, where the critical wave number $k_c = \frac{2\pi}{t_p}$ corresponds to the maximum wavelength that can be sustained on a interface of length t_p (the plate thickness), and γ is the perforation ratio (fraction of plate area occupied by holes). Bankoff suggests $m = 1$ and c of the form

$$c = 1.07 + 4.33 \times 10^{-3} D^* \quad D^* < 200 \quad (2.3-4)$$

$$c = 2 \quad D^* \geq 200 \quad (2.3-5)$$

where D^* is a Bond number defined as

$$D^* = n\pi D \left(g \frac{\rho_f - \rho_g}{\sigma} \right)^{1/2} \quad (2.3-6)$$

and n is the number of holes. Tien uses the Kutateladze form ($\beta = 1$), but the form of c allows the Wallis form also to be invoked for small diameters. He suggests c of the form

$$c = c_7 [\tanh c_8 (D^*)^{1/4}] \quad (2.3-7)$$

where D^* is a Bond number defined differently from Equation (2.3-6) as

$$D^* = D \left[\frac{g(\rho_f - \rho_g)}{\sigma} \right]^{1/2} \quad (2.3-8)$$

The values of m , c_7 , and c_8 Tien found for four different conditions are provided in **Table 2.3-1**. With regard to guidelines for plant-specific geometry (i.e., tie plates, support plates, etc.), flooding data obtained in measurements from the plant geometry should be used to generate an appropriate CCFL model that can be input with CCFL junction data cards.

Table 2.3-1 Values of m , c_7 , and c_8 for Tien's CCFL Correlation Form

Tests	m	c_7	c_8
Nozzle air supply with tapered inlet	0.8	2.1	0.9
Nozzle air supply with sharp edge inlet	0.8	2.1	0.8
Indirect air supply with tapered inlet and sharp edge output	0.65	1.79	0.9
Indirect air supply with sharp edge inlet and tapered output	0.65	1.79	0.8

Wallis,^{2.3-5} Bankoff,^{2.3-6} and Tien^{2.3-7} discuss the effects of viscosity, surface tension, and subcooling on the correlations. At the present time, these effects have not been directly incorporated into the form of the CCFL correlation used in RELAP5. We anticipate that these, particularly the subcooling effects, will be addressed in future modifications to the code.

2.3.8 Level Tracking Model

The volume control flag 1 in tlpvbfe is used to activate the level tracking model as described in Volume I. If more than one junction is connected to the top of the volume or if more than one junction is connected to the bottom of the volume, the mixture level model is not used and is turned off.

If the volume control flag is set, the major edit will print out parameters associated with the mixture level in the hydrodynamic volume. The parameters are voidla, the void above the level; voidlb, the void below the level; vollev, the location of the level within the volume; and vlev, the velocity of the level movement. The parameters voidla, voidlb, and vollev, can also be written to the restart-plot file if a 2080XXXX card is used and can be used in minor edit requests.

2.3.9 Thermal Stratification Model

The volume control flag $\bar{t}$ in tlpvbfe is used to turn on the subcell resolution scheme in the model. The model is invoked if the detection criteria are satisfied. The model is intended for one-dimensional components only.

The thermal stratification model should be used to improve the accuracy of calculations where there is a warm fluid layer appearing above a cold fluid in a vertical stack of cells. A complete description of the model is presented in Section 3 of Volume I.

2.3.10 Energy Conservation at an Abrupt Change

The junction control flag $\bar{e}$ in jefvcahs is used to activate the modification to the energy flux term described in Volume I. This model is recommended for break junctions that connect to containment volumes that are modeled using regular volumes (not time-dependent volumes).

2.3.11 Jet Junction Model

A single-junction (sngljun) component may be flagged as a jet junction. A $\bar{j}$ value of 1 on the junction flags (jefvcahs) labels the junction as a jet junction. Jet junctions are used where subcooled water is injected into the bottom of a stratified pool. The flag activates logic to increase the condensation rate on the surface of the pool. Condensation is only enhanced in the volume above the jet when the volume is in vertical stratification or a level exists from the level model. Normally the letters VST appear in the major edits and 13 appears for the flow regime number (florege) in the minor edits and on the restart-plot file when vertical stratification exists in a volume and the letters LEV appear on the major edits and 14 appears for the flow regime number (florege) in the minor edits and on the restart-plot file when a level exists in a volume from the level model. However, when the liquid interfacial coefficient has been altered by the jet junction model and the volume is either in vertical stratification or a level exists from the level model, the letters JET appear in the major edits and 15 appears for the flow regime number (florege) in the minor edits and on the restart-plot files.

2.3.12 References

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- 2.3-2. *Ibid.* pp. 61-63.
- 2.3-3. K. E. Carlson. *Improvements to the RELAP5/MOD3 Noncondensable Model*. EGG-EAST-8879. Idaho National Engineering Laboratory. January 1990.
- 2.3-4. W. H. Grush. *Pygmalion*. Idaho National Engineering Laboratory software. 1994.
- 2.3-5. G. B. Wallis. *One-Dimensional Two-Phase Flow*. New York: McGraw-Hill. 1969. pp. 336-341.

- 2.3-6. S. G. Bankoff, R. R. Tankin, M. C. Yuen, and C. L. Hsieh. "Countercurrent Flow of Air/Water and Steam/Water Through a Horizontal Perforated Plate." *International Journal of Heat and Mass Transfer*. 24. 1981. pp. 1381-1385.
- 2.3-7. C. L. Tien, K. S. Chuent, and C. P. Lin. *Flooding in Two-Phase Countercurrent Flow*. EPRI NP-984. February 1979.

2.4 Hydrodynamic Components

The basic two-fluid model is applied uniformly to all volumes and junctions. Thus, the programming design of the hydrodynamic calculation is primarily organized on volumes and junctions. Components are organized collections of volumes and junctions and, to a lesser extent, the program is organized on components. Components are designed for either input convenience or to specify additional specialized processing. A pipe component is an example of a component designed for input convenience, since by taking advantage of typical features of a pipe, several volumes and junctions can be described with little more data than for one volume. Pump and valve components are examples of components requiring additional processing. A pump component includes data defining pump head and torque characteristics for single-phase and two-phase conditions as a function of pump angular velocity. A pump component requires additional processing to advance the differential equation defining pump angular velocity. A valve component requires additional data defining its characteristics and additional processing to calculate the junction flow area as a function of valve position.

Components are numbered with a three-digit number, 001 - 999. Components need not be in strictly consecutive order so that changes to a model of a hydrodynamic system requiring addition or deletion of components are easily made. Volumes and junctions within a component are numbered by appending a six digit number to the component number, CCCXXYYZZ. The CCC is the component number. At present, YYZZ are zeros and XX is numbered consecutively starting at 01 for the volumes and junctions in the one-dimensional components presently defined.

2.4.1 Common Features of Components

Each volume's flow area, length, and volume must be supplied as input. As noted above, each one-dimensional volume has a x-coordinate direction along which fluid flows in a positive or negative direction, and may have y- and z-coordinate directions if crossflow connections are made to the volume. The x-volume flow area is the volume cross-sectional area perpendicular to the x-coordinate direction. The x-volume length is the length along the x-coordinate direction and similarly for the y- and z-coordinate directions. The hydrodynamic numerical techniques require that the volume be equal to the volume flow area times the length for each coordinate direction. This requirement is easily satisfied for constant area volumes, but poses difficulties for irregular shaped volumes. Since it is very important that such a systems code as RELAP5 conserves mass and energy, with momentum being an important but lesser consideration, we recommend that an accurate volume be used; that the volume flow area be the cross-sectional area averaged over the actual length of the volume; and the volume length be the quotient of the volume and the flow area. The component input routines permit the volume, flow area, and length of each volume to be

entered as three nonzero positive numbers or two nonzero positive numbers and a zero. If three nonzero quantities are entered, the volume must equal the flow area times length within a relative error of 0.000001. If one quantity is zero, that quantity is computed from the other two. The user need not be concerned with y- or z-coordinate data unless crossflow connections are made, and even then only if the default data for those coordinates are not satisfactory.

The volume horizontal angle specifies the orientation of the volume in the horizontal plane. The code numerics have no requirement for this quantity; they were entered so a graphics package could be developed to show isometric views of the system as an aid in model checking. Such a graphics package is not included in RELAP5/MOD3.3lo. The horizontal angle is checked to verify that its absolute value is less than or equal to 360 degrees, but no further use is made of the quantity.

The volume vertical angle specifies the vertical orientation of the volume. This quantity would also be used in the graphics package and, in addition, specifies the vertical orientation of the volume coordinate direction. The vertical angle must be within the range 90 to -90 degrees. The angle 0 degrees means the x-coordinate direction is in the horizontal plane; a positive angle means that the coordinate direction is directed upward; a negative angle means it is directed downward. Slanted vertical orientation, such as an angle equal to 45 degrees, is permitted. Note that as the vertical angle changes from zero, the y-coordinate is always in the horizontal plane, and that the x- and z-coordinates, and their associated faces move out of their original horizontal and vertical planes, respectively.

The coordinate direction implies the position of the inlet and outlet ends of the volume. The terms inlet and outlet are convenient mnemonics relative to the coordinate direction but do not necessarily have any relation to the fluid flow. The direction of fluid flow is indicated by the sign of the velocity relative to the coordinate direction. For input convenience or ease in output interpretation, the coordinate direction should be an easily remembered direction, such as the normal flow direction as opposed to the flow in an accident situation.

As noted in the discussion of **Figure 2.1-5**, and described further below, a junction connects a specified end of one volume to the specified end of another volume. This, in turn, establishes relative positioning of the volumes. Because of gravity heads, the relative position is important to any volume with a nonzero vertical component of a volume coordinate direction. If the coordinate direction in a volume with a vertical component is reversed but no other changes are made, the inlet and outlet ends of the volumes are also reversed. The physical problem is changed since the relative vertical positions of the volumes are changed. If appropriate changes are made to junctions connecting the reversed volume, such that the physical problem remains unchanged, the only change in the problem results would be a reversal in the sign of the vector quantities associated with the volume. Furthermore, given a stack of vertically oriented volumes, the proper gravity head is computed whether the direction coordinates are all upward, all downward, or any random distribution. This assumes that junction connections are such that a vertical stack is specified. As shown in **Figure 2.1-5**, a change in junction specification can change the relative position of two volumes from two, vertically stacked volumes to a U-tube configuration.

Input for a volume includes the elevation change in a volume. For a straight pipe, the elevation change (Δz) is related to the volume length (Δx) and the vertical angle (ϕ) by

$$\Delta z = \Delta x \sin \phi \quad (2.4-1)$$

Note that the elevation change associated with the x-coordinate has the same sign as the vertical angle. To allow for irregularly shaped and curved volumes, the input elevation change is used for gravity head calculations. Input checks are limited to the following: the magnitude of the elevation change must be equal to or less than the volume length; and for the x-direction, the elevation change must be zero if the vertical angle is zero; and the elevation change must be nonzero and have the same sign as the vertical angle if the vertical angle is nonzero. The volume input does not need the elevation height of a volume relative to an arbitrary base. The elevation change data performs the same function in determining gravity heads.

If the hydrodynamic system has one or more loops, the user must ensure that the sum of the elevation changes of the volumes in each loop is zero. A loop is any hydrodynamic flow path starting at a volume, passing through one or more other volumes, and returning to the starting volume. If the net elevation change in a loop is not zero, an incorrect gravity head exists; this is comparable to having an undesired pump in the loop. This error is checked by the program. If closure is not within the error criterion of 0.0001 m, an input error will result.

The horizontal flow regime map is used rather than the vertical flow regime map when ϕ is less than or equal to 45 degrees. Horizontal flow calculations include a horizontal stratified flow capability and a horizontal stratified entrainment model.

Wall friction effects are computed from pipe roughness and hydraulic diameter data entered for each volume. If the input x-direction hydraulic diameter ($D_{h,x}$) is zero, it is computed from the x-direction volume flow area ($A_{v,x}$),

$$D_{h,x} = \sqrt{\frac{4A_{v,x}}{\pi}} \quad (2.4-2)$$

If the input y-direction hydraulic diameter ($D_{h,y}$) is zero, it is computed from the y-direction volume flow area ($A_{v,y}$) and the x-direction volume flow area ($A_{v,x}$), using the formula

$$D_{h,y} = \frac{4A_{v,y}}{(\pi A_{v,x})^{0.5}} \quad (2.4-3)$$

If the input z-direction hydraulic diameter ($D_{h,z}$) is zero, it is computed from the z-direction volume flow area ($A_{v,z}$) and the x-direction volume flow area ($A_{v,x}$), using the formula

$$D_{h,z} = \frac{4A_{v,z}}{(\pi A_{v,x})^{0.5}} \quad (2.4-4)$$

The thermal stratification (t), level model (l), water packer (p), vertical stratification (v), and equilibrium (e) flags can be entered only for the x- coordinate direction. The friction (f) flag can be entered for the x-, y-, and z-coordinate directions.

A check is made that the pipe roughness is less than half the hydraulic diameter.

Most volumes allow seven control flags (tlpvbfe): the t flag is the thermal stratification flag; the l flag is the level model flag; the p flag is the water packer flag; the v flag is the vertical stratification flag; the b flag is the bundle interphase friction flag, recommended for cores and steam generators; the f flag determines whether wall friction from the volume is to be included or neglected; the e flag controls whether a nonequilibrium (two phases permitted to have unequal temperature) or an equilibrium (two phases forced to have equal temperatures) calculation is used. Generally we recommend that wall friction be computed in the x-direction and usually not in the y-direction, and that the nonequilibrium equation of state be used.

System volumes require initial thermodynamic state conditions, and time-dependent volumes require state conditions as a function of time or a time-advanced quantity. Seven options, numbered 0 through 6, are available to specify state conditions. Options 0 through 3 specify water-only conditions and do not allow a noncondensable gas. Option 0 requires pressure, liquid specific internal energy, vapor specific internal energy, and void fraction. Options 1 and 2 always specify saturation conditions. Option 1 requires saturation temperature and equilibrium quality. Option 2 requires saturation pressure and equilibrium quality. Two phases are present if the quality is neither 0 nor 1. Option 3 always specifies single-phase conditions and requires pressure and temperature. Options 1 through 3 can only specify equilibrium conditions even if the nonequilibrium equation of state is requested. The next three options can specify the presence of a noncondensable gas. Option 4 requires pressure, temperature, and equilibrium quality. Equilibrium conditions are assumed, and the vapor consists of air and water vapor at 100% humidity. Option 5 requires temperature, equilibrium quality, and noncondensable quality. Option 6 requires pressure, liquid specific internal energy, vapor specific internal energy, void fraction, and noncondensable quality.

For options 0 through 6, the boron concentration is assumed to be zero. If 10 is added to the above option numbers, a boron concentration is required. See Section 3 of Volume I for a complete description of the boron transport model. Boron is assumed to be only present in and to be convected by liquid water. If a volume with liquid and boron has the liquid water removed by convection, the boron is also removed. If the liquid water is evaporated, the boron remains. This is analogous to boron precipitating out as water is evaporated. Infinite solubility of boron in water is assumed and boron remains in solution regardless of its

concentration until all of the liquid water disappears. Boron instantly redissolves the instant the quality becomes less than 1. Boron concentrations are computed using only a boron continuity equation for each volume. Boron is assumed to have no momentum, no internal energy, and to have no effect on the equation of state.

Junctions connect two volumes by specifying a connection code for each volume. The connection code specifies both the volume and a specific face of the volume. Except for a pipe component, current components have only one volume, and a component reference is essentially a volume reference. The connection codes for each component type are described in the beginning of the input description for that component, as well as in Section 2.1.

The junction coordinate direction is from one volume end to another volume end and the input description use the words FROM and TO to identify the connections. If a junction is reversed, the sign of the vector quantities associated with the junction are reversed. To maintain the same physical problem, no further changes are needed in other components. The initial velocities in reversed system junctions or time-dependent velocities in time-dependent junctions should also be reversed.

Two quantities, the junction area and the junction area ratio, are defined from the user-supplied junction area. These are printed in the major edit as JUN.AREA and THROAT RATIO. Junctions can connect two volumes with possibly different volume flow areas, and the junction can also have a different flow area. Two options are provided for calculating area change effects as the fluid flows through the upstream volume flow area, the junction flow area, and the downstream volume flow area.

The smooth area change option uses only the stream tube form of the momentum equation that includes spatial acceleration and wall friction terms. This option should be used when there are no area changes or when the area changes are smooth, such as in a venturi. There are no restrictions on the user-supplied junction area for smooth area changes, and the junction area may be smaller than, larger than, or between the adjacent volume flow areas. The junction flow area is set to the user-supplied flow area and the junction area ratio is set to 1.0.

The abrupt area change option ($a = 1$) provides for area apportioning at branches, restricted junction area, extra interphase drag (see Volume IV), and additional losses resulting from abrupt expansions, abrupt contractions, orifices, and vena-contracta effects. The abrupt area change option ($a = 2$) includes everything in the $a = 1$ option except for the losses. The full abrupt area change model ($a = 1$, code calculated losses) is recommended for sudden (i.e., sharp, blunt) area changes, while the partial abrupt area change model ($a = 2$, no code calculated losses, user input losses are to be used) is recommended for rounded or beveled area changes. The extra interphase drag term (see Volume IV) in the abrupt area model ($a = 1$ or 2), helps ensure more homogeneous flow that would be expected through a sudden area change.

The user-supplied junction area (for $a = 1$ or $a = 2$) must be equal to or less than the minimum of the adjacent volume flow areas for an abrupt area change. The junction area is then set to the minimum of the adjacent volume flow areas, and the junction area ratio is set to the ratio of the user-supplied junction area to the minimum of the adjacent volume flow areas. When the user-supplied flow area equals the minimum

of the adjacent volume flow areas, the junction area ratio is 1.0, and the junction is a contraction/expansion. Program logic checks flow direction, and an expansion with flow in one direction is treated as a contraction when flow reverses, and vice versa. If the user-supplied junction area is less than the minimum of the adjacent volume flow areas, an orifice is indicated, and the junction area ratio is less than one. Specifying the abrupt area change options ($a = 1$ or $a = 2$) when there is no area change gives the same result as specifying the smooth area change option, but slightly more computer time is required. Note, valve junctions using any area change options vary the junction area ratio as the valve opens and closes.

Junction velocities correspond to the junction flow area. Thus, the flow rate of a phase is the product of the appropriate junction void fraction, the junction density (donored quantities), the junction velocity, and the junction flow area. For orifices and valves, the actual flow velocity is higher at the minimum area, which is the junction area multiplied by the junction area ratio. The junction area ratio is used to compute the velocity at the minimum area when needed, such as in the choked flow model.

$$V_{\text{orifice}} = \frac{V_{\text{junction}} A_{\text{junction}}}{A_{\text{orifice}}} \quad (2.4-5)$$

For user convenience, if the user-supplied junction area is zero, it is set to the minimum of the adjacent volume flow areas regardless of the area change option. This is the proper default value for most junctions, and only smooth area changes (where there is an area change) and orifices need nonzero user-supplied junction areas.

Junction user-inputted loss coefficients can be entered when additional losses above or instead of the wall friction and abrupt area change losses are needed. These losses could arise from pipe bends, irregularly shaped volumes, entrance or exit losses, or internal obstructions. Since the abrupt area change loss model ($a = 1$) is only for sudden (i.e., sharp, blunt) area changes, it is in general too high and not applicable for rounded or beveled area changes. The user input loss coefficients without using $a = 1$, should be used for rounded or beveled area changes. Six coefficients are entered. Three are for forward (positive) flow, where one is for the Reynolds number independent part and two are for the Reynolds number dependent part. The other three are for reverse (negative) flow, where one is for the Reynolds number independent part and two are for the Reynolds number dependent part. The coefficients are applied to the junction dynamic pressure. Zero coefficients mean no additional losses are computed.

Table 2.4-1 shows the junction flow area, abrupt area change model loss, and user-input loss for area change options $a = 0, 1$, and 2 . The user input loss is optional in all three cases. For area change option $a = 1$, the abrupt area change model loss and the optional user-input loss are additive.

Table 2.4-1 Area Change Options

Area change option (a)	Junction flow area	Abrupt area change model loss	User-input loss
0	User-supplied flow area	No	Optional
1	Minimum of adjacent volume flow areas	Yes	Optional
2	Minimum of adjacent volume flow areas	No	Optional

If junction user-supplied loss coefficients are entered and the abrupt area change option ($a = 1$ or $a = 2$) is selected, the user should note that the junction area is set to the minimum of the adjacent volume flow areas rather than the user-supplied flow area. Thus, the user-supplied loss coefficient needs to be adjusted for the difference between the minimum of the adjacent volume flow areas and the user-supplied junction flow area. For example, if the loss coefficient associated with the user-supplied flow area is K_j , then the adjusted coefficient K'_j would be computed as:

$$K'_j = K_j \frac{A_v^2}{A_j^2} \quad (2.4-6)$$

where A_j is the user-supplied junction flow area and A_v is the minimum of the adjacent volume flow areas.

It is important to note that RELAP5/MOD3.3lo computes interfacial drag at junctions rather than within volumes. This has important implications with respect to modeling reactor core bundles and steam generator bundles. In these instances, the user should invoke the bundle interfacial drag model by specifying $b = 1$ on the pipe volume control flags data card. In addition, in modeling grid spacers as junctions within the core or steam generator bundle, the user should specify the junction area and hydraulic diameter as equal to that for the bundle, rather than those characteristics of the grid spacer. The reason for this is that the bundle interfacial drag model was formulated on the basis of bundle geometry. In order to achieve the correct pressure drop at each grid spacer junction, the user should input a loss coefficient that is adjusted for the difference between specifying bundle geometry rather than grid geometry. For example, if

the loss coefficient associated with the grid spacer is K_j , then the adjusted loss coefficient K'_j would be computed as

$$K'_j = K_j \frac{A_b^2}{A_g^2} \quad (2.4-7)$$

where A_g is the flow area of the grid spacer and A_b is the flow area of the bundle. In modeling the upper core tie plate, the user should specify the junction area and hydraulic diameter characteristics of the tie plate. The reason is the CCFL model was formulated based on actual geometry.

Eight control flags are associated with junctions (jefvcahs). The j flag is the jet junction flag, recommended at junctions where subcooled water is injected into the bottom of a pool. The e flag is the energy correction flag, recommended at break junctions into a containment. The f flag is the CCFL model flag, recommended for tie plates, downcomer annulus, etc. The v flag is the horizontal stratification entrainment/pullthrough model flag, recommended at break junctions connected to horizontal volumes. The c flag controls applications of the choking model.

The recommended input junction flags when the choking model is on ($c = 0$) are smooth area ($a = 0$) and nonhomogeneous ($h = 0$). Since the smooth area change option is recommended where the choking flag is active, the user will need to specify loss factors appropriate for the geometry of the choking plane (i.e., orifice, nozzle, etc.). The smooth and abrupt area options are discussed in Section 2.4.1. With regard to the nonhomogeneous ($h = 0$) option, it is generally recommended that $h = 0$ be used. There may be rare situations where the combined interphase drag is too low, resulting in too much slip and too low mass flow. For this situation, the homogeneous option ($h = 1$ or 2) is recommended. Note, the user should monitor the calculated results for nonphysical choking. If this occurs, the user should turn choking off ($c = 1$) at junctions where this occurs.

The a flag is for the area change option, and has already been discussed. The h flag controls the type of momentum treatment; two-velocity, or one-velocity models. The two-velocity model is recommended except as indicated above. The s flag controls whether the momentum flux is to be used.

System junctions require initial velocities, and time-dependent junctions require velocities as a function of time. Two options are available to specify the velocities. One option requires the velocities; the other requires mass flow rates from which the velocities are computed. If the flow is single-phase, the velocity of the missing phase is set to that of the flowing phase. This matches the transient calculation that computes equal phasic velocities when one phase is missing. The velocity conditions also require an interface velocity. This input is for future capability involving moving volume interfaces. For now, the interface velocity must be set to zero.

2.4.2 Time-Dependent Volume

A time-dependent volume must be used wherever fluid can enter or leave the system being simulated. The geometry data required are similar to system volumes, but during input processing the volume's length, elevation change, and volume are set to zero. With the staggered mesh, the pressure boundary would be applied in the center of the time-dependent volume. Setting these quantities to zero moves the boundary to the edge of the system volume.

The state conditions as a function of time or some time-advanced quantity are entered as a table, with time or the time-advanced quantity as the independent or search variable. The table must be ordered in increasing values of the search variable, and each succeeding value of the search variable must be equal to or greater than the preceding value. Linear interpolation is used if the search argument lies between search variable entries. End point values are used if the search argument lies outside the search variable entries. If constant state values are desired, only one set of data consisting of any search value and the associated constant data needs to be entered. The program recognizes when only one set of data is entered, and computer time is saved since the equation of state is evaluated only once rather than every time advancement. Step changes can be accommodated by entering two adjacent sets of data with the same time or an extremely small time difference.

The default search argument is time. If no trip number is entered, or if the trip number is zero, the current advancement time is used as the search argument. When a nonzero trip number is entered, a unit step function based on the time the trip was last set is applied. If the trip is false, the search argument is -1.0. When the trip is true, the search argument is the current advancement time minus the last time the trip was set. Thus, the search argument is always -1.0 when the trip is false and can range between zero and the problem time when the trip is true. A time-dependent volume could have some constant condition when the trip is false. In order to ensure proper operation for the constant condition, both a card with a negative search argument and a card with a zero search argument need to be entered (see example in Volume V). When the trip is true, it could follow a prescribed function of time where the time origin is the time of the trip not the start of the transient.

Through an input option, nearly any time-advanced quantity can be specified as the search argument. The allowed quantities are listed in the input description. The search variables in the table are assumed to have the same units as the search argument, and the table lookup, interpolation, and treatment of out of range arguments are identical to those described for the default time argument. However, handling of trips is different. If the trip number is zero, the current value of the specified time-advanced variable is used. If the trip number is nonzero, the time delay cannot be applied as for the default time case, since the search argument may not be time. Thus, if the trip is false, the search argument is -1.0E75; if the trip is true, the current value of the specified variable is the search argument.

When time is the search argument, the current value is the value at the end of the time step; for any other variable, the current value is the value at the beginning of the time step. Time is the default search argument, but time can also be specified as the search argument through the input option of naming a time-

advanced variable. These two uses of time as the search argument are different if a trip is used, since the default method can apply a time delay and the other cannot.

2.4.3 Time-Dependent Junction

Time-dependent junctions can be used whenever the phasic velocities or phasic mass flow rates are known as a function of time or other time-advanced quantity. Time-dependent junctions can connect any two system volumes, or a system volume and a time-dependent volume. Phasic mass flow rates are converted to phasic velocities using the upstream phasic densities. Examples of their use would be to model a constant displacement pump in a fill system, a pump or a valve (or both), by using an associated control system or measured experimental data. Time-dependent junctions are also used frequently in test problems to check code operation.

The phasic velocities or mass flows in the time-dependent junction as a function of time or the time-advanced quantity are entered as a table, with time or the time-advanced quantity as the independent or search variable. The requirements, interpolation, and trip logic are identical to that for time-dependent volumes.

The capability of using time-advanced quantities as search arguments can be used to model pressure-dependent water injection systems. If the injection flow is a function of the pressure at the injection point, the volume pressure at that point is used as the search argument. A trip is defined to be true when the injection system is actuated. Entry of table data with a negative pressure and zero flows causes the flow to be zero when the trip is false. In order to ensure proper operation for zero flow, both a card with negative pressure and a card with a zero pressure need to be entered (see example in Volume V). The remaining table entries define the injection flow as a function of positive pressures. The source of injection water is a time-dependent volume. The pressure of the water supplied by the time-dependent volume could also be a function of the pressure at the injection volume to represent the work of pumping the water into the system. If the injection flow is a function of a pressure difference, the pressure difference can be defined by a control system variable, and that control variable is then defined as the search argument.

Some uses of time-dependent junctions can cause modeling difficulties. When using a time-dependent junction to specify flow from a time-dependent volume into the system, the incoming phasic densities, void fractions, phasic velocities, phasic mass flow, and phasic energies can be specified. But when using a time-dependent junction to specify flow out of a system, the densities, void fractions, and energies of the fluid leaving the system are not known in advance. Thus, use of time-dependent junctions to control outflow is not recommended. The following is one example of a modeling problem. The user anticipates that a volume will contain only vapor and accordingly sets a time-dependent junction to a nonzero vapor flow and zero liquid flow. If the user anticipated incorrectly and liquid condenses or is carried into the volume, the liquid will accumulate unrealistically since it cannot leave the system.

In a simple pipe modeling application, a time-dependent volume and junction can be used to specify the inlet flow. Likewise, a time-dependent volume and junction can model the feedwater flow into a reactor steam generator. Controlling the fluid flow out of the pipe or controlling the water/steam flow out

of the steam generator through a time-dependent junction is not recommended. If a system junction (flows computed by the simulation rather than specified as boundary conditions) connected to a time-dependent volume is not sufficient, perhaps a servo valve can provide the required simulation.

2.4.4 Single-Volume

A single-volume component is simply one system volume. A single-volume can also be described as a pipe component containing only one volume. This single-volume component uses fewer input cards and fewer data items than does a pipe component. However, if the single-volume might be divided into several volumes for nodalization studies, we suggest the pipe component, since such changes are quite easy for pipes.

2.4.5 Single-Junction

A single-junction component is simply one system junction. It is used to connect other components such as two pipes. Initial junction conditions can be phasic velocities or phasic mass flow rates.

2.4.6 Pipe

A pipe component is a series of volumes and interior junctions, the number of junctions being one less than the number of volumes, and the junctions connect the outlet of one volume to the inlet of the next volume. Pipe components can be used for those portions of the system without branches. Pipe components offer input conveniences, since most characteristics of the volumes and junctions in a pipe are similar or change infrequently along the pipe, and input data requirements can be reduced accordingly. Because of the sequential connection of the volumes, junctions are generated automatically rather than being individually described. Although the input is designed to assume considerable similarity in volume and junction characteristics, any of the volume and junction features (such as flow area, orientation, pipe roughness, or control flags) can be changed at each volume or junction.

2.4.7 Branch

Branch components are provided to model interconnected piping networks. The branching model is based on one-dimensional fluid flow, which is adequate for most cases of branching and merging flow. Such situations include wyes, parallel flow paths from upper and lower plenums, and any branch from a vessel of large cross-section. For branching situations where phase separation effects caused by momentum or gravity are important, an approximate mapping technique can be used to map the two-dimensional situation into the one-dimensional space of the fluid model.

A branch component consists of one system volume and zero to nine junctions. The limit of nine junctions is due to a card numbering constraint. Junctions from other components, such as single-junction, pump, other branch, or even time-dependent junction components, may be connected to the branch component. The results are identical whether junctions are attached to the branch volume as part of the branch component or are in other components. Use of junctions connected to the branch, but defined in

other components, is required in the case of pump and valve components and may also be used to attach more than the maximum of nine junctions that can be described in the branch component input.

2.4.8 Pump

The pump component model can be separated into models for hydrodynamics, pump-fluid interaction, and pump driving torque. The pump component input provides information for the hydrodynamic and pump-fluid interaction models and may optionally include input for an electric motor to drive the pump. A pump may also be connected to a shaft that is a specialized component within the control system. A shaft component is used when the pump is driven by a turbine or by an electric motor with a control system to regulate speed.

2.4.8.1 Pump Model Description

The hydrodynamic model of a pump component consists of one volume and two associated junctions. The coordinate directions of the junctions are aligned with the coordinate direction of the volume. One junction is connected to the inlet and is called the suction junction; the other junction is connected to the outlet and is called the discharge junction. The pump head, torque, and angular velocity are computed using volume densities and velocities. The head developed by the pump is divided equally and treated like a body force in the momentum equations for each junction. With the exception of the head term, the hydrodynamic model for the pump volume and junctions is identical to that for normal volumes and junctions.

2.4.8.1.1 Pump Performance Modeling

Interaction of the pump and the fluid is described by empirically developed curves relating pump head and torque to the volumetric flow and pump angular velocity. Pump characteristic curves, frequently referred to as four-quadrant curves, present the information in terms of actual head (H), torque (τ), volumetric flow (Q), and angular velocity (ω or N). These data are generally available from pump manufacturers. For use in RELAP5, the four-quadrant curves must be converted to a more condensed form, called homologous curves, which use dimensionless quantities. The dimensionless quantities involve the head ratio, torque ratio, volumetric flow ratio, and angular velocity ratio, where the ratios are actual values divided by rated values. The rated values are also required pump component input and correspond to the design point or point of maximum efficiency for the pump.

The homologous curves are entered in tabular form, and the dependent variable is obtained as a function of the independent variable by a table search and linear interpolation scheme. There is a separate set of curves for head and torque, and each set is composed of eight curves. Not all the regimes need be described by the input, but a problem is terminated if an empty table is referenced. Both head and torque data must be entered for the regimes that are described with input.

The homologous curves for pump head and torque are for single-phase operation. These same tables are used by the two-phase pump model, but additional data must be input to model two-phase degradation effects.

Pump head data are always used in the momentum equations. Torque data may or may not be used in computing pump rotational velocity, depending on the pump motor model selected and if it is energized or not. However, both head and torque are used to determine pump energy dissipation, and consistent data must therefore be entered. The pump homologous data should be checked by computing pump efficiency from the homologous data. No such checking is currently included in RELAP5, nor is the operating efficiency edited on major edits.

The sign conventions for various pump quantities are as follows: a pump operating in the normal pump regime has a positive angular velocity; the volumetric flow is positive if it is in the same direction as the volume coordinate direction; the head is positive if it accelerates the flow in the volume coordinate direction; and the torque is that exerted by the fluid on the pump and is negative if it tends to decelerate the pump. In normal pump regimes and in steady-state, this torque is negative and is balanced by the positive torque from the pump motor.

2.4.8.1.2 Pump Data Homologous Representation

The use of pump performance data in terms of nondimensional homologous parameters is often confusing. The purpose of this discussion is to briefly outline rules for a procedure to properly use the homologous data.

The homologous parameters for pumps are obtained from dimensional analysis that can only provide the conditions for similarity. Three independent parameters are obtained from application of Buckingham's Pi theorem.^{2.4-1} They are

$$\pi_1 = \frac{Q}{vD} \quad (2.4-8)$$

$$\pi_2 = \frac{NQ^{1/2}}{(gH)^{3/4}} \quad (2.4-9)$$

$$\pi_3 = \frac{Q}{ND^3} \quad (2.4-10)$$

A fourth parameter that is commonly used can be obtained by a combination of π_2 and π_3 to yield

$$\pi_4 = \frac{gH}{N^2 D^2} \quad (2.4-11)$$

The first parameter, π_1 , is analogous to a Reynolds number and is the only parameter involving the fluid kinematic viscosity, ν . Experience with pump design and scaling has shown that viscous effects caused by skin friction are small, especially for high Reynolds number flows, and, in practice, the requirement to maintain π_1 constant is not used. The use of π_2 , π_3 , and π_4 to correlate pump performance has proven quite useful. The parameter π_2 is called the specific speed and is often used as a single parameter to characterize the type of pump impeller best suited for a particular application. In practice, the acceleration of gravity, g , is omitted, and the specific speed is simply defined as

$$N_s = \frac{NQ^{1/2}}{H^{3/4}} \quad (2.4-12)$$

where the speed N is in rpm, the capacity Q is in gpm, and the head H is in ft. In this form, N_s is not dimensionless but has a history of usage that still persists.

Two performance parameters that are used for pump modeling are the specific nondimensional capacity,

$$Q_s = \frac{Q}{ND^3} \quad (2.4-13)$$

and the specific head (dimensional due to omission of the gravitational acceleration constant),

$$H_s = \frac{H}{N^2 D^2} \quad (2.4-14)$$

The D that appears in Equations (2.4-13) and (2.4-14) is a characteristic dimension of the pump and is assumed to be the impeller diameter. When scaling pump performance using homologous parameters, the implication is that all pump dimensions are geometrically similar (i.e., changing D implies a proportional change in impeller width, leakage paths, and in all linear dimensions of the pump).

When the pump torque performance is included, one additional dimensionless parameter is obtained from dimensional analysis and is

$$\pi_5 = \frac{\tau}{\rho N^2 D^5} \quad (2.4-15)$$

where π_5 is the nondimensional specific torque.

Generally, constant density is assumed, so the dimensional specific torque used in constructing the homologous representation is reduced to

$$\pi_s = \frac{\tau}{N^2 D^5} \quad (2.4-16)$$

Homologous states are states for which specific capacity, head, and torque are all constant. Thus, at any state it is possible, within the limitations of similarity theory, to predict the performance for other combinations of speed, head, and flow that have the same homologous state. It is also possible to scale pump performance with reasonable accuracy to account for changes in physical pump size through the diameter D by keeping the homologous parameters fixed.

Pump performance data are usually displayed by plotting head and torque as functions of speed and volumetric flow. **Figure 2.4-1** is a four-quadrant pump curve for the Semiscale MOD1 pump, and has speed and flow as independent variables with lines of constant head.

Figure 2.4-2 is a comparable four-quadrant plot of the Semiscale MOD1 pump torque data. All possible operating states of the pump are represented on such plots. These data for a particular pump can be approximately collapsed into a single curve by nondimensionalizing specific head and capacity parameters for corresponding homologous operating points using the design point values for head, capacity, and speed.

All points on **Figure 2.4-1** having the same specific capacity are straight lines passing through the origin (lines of constant Q/N). The impeller diameter is omitted from the homologous parameters since it is constant for a particular pump. The design operating point is indicated by the cross. The line of constant Q/N , passing through the design point and its reflection about the ordinate divides each quadrant into two

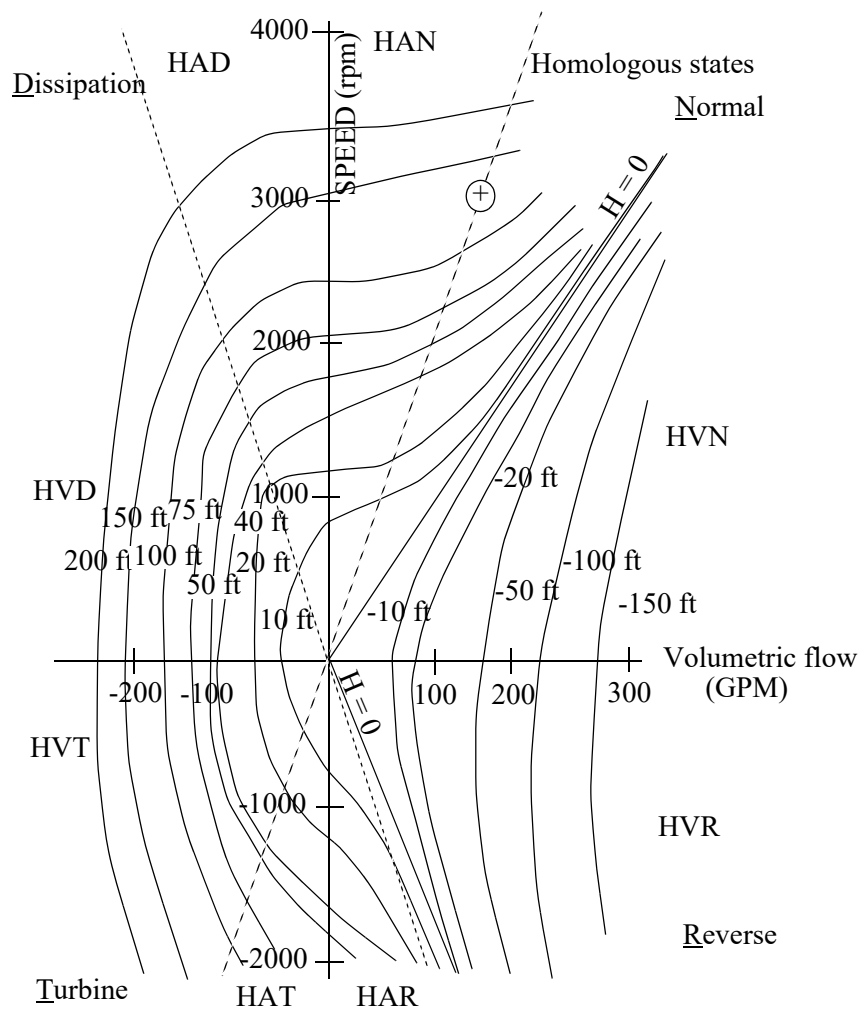


Figure 2.4-1 Four-Quadrant Head Curve for Semiscale Mod1 Pump (ANC-A-2083)

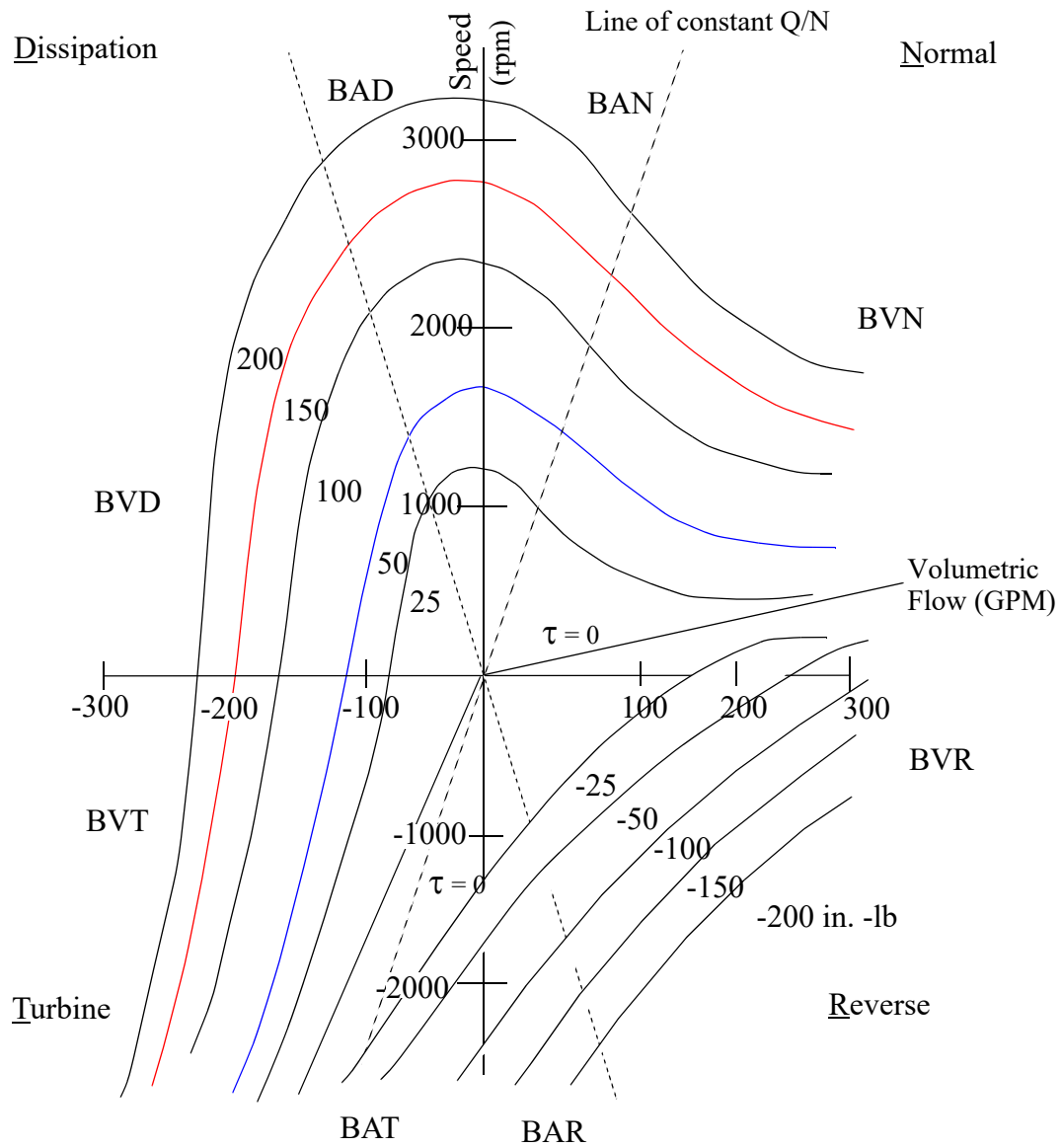


Figure 2.4-2 Four-Quadrant Torque Curve for Semiscale Mod1 Pump (ANC-A-3449)

octants. Each of these eight octants is named according to the convention listed in **Table 2.4-2**, for the purpose of constructing the homologous representation.

Table 2.4-2 Pump Homologous Curve Definitions

Regime number	Regime mode ID name	α	v	v/α	Independent variable ^a	Dependent ^a variable head	Dependent ^a variable torque
1	HAN BAN Normal pump	> 0	≥ 0	≤ 1	v/α	h/α^2	β/α^2
2	HVN BVN Normal pump	> 0	≥ 0	> 1	α/v	h/v^2	β/v^2
3	HAD BAD Energy dissipation	> 0	< 0	≥ -1	v/α	h/α^2	β/α^2
4	HVD BVD Energy dissipation	> 0	< 0	< -1	α/v	h/v^2	β/v^2
5	HAT BAT Normal turbine	≤ 0	≤ 0	≤ 1	v/α	h/α^2	β/α^2
6	HVT BVT Normal turbine	≤ 0	≤ 0	> 1	α/v	h/v^2	β/v^2
7	HAR BAR Reverse pump	≤ 0	> 0	≥ -1	v/α	h/α^2	β/α^2
8	HVR BVR Reverse pump	≤ 0	> 0	< -1	α/v	h/v^2	β/v^2

a. α = rotational ratio; v = volumetric flow ratio; h = head ratio; and β = torque ratio.

The four-quadrant pump head and torque maps in **Figure 2.4-1** and **Figure 2.4-2** can be reduced to the homologous representation curves in two steps. First, the maps are made dimensionless by using the rated head, H_R , flow, Q_R , speed, N_R , and torque, τ_R , to form corresponding dimensionless parameters $h = \frac{H}{H_R}$, $v = \frac{Q}{Q_R}$, $\alpha = \frac{N}{N_R}$, and $\beta = \frac{\tau}{\tau_R}$, respectively. Second, the data are plotted in terms of the homologous parameter h/α^2 or h/v^2 , v/α or α/v , and β/α^2 or β/v^2 . The parameter used depends upon the

octant in which the curve is being plotted. The choice is made so that the values are bounded (i.e., the denominators never vanish and, in the case of the capacity parameter, the range of variation is confined between plus and minus 1.0). **Figure 2.4-3** is the homologous head curve that is obtained from the head map in **Figure 2.4-1**. Note that not all points fall on a single curve. This is a result of the inexact nature of the similarity theory. Real pumps do not perform exactly according to the similarity relations due to leakage, viscous effects, etc.; however, the correspondence is surprisingly close, as evidenced by the tight clustering of points. The homologous curve for the torque data of **Figure 2.4-2** is shown on **Figure 2.4-4**. Since the data do not form a single curve, the design operating usual approach is to use least squares or other smoothing techniques to obtain curves passing through the point (1.0, 1.0). These curves must also be continuous at the point v/α or α/v equal to + 1.0. The legends on **Figure 2.4-3** and **Figure 2.4-4** have a key indicating which of the homologous parameters are used in each octant. All combinations of head, flow, speed, and torque can now be located on a corresponding segment of the homologous curve. Note that the impeller diameter parameter that appears in the dimensionless similarity parameters is not used in the homologous reduction of the four-quadrant representation; thus, special considerations are necessary for application of the data to a larger but geometrically similar pump. The advantage of using the homologous pump performance data representation in a computer code is obvious. Two-dimensional data arrays and two-dimensional interpolation are avoided, and, only two parameter tables and one-dimensional interpolation are required.

2.4.8.1.3 Homologous Data and Scaling

In most system simulation tasks, complete pump performance data are not available. Usually, only first-quadrant data are available (normal operation), and sometimes only the design or rated values are known. In the case of full-scale nuclear power plant pumps, it is difficult to test the pumps in all octants of operation or even very far from design conditions. The usual approach to obtain data for such systems is through the use of scaled-down pump tests.

The scaled pump test data can be for the same physical pump operated at reduced speed or for a pump scaled in size such that similarity is preserved. For the case of a pump scaled in size, it is necessary to maintain the similarity in specific head, capacity, speed, and torque parameters. (Note that the diameter was dropped in the development of the homologous performance model since a fixed configuration was considered.) Consideration of the diameter change must be implicitly included in the selection of “rated” parameters to properly account for changes in geometric scale. The homologous parameters, including the impeller diameter, are given in Equations (2.4-13), (2.4-14), and (2.4-16). When a change in scale is considered, an additional degree of freedom is possible, since only two parameters, the rated specific head and capacity, must be held constant. The specific speed is also held fixed whenever both specific head and capacity are kept fixed. There are many combinations of N and D for which this is possible.

The usual situation encountered in applications work is that homologous data exist for a similar pump, and the question arises, “Can we use these data to simulate our pump by adjusting the rated parameters?” The question can best be answered by the following statements. First, the best approach is to use the rated conditions corresponding to the pump used to generate the data. Second, the rated condition can be changed if the specific head and capacity are kept the same as for the pump used to generate the

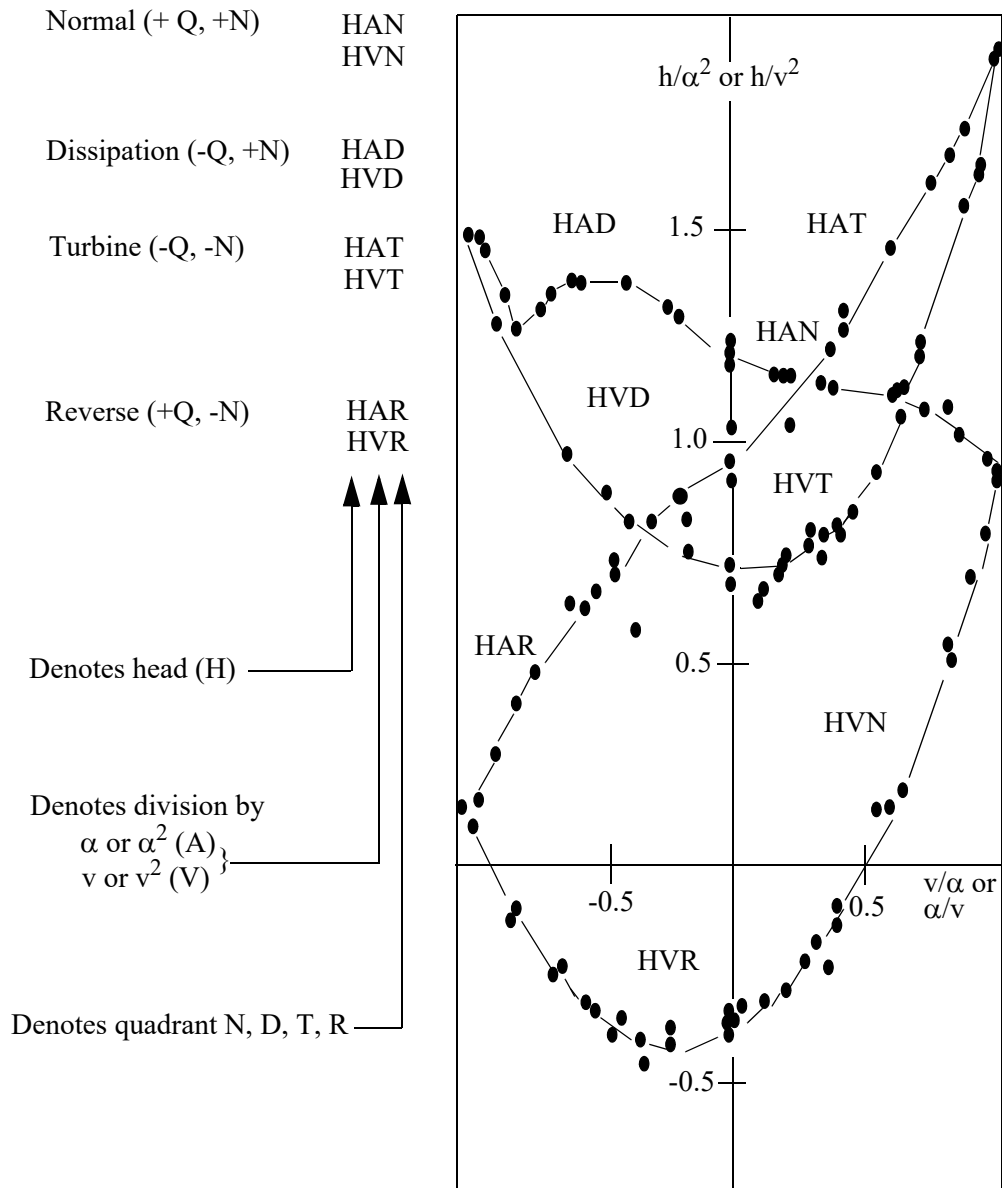


Figure 2.4-3 Homologous Head Curve

data. Similarity is assured since the modeled pump will have the same specific speed. The rated conditions by definition locate the region of pump operation on the homologous performance curve, at the design point or point of maximum efficiency. The rated conditions can then be safely adjusted in this way. They can also be adjusted using the impeller diameter as an additional parameter while still maintaining the rated specific speed, head, and capacity constant. However, this type of scaling implies a change in pump geometry and the extrapolation depends more heavily on the validity of the pump similarity relationships, which are only approximate. In all cases, the rated conditions must correspond to the same specific speed

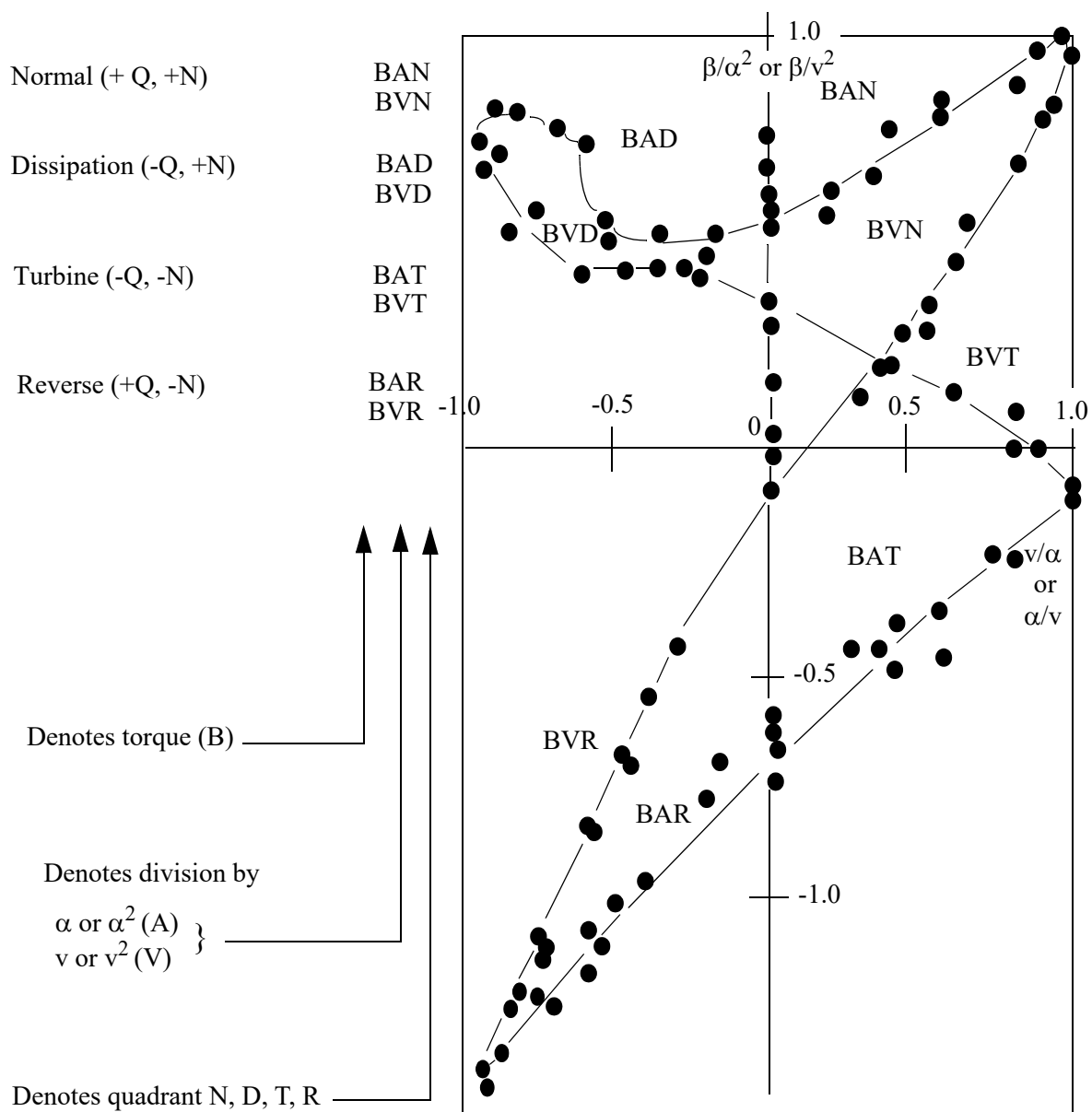


Figure 2.4-4 Homologous Torque Curve

as the pump used to produce the data. However, the operating point does not have to correspond to the rated conditions. In such a case, the pump will operate at off-design conditions and efficiency will be less than the design value. Such off-design adjustments may be desirable to better match the modeled pump head-flow characteristics at the system operating point.

2.4.8.1.4 Two-Phase Performance Representation

The discussion above applies to a pump operated with a single-phase fluid of constant density. When pump performance operation with a two-phase fluid is considered, the homologous representation of performance data has a less firm basis. An empirical modification of the homologous approach has therefore been developed based on Semiscale and Westinghouse Canada Limited (WCL) data. The RELAP5 two-phase pump model is the same as that developed for RELAP4.^{2.4-2} The approach is one in which the two-phase performance data are plotted, and a lowest performance envelope is constructed. This curve is called the fully degraded two-phase performance. The fully degraded performance and the single-phase performance data are used to form two-phase difference homologous performance curves for head or torque. The pump performance is then expressed in terms of the single-phase data and the difference data using a two-phase multiplier that is a function of void fraction. The pump head is expressed as

$$H = H_{1\phi} - M_H \alpha_g \Delta H \quad (2.4-17)$$

where ΔH is the head difference obtained from the single-phase to two-phase difference homologous curve. The function $M_H(\alpha_g)$ is the two-phase multiplier, defined such that it is zero for the void fraction, α_g , equal to 0.0 and 1.0. The pump torque is expressed in a similar way. Very little advice can be offered with respect to scaling of the two-phase performance data. Generally, it is assumed that the same similarity principles used for single-phase performance also hold for two-phase performance. A complete set of data was generated for a Semiscale pump, and these data are widely used for predicting two-phase performance of other pumps.

2.4.8.1.5 Pump Velocity Modeling

The pump computation for a time step begins by computing pump head and torque from the homologous data using pump angular velocity and volume conditions at the beginning of the time step. The head is used in the momentum equations. The remaining pump calculation determines the pump angular velocity at the end of the time step. The logic for computing pump angular velocity is complex, since stop logic, friction, an initializing calculation, the presence or absence of two tables, and two trips are involved. Additional capability is provided if the pump is associated with a shaft component. An optional card in the pump component input data specifies whether the pump is associated with a shaft. The remainder of this section defines pump capability when not associated with a shaft. In Section 4.2.3, the available shaft component capabilities are described and user suggestions are given.

Pump frictional torque (τ_{fr}) is modeled as a cubic function of the pump rotational velocity and is given by

$$\tau_{fr} = \pm \left(\tau_{fr0} + \tau_{fr1} \left| \frac{\omega}{\omega_R} \right| + \tau_{fr2} \left| \frac{\omega}{\omega_R} \right|^2 + \tau_{fr3} \left| \frac{\omega}{\omega_R} \right|^3 \right) \quad (2.4-18)$$

where ω is the pump rotational velocity; ω_R is the rated pump rotational velocity; and τ_{fr0} , τ_{fr1} , τ_{fr2} , and τ_{fr3} are input data. The pump friction torque is negative if $\frac{\omega}{\omega_R} > 0$, and it is positive if $\frac{\omega}{\omega_R} < 0$.

The pump model has special capabilities to accommodate experimental systems. For example; the LOFT system primary pumps use a motor-generator, flywheel, fluid coupling, and an active control system in order to better represent full-size PWR pumps. Allowing a variable pump inertia provides a simple model of the LOFT pump rotational behavior. To facilitate LOFT usage, pump input provides for constant inertia or optionally allows input of variable inertia data. The variable pump inertia (I_p) is given by

$$I_p = I_{pn} \quad \text{for } \left| \frac{\omega}{\omega_R} \right| < S_L \quad (2.4-19)$$

$$I_p = I_{p0} + I_{p1} \left| \frac{\omega}{\omega_R} \right| + I_{p2} \left| \frac{\omega}{\omega_R} \right|^2 + I_{p3} \left| \frac{\omega}{\omega_R} \right|^3 \quad \text{for } \left| \frac{\omega}{\omega_R} \right| \geq S_L \quad (2.4-20)$$

where ω is the pump rotational velocity, ω_R is the rated pump rotational velocity, and I_{pn} , I_{p0} , I_{p1} , I_{p2} , I_{p3} , and S_L are input data.

A pump stop card containing limits on problem time, forward pump angular velocity, and reverse angular velocity may optionally be entered. The pump angular velocity is set to zero and remains zero for the remainder of the problem if any of the limits are exceeded. Selected tests can effectively be disabled by entering a very large number for the limits. If the problem time limit = 0, then the problem time test is ignored.

A time-dependent pump velocity table and an associated trip number may be entered. If the table is entered and the trip number is zero, the pump angular velocity is always determined from this table. If the trip number is nonzero, the table is used only when the trip is true. The default search variable for the time-dependent pump velocity table is time, but time-advanced quantities may be specified as the search variable. When time is the search variable by default, the search argument is time minus the time of the trip. When a time-advanced variable is specified as the search variable (even if it is time), the search argument is just the specified variable. The use of the pump velocity implies a pump motor to drive the pump at the specified velocity.

The following is a possible example of the use of a time-advanced variable as the search argument in the pump velocity table. The motor and its control system that drives a BWR recirculation pump could be modeled using the control system with one of the control variables representing the rotational velocity of the motor. The recirculation pump would be modeled as a hydrodynamic pump component. The torque exerted by the water on the pump would be one of the input variables to the control system model. Motor velocity would be supplied to the pump component by specifying the motor velocity as the search argument of the time-dependent pump velocity table. The table would relate the motor rotational velocity

to the pump rotational velocity. If the motor and pump were directly coupled, the search variables and dependent variables would be the same.

Whenever the time-dependent pump angular velocity table is not being used, the pump angular velocity is determined by the advancement in time of the differential equation relating pump moment of inertia, angular acceleration, and net torque. The net torque is the pump motor torque minus the homologous torque value and the frictional torque. If the pump trip is false, electric power is being supplied to the pump motor; if the trip is true, electric power is disconnected from the pump motor and the pump motor torque is zero. If a table of pump motor torque as a function of pump angular velocity is entered, the pump motor is directly specified and motor torque is obtained from the table, interpolating when needed. If the table is missing, the pump motor is implied and torque is assumed to be such that the net torque is zero. This is implemented in the program by simply setting the pump angular velocity at the end of the time step equal to that at the beginning of the time step. This latter option is usually used when the problem starts with the pump at its normal steady-state velocity; the pump is assumed to remain at this velocity until the pump trip, and the trip, once true, remains true for the rest of the problem.

2.4.8.2 Pump Modeling Examples

Two examples are discussed to illustrate pump operation. Consider a pump in a closed loop filled with liquid water. At the start of the transient, all the water in the loop is at zero velocity but the pump is rotating in the positive direction. No pump motor torque table is used, the pump trip is initially false, and thus, the pump angular velocity is constant at the initial value until the pump trip becomes true. With the pump rotating at a constant angular velocity but the water at rest, the head is high and the water is accelerated. As the velocity of the water increases, wall friction and area change losses increase because of the dependence of these losses on water velocity. At the same time, the pump head obtained from the homologous data will decrease as the volumetric flow increases. A steady-state will be reached when the pump head and the loss effects balance. If no wall friction options are selected for the loop piping and no area losses are present, the water will accelerate until the pump head is zero. When steady-state is reached and the pump trip is then set true, the pump will begin to decelerate because the pump friction torque and the torque exerted by the water on the pump are no longer balanced by the pump motor torque. The water also begins to decelerate owing to loss effects. The interaction between the water and pump depends on the relative inertias and friction losses between the two. If the water tends to decelerate more rapidly than the pump, the pump will use its rotational kinetic energy to maintain water velocity. If the pump tends to decelerate more rapidly than the water, the pump, depending on its design as reflected in the homologous data, may continue to act as a pump or the kinetic energy of the water may tend to maintain pump angular velocity.

The second example is similar to the first example except that the initial pump angular rotational velocity is zero and a pump motor torque curve for an induction motor is used. From the curve, the torque is positive at zero angular velocity and increases slowly as the velocity increases to a value slightly below the synchronous speed. Then, the torque decreases sharply to zero at the synchronous speed and continues to negative torque. At the initial conditions, the net torque is positive, the pump angular velocity increases, and the water is accelerated. If the pump torque is sufficiently high, the pump velocity increases to slightly

below the synchronous speed where the developed torque matches the frictional torque and the torque imposed by the water. As the water accelerates, the angular velocity decreases slightly to meet the increased torque requirements. The angular velocity decrease is very small owing to the steep slope of the torque versus angular velocity near the synchronous speed. Thus, once the pump approaches the synchronous speed, the transient behavior of the second example is similar to the first example.

2.4.8.3 Built-in Pump Data

RELAP5 contains built-in, single-phase homologous data for a Bingham Pump Company pump with a specific speed of 4,200 and a Westinghouse Electric Corporation pump with a specific speed of 5,200. Two-phase difference homologous data are also associated with these pumps, but the data curves are identical and were obtained from two-phase tests of the Semiscale pump. (The data curves are stored as data statements in subroutine RPUMP.) No built-in, two-phase multiplier tables are entered. Specification of built-in, single-phase homologous data does not require specification of the built-in, two-phase difference homologous data, or vice versa.

If multiple pump components are used and some tables are common to more than one component, then user effort and computer storage can be saved by entering the data for only one component and specifying that other components use that data. This holds true for built-in data, since built-in data are treated as input data and stored in the pump component data when requested. There are no component ordering restrictions when one pump component references tables in another pump component. Thus, a pump component may reference a pump component numbered higher or lower than itself. Also, a pump component may reference another pump component that references another pump component, as long as a pump component with data entered is eventually reached.

Pump Edit Parameters. Major output edits include pump performance information in addition to the quantities common to all volumes and junctions. Pump angular velocity, head, torque, octant number, and motor torque are edited. Pump angular velocity, head, torque, motor torque, and inertia are available as minor edit variables. The pump torque is the sum of torque from homologous data and friction effects. Pump motor torque is zero if the motor is tripped or if no motor is directly specified or implied.

2.4.9 Jet Pump

A jet pump is modeled in RELAP5 using the JETMIXER component. In a jet pump, the pumping action is caused by the momentum mixing of the high-speed drive line flow with the slower suction line flow. **Figure 2.4-5** contains a schematic showing the typical nodalization used for a jet pump mixing section.

2.4.9.1 Input Requirements

The input for a JETMIXER component is the same as that for a BRANCH component, with the following modifications:

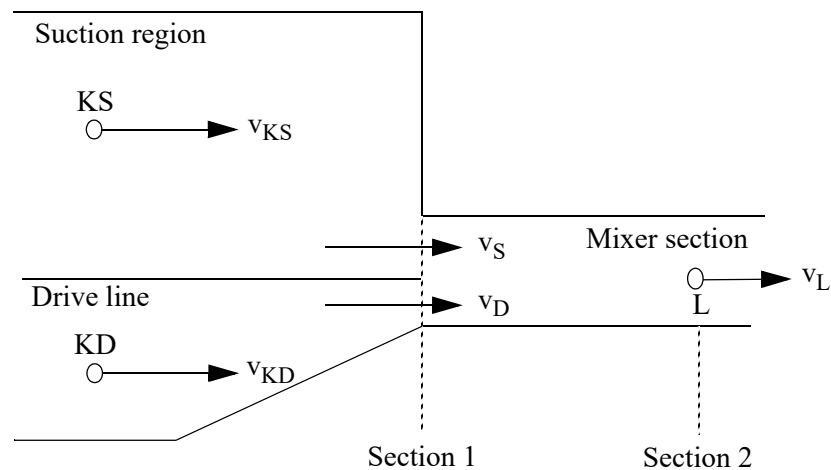


Figure 2.4-5 Schematic of Mixing Junctions

1. For a BRANCH component, the junctions connected to that branch can be input with the branch or as separate components. For a JETMIXER, three (and only three) junctions, representing the drive, suction, and discharge, must be input with the JETMIXER component, i.e., $NJ = 3$. If NJ is not equal to 3, an input error message is printed.
2. The three junction card sequences must be numbered as follows: Cards CCC1101 and CCC1201 represent the drive junction. Cards CCC2101 and CCC2201 represent the suction junction. Cards CCC3101 and CCC3201 represent the discharge junction in the mixing section.
3. The drive and suction junctions must have their TO connection codes referring to the JETMIXER volume, and the discharge junction must have its FROM connection code referring to the JETMIXER volume. If this is not the case, an input error message is printed. The drive and suction junctions must be connected to the inlet side of the JETMIXER volume, and the discharge junction must be connected to the outlet of the JETMIXER volume. If this is not the case, an input error message is printed.

2.4.9.2 Recommendations

Although the junction and volume areas for a JETMIXER are not restricted, the JETMIXER will properly model a jet pump only if the drive and suction junctions flow areas sum to the JETMIXER volume area.

The drive and suction junctions can be modeled with smooth or abrupt area changes. If they are modeled as smooth junctions, then the appropriate forward and reverse loss coefficients must be input by

the user. They should be obtained from standard references for configurations similar to those of the jet pump being modeled. The use of smooth junctions gives the user more explicit control over the resistance coefficients. In either case, it should be remembered that the turning losses associated with reverse flow through the suction junction are automatically included in all code calculations.

The JETMIXER component volume is intended to represent the mixing region of the jet pump. The diffuser section of a jet pump normally follows the mixing section. The diffuser section is not an integral part of the JETMIXER component and must be modeled using one or more additional volumes. Several volumes with slowly varying cross-sections and the smooth junction option can be used to model the diffuser region.

2.4.9.3 Additional Guidelines

It has been customary to identify jet pump operations in terms of two dimensionless parameters. These are the M and N parameters, defined as follows:

The M ratio (flow ratio) is the suction flow rate, W_S , divided by the drive flow rate, W_D ,

$$M = \frac{W_S}{W_D} \quad (2.4-21)$$

The N ratio (head ratio) is the increase in dynamic pressure for the suction-discharge path divided by the loss of dynamic pressure for the drive-discharge path,

$$N = \frac{\left(P + \frac{1}{2}\rho v^2 + \rho gH\right)_{Dis} - \left(P + \frac{1}{2}\rho v^2 + \rho gH\right)_S}{\left(P + \frac{1}{2}\rho v^2 + \rho gH\right)_D - \left(P + \frac{1}{2}\rho v^2 + \rho gH\right)_{Dis}} \quad (2.4-22)$$

Figure 2.4-6 shows an expanded view of the normal operating region (first quadrant) with several curves representing different flow resistances. This figure can be used as a guide for modeling different jet pump geometries. Each curve shows the M - N performance generated with base-case loss coefficients plus a single additional loss coefficient ($K = 0.2$) added to either the drive, suction, or discharge junction. This figure gives an indication of the quantitative change in performance caused by the respective drive, suction, or discharge losses. Using this figure, one can, with a few preliminary runs, design a code model for a specific jet pump if the performance data are available. If no specific performance data are available, we recommend that standard handbook losses be applied.

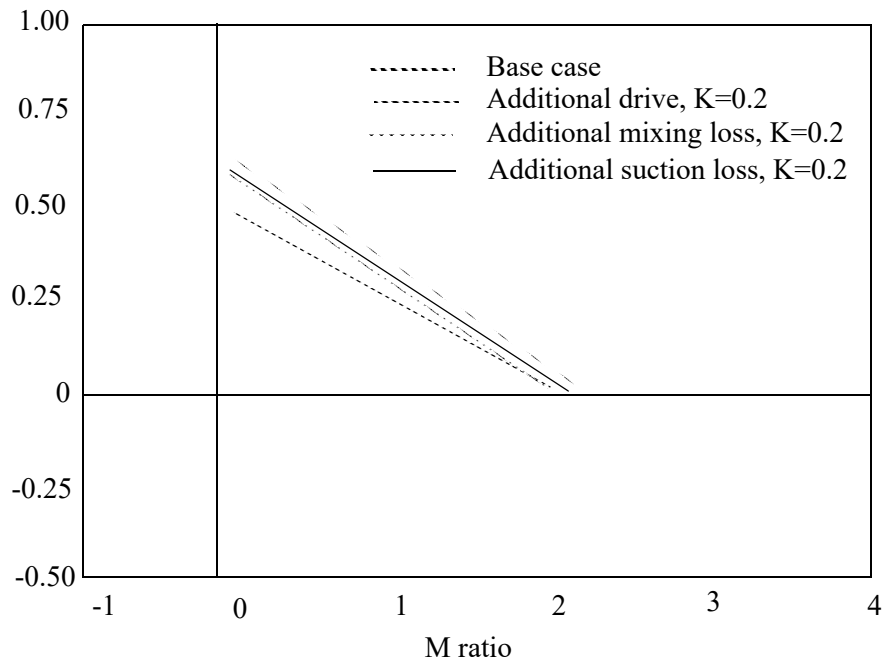


Figure 2.4-6 Jet Pump Model Design

2.4.9.4 Output

There is no special output printed for the JETMIXER component. We recommend that control variables be used to set up the M and N parameters for minor edit purposes and that these parameters be printed with every edit.

2.4.10 Valves

In RELAP5, nine valves are modeled that are of seven types. The types of valves provided are check valves, trip valves, inertial swing check valves, motor valves, end valves, servo valves, and relief valves. A single model for each type of valve is provided except for the check valves. For check valves, three models are provided, each of which has different hysteresis effects with respect to the opening/closing forces. Of the seven types of valves, the check valves and trip valves are modeled as instantaneous *on/off* switches. That is, if the opening conditions are met, then the valve is instantly and fully opened; if the closing conditions are met, the valve is instantly and fully closed. The remaining five types of valves are more realistic models in that opening/closing rates are considered. In the case of the inertial swing check valve and the relief valve, the dynamic behavior of the valve mechanism is modeled.

Fundamentally, a valve is used to regulate flow by varying the flow area at a specific location in a flow stream. Hence, in the RELAP5 scheme, a valve is modeled as a junction component that gives the user a means of varying a junction flow area as a function of time and/or hydrodynamic properties. Valve action is modeled explicitly and therefore lags the hydrodynamic calculational results by one time step. In order for the user to more fully use the valve models, some characteristics and recommendations for each valve are discussed in the following subsections.

2.4.10.1 Check Valves

Check valves are on/off switches, and the on/off action is determined by the formulation presented in Volume 1 of this manual. In turn, it is the characteristic of these formulations that determines the kind of behavior modeled by each type of check valve.

2.4.10.1.1 Static Pressure Controlled Check Valve

The check valve logic in Section 3 of Volume I describes the operation of a static pressure controlled check valve. If the equation is positive, the valve is instantaneously and fully opened, and the switch is *on*. If the equation is negative, the valve is instantaneously and fully closed, and the switch is *off*. If the equation is zero, an equilibrium condition exists, and no action is taken to change the existing state of the valve. Hence, in terms of pressure differential, there is no hysteresis. However, because the valve model is evaluated explicitly in the numerical scheme, the actual valve actuation will lag one time step behind the pressure differential. In terms of fluid flowing through the valve in a transient state, it is obvious that if the valve is closed and then opens, the flow rate is zero; but when a pressure differential closes the valve, the flow rate may be either positive, negative, or zero. Hence, with respect to flow, a hysteresis effect will be observed. Also, in the strictest sense, this type of valve is not a check valve, since the model allows reverse flow.

2.4.10.2 Flow Controlled Check Valve

Section 3 in Volume 1 shows the model of a check valve in the strictest sense in that flow is allowed only in the positive or forward direction, and the model is again designed to perform as an *on/off* switch. If the valve is closed, it will remain closed until the static pressure differential becomes positive, at which time the valve is instantaneously and fully opened, and the switch is *on*. Once the valve is opened, it will remain open until flow is negative or reversed, regardless of the pressure differential. Hence, with respect to pressure differential, a hysteresis effect may be observed. With respect to flow, it defines a negligible hysteresis effect, since flow is zero when the valve opens, and closes if flow becomes infinitesimally negative. However, since valve actuation lags one time step behind the pressure and flow calculation, a significant flow reversal may be calculated before the valve model completes a closed condition.

2.4.10.3 Dynamic Pressure Controlled Check Valve

Section 3 in Volume 1 shows the model of a dynamic pressure-actuated valve also designed to perform as an *on/off* switch. If the valve is closed, there is no flow through the valve, hence the valve must

be opened by static pressure differential. For this condition, the valve is opened instantaneously and fully, and the switch is *on*. Once the valve is opened, the fluid is accelerated, flow through the valve begins, and the dynamic pressure aids in holding the valve open. Since the valve cannot close until the closing back pressure, PCV, exceeds the junction static and dynamic pressure, there is a hysteresis effect both with respect to the opening and closing pressure differential and with respect to the fluid flow. These hysteresis effects are also determined by the sign of PCV, as input by the user. If PCV is input as positive, positive or forward flow through the valve will be allowed, and negative or reverse flow will be restricted. In this sense, the valve performs as a check valve. However, if PCV is input as negative, it will aid in opening the valve, and significant negative or reverse flow must occur before the valve will close. In this sense, the valve will not perform as a check valve. In addition, valve actuation lags one time step behind the pressure and flow calculations in the numerical scheme.

2.4.10.3.1 Check Valve Closing Back Pressure Term PCV

In Section 3 in Volume 1, the term PCV is used; in the input requirements, this term is designated as the closing back pressure. However, to be precise, PCV is a constant representing an actuation set point. If positive, PCV behaves as a back pressure acting to close the valve. In both the static and dynamic pressure-controlled valves, PCV acts both as an actuation set point for opening a closed valve and as a closing force for closing an open valve. For the flow-controlled valve, the back pressure acts only as an actuation set point for opening a closed valve.

2.4.10.4 Trip Valve

The trip valve is also an on/off switch that is controlled by a trip such that when the trip is *true* (i.e., *on*), the valve is *on* (i.e., instantly and fully open). Conversely, when the trip is *false* (i.e., *off*), the valve is *off* (i.e., instantly and fully closed).

Since trips are highly general functions in RELAP5 and since trips can be *driven* by control systems, the on/off function of a trip valve can be designed in any manner the user desires. The user should remember, however, that trips, control systems, and valves are explicit functions in RELAP5 and hence lag the calculational results by one time step.

2.4.10.5 Inertial Swing Check Valve

The inertial valve model closely approximates the behavior of a real flapper-type check valve. To direct the model to neglect flapper mass and inertial effects, the user simply inputs the flapper mass and moment of inertia as small numbers. Flapper open angles are positive in the positive junction flow direction. The code assumes that gravity always acts in the vertically downward direction, so that gravity can act to either open or close the valve, depending on the implied junction direction. The minimum flapper angle must be greater than or equal to zero.

2.4.10.6 Motor, End, and Servo Valves

The interaction of motor, end, and servo valves with fluid flow are identical but the means of positioning the valves are different. All three valves use a normalized stem position to position the valve. The normalized stem position ranges between 0.0 for the closed position to 1.0 for the fully open position. The flow area corresponding to a normalized stem position is determined from the normalized flow area, which also ranges from 0.0 for fully closed to 1.0 for fully open. A general table can be used to describe the normalized flow area for a given normalized stem position. If the general table is not used, the normalized flow area is set to equal the normalized stem position. Two models are provided to effect flow changes based on valve flow area. If the abrupt area change flag is set, the abrupt area change model is used to determine flow losses, and the valve flow area is treated as the orifice area in the abrupt area change model. If the abrupt area change flag is not set, a CSUBV table must be entered. This table contains forward and reverse flow coefficients as a function of normalized valve area or normalized stem position. The model using CSUBV coefficients should usually be used when the valve is designed for regulating flow. The user is referred to Volume 5, Section 4.6.7.6 for a description of CSUBV table input development and use.

The motor valve assumes that the valve stem is positioned by a motor. The valve position can be stationary, or the valve can be moving at a constant rate in the opening or closing direction. The rate is a user-input quantity in terms of normalized position change per second. The motor is controlled by an open trip and a closed trip. The valve stem position is stationary when both trips are false; when the open trip is true, the valve stem moves in the open position; when the close trip is true, the valve moves in the closing direction. The code terminates if both trips are simultaneously set true. Section 4.1.4 shows trip logic for the open trip that could be used to position a valve to regulate flow such that an upstream pressure is held within a set range.

The end valve is a motor valve that once it starts opening or closing, it opens or closes all the way to the end before it can reverse, even if the signal to reverse it is received.

The servo valve uses the value of a control variable to indicate the normalized valve stem position. A typical application would be regulating steam flow to the turbine to maintain a desired quantity such as primary system temperature or secondary side steam generator pressure. The control system, perhaps using a STEAMCTL (specialized proportional-integral controller), would compare the current value of the primary temperature or steam generator pressure to the desired value, and from the difference of the values compute an appropriate valve position. The servo valve, using the control output, would position the valve and thus regulate the flow.

2.4.10.7 Relief Valve

A scheme was designed to input the terms required to define a typical relief valve geometry and dynamic parameters. This scheme is consistent with the RELAP5 input philosophy in that extensive checking is performed during input processing, and error flags are set to terminate the problem if input errors are encountered. Error messages are also printed to inform the user that the data entered were in error. The specific input description is detailed in Appendix A.

2.4.11 Separator

Figure 2.4-7 contains a schematic showing the typical nodalization used for a separator and the adjoining bypass and downcomer regions. If there is any possibility of recirculation flow through a bypass region, we recommend that this flow path be included. In general, there will be a mixture level at some location in the downcomer volumes.

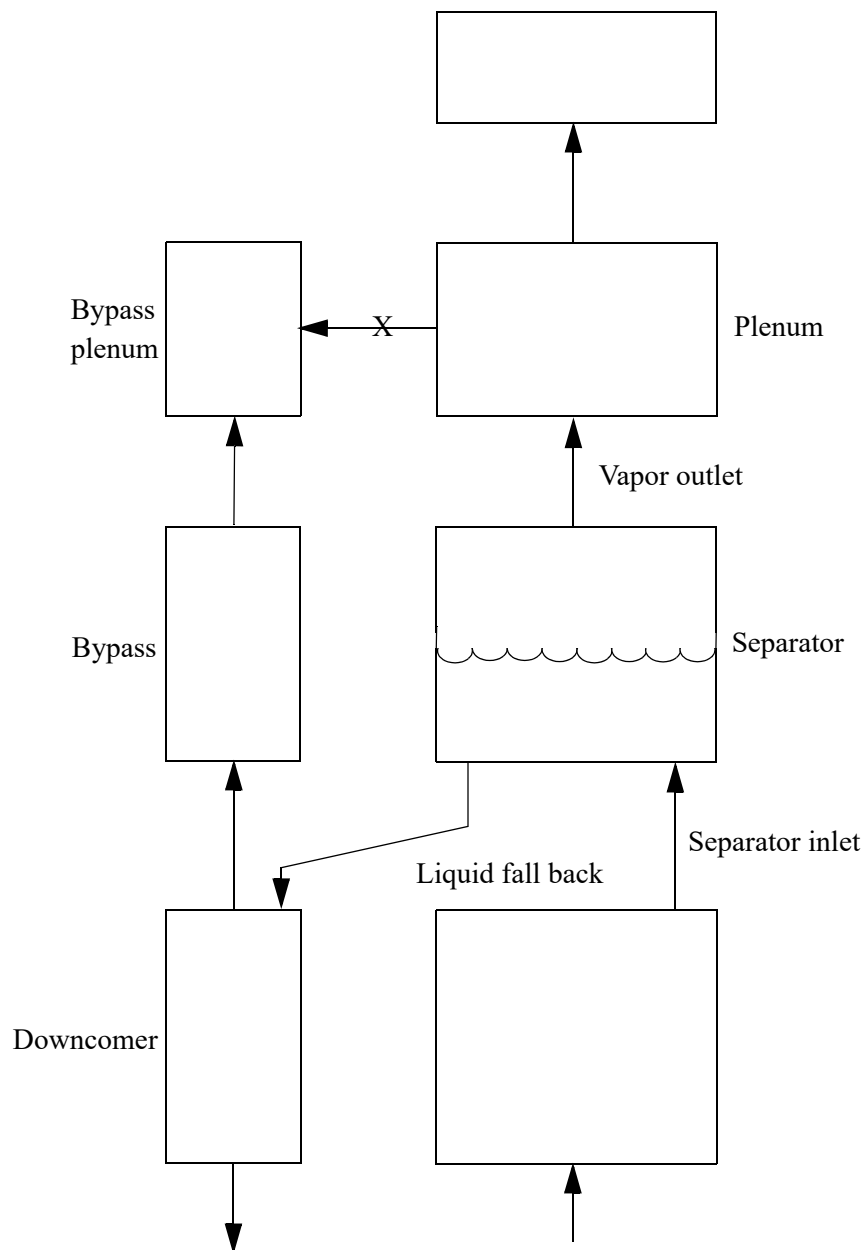


Figure 2.4-7 Schematic Of Separator

The RELAP5 separator model can be looked on in the following way. The flux through the liquid outlet is all water when the volume fraction of liquid is above a critical value. Below that critical value, a mixture of steam and water are fluxed out the liquid outlet. A similar thing occurs for the steam outlet. When the volume fraction of steam is above a critical value, only steam is fluxed through the steam outlet. When the volume fraction of steam is below this critical value, a mixture of steam and water is fluxed through the steam outlet. The critical values are given names VUNDER for the water outlet, and VOVER for the steam outlet. This behavior can be shown in **Figure 2.4-8**.

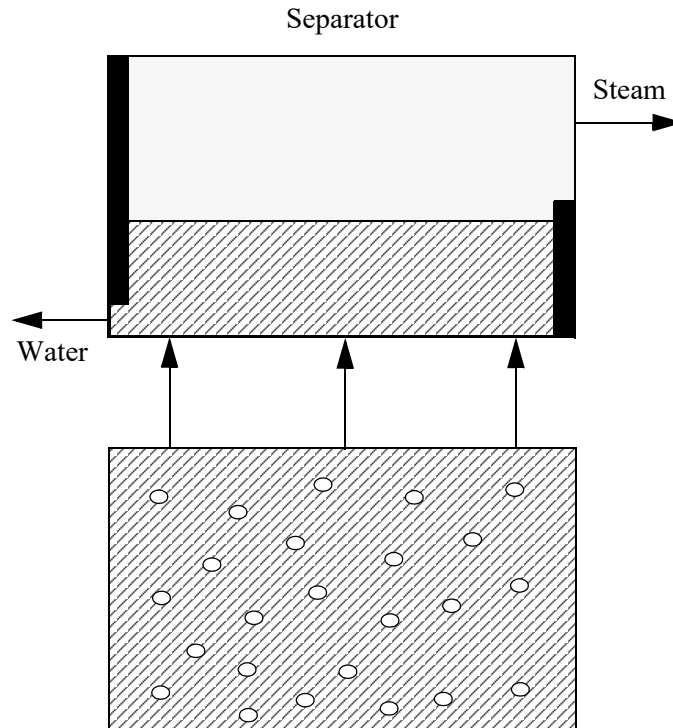


Figure 2.4-8 Physical Picture of a Separator

The volume on the bottom furnishes a mixture of steam and water to the separator. For the separator water outlet, notice that if the water level drops below the outlet baffle on the left, steam will also come out the junction. And for the separator steam outlet, notice that if the water level rises above the outlet baffle on the right, that water will also come out the junction. Thus the critical values of VUNDER and VOVER are given by the following formulas,

$$\text{VUNDER} = \frac{A_{fj}}{A_t} \text{ and } \text{VOVER} = \frac{A_{gj}}{A_t} \quad (2.4-23)$$

where

A_{fj} = the area open to the water

A_t = the total area

A_{gj} = the area open to the steam.

When the water level drops below the baffle on the left, the volume fraction of steam that is fluxed through the water outlet junction is a linear function of the water level height. A similar relationship is used for the steam outlet when the water level rises above the baffle on the right. These linear relationships are such that if the separator is empty of water, then pure steam comes out the liquid outlet, and conversely, if the separator is full of water, then pure water comes out the steam outlet. The behavior of the separator can now be characterized by **Figure 2.4-9** and **Figure 2.4-10** where the y-axis shows the volume fraction fluxed out the outlet (water or steam) and the x-axis shows the volume fraction (water or steam) in the separator volume.

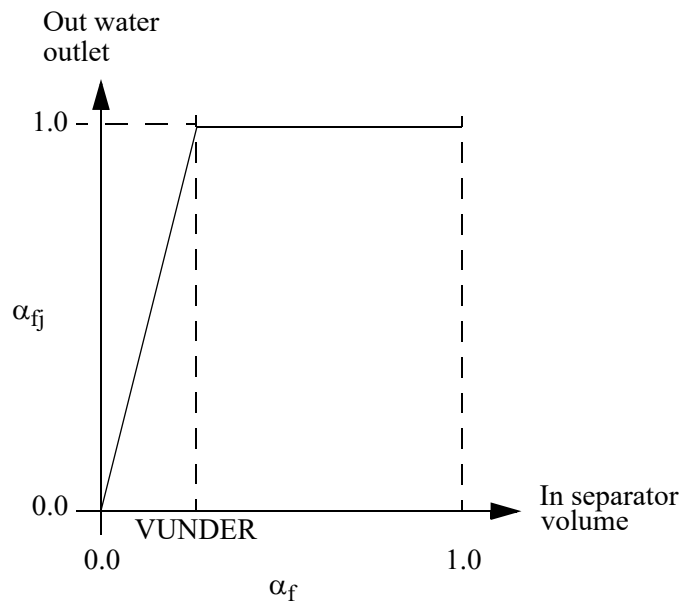


Figure 2.4-9 Separator Volume Fraction of Water Fluxed Out the Water Outlet

2.4.11.1 Input Requirements

The input for a SEPARATR component is the same as that for a BRANCH component, with the following modifications:

1. For a BRANCH component, the junctions connected to the branch can be input with the branch or as separate components. For a SEPARATR, the three junctions,

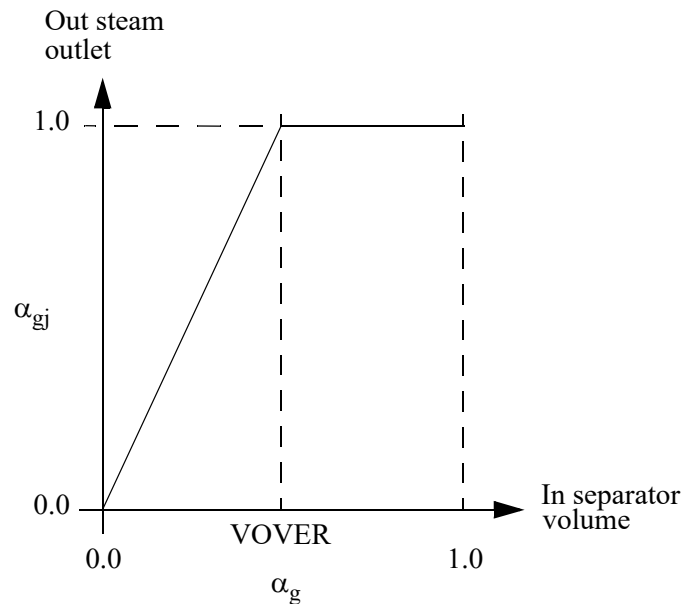


Figure 2.4-10 Separator Volume Fraction Of Steam Fluxed Out The Steam Outlet

- representing the vapor outlet, liquid fall back, and separator inlet, must be input with the SEPARATR component, i.e., NJ = 3.
2. The three junction card sequences must be numbered as follows: Cards CCC1101 and CCC1201 represent the vapor outlet junction, Cards CCC2101 and CCC2201 represent the liquid fall back junction, and Cards CCC3101 and CCC3201 represent the separator inlet junction.
 3. The FROM connection for the vapor outlet junctions must refer to the outlet of the separator (old format is CCC010000, and expanded format is CCC010002). The FROM connection for the liquid return junction must refer to the inlet of the separator (old format is CCC000000, and expanded format is CCC010001). The inlet junction must also be connected to the separator inlet side (old format is CCC000000, and expanded format is CCC010001).
 4. The type of separator desired is specified on Card CCC0002. If the mechanistic separator or dryer options are chosen, additional input data may be entered on the CCC050X cards for the mechanistic separator model and on the CCC0600 card for the dryer model. Default data are provided for both the mechanistic separator and dryer models.
 5. A word, W7(R), is added to the SEPARATR component junction geometry Cards CCCN101 for the simple separator option. For the vapor outlet, Word W7(R) specifies

VOVER. For the liquid fall back junction, Word W7(R) specifies VUNDER. No input should be entered for Word W7(R) on the separator inlet junction.

6. During input processing, if the specified velocities/flows will result in more than 90% of the vapor at the vapor outlet junction or more than 90% of the liquid at the liquid outlet junction being extracted on the first time step, an input error occurs.

2.4.11.2 Recommendations for the Simple Separator Option

The smooth or abrupt junction option can be used for the separator. Separators in general have many internal surfaces that lead to flow resistances above that of an open region. For this reason, additional energy loss coefficients may be required at the appropriate separator junctions. These should be obtained from handbook values or adjusted to match a known pressure drop across the separator. In some cases, it is necessary to use large loss coefficients (~ 100) in order to remove void oscillations in the separator volume. In addition, we recommend that choking be turned off for all three junctions. The nonhomogeneous options should be used for the vapor outlet and liquid fall back junctions.

An important parameter that influences the operation of any heat exchanger/separator combination is the equivalent mixture level in the downcomer region. This level is primarily determined by the rate of flow in the liquid return junction, which in turn is affected by the water level in the separator and the vapor flow out of the separator. The liquid return flow and water level in the separator are affected by the user-input void limits VOVER and VUNDER that determine the range of ideal separation. Because of the simple black-box nature of the separator, these limits should be adjusted to obtain the desired operating mixture level in the downcomer region. The default void limits (VOVER = 0.5 and VUNDER = 0.15) for ideal separation are intended to be preliminary.

The black-box nature of the separator, along with the use of VOVER and VUNDER, may result in some changes to the user-input initial conditions. If the user inputs a mass flow rate for both the vapor outlet and liquid fall back junctions, the code will in many cases alter the mass flow rates so that they no longer match those inputted. This is due to the use of the piecewise linear donor junction voids used (see Volume 1 of this manual). Depending on the relations of α_{gK} and VOVER as well as α_{fK} and VUNDER, it may be necessary to scale back the mass flow rates to achieve the desired input mass flow rates. Once the transient calculation begins, the mass flow rates and voids will most likely change from the initial value and some adjustment of VOVER and VUNDER may be required.

The final recommendation concerns the use of a bypass volume. If there is any possibility of a recirculation flow through a bypass-like region, we recommend such a flow path be included. The inclusion of such a flow path has generally improved the performance predictions. The use of a crossflow junction between the separator plenum and a bypass plenum instead of a normal junction generally provides a better model for the recirculation flow.

2.4.11.3 Recommendations for the Mechanistic Separator and Dryer Options

The mechanistic separator and dryer models are new in RELAP5/MOD3.3lo. They are intended to model the centrifugal separator and chevron dryer components in a BWR reactor. Until a base of user experience is obtained, we recommend default input data for the models be used. The user should explicitly model the separator standpipe as a separate volume or set of volumes because the separator component volume is intended to model the volume within the separator barrel and discharge passages. Likewise, the dryer volume should encompass the physical volume inside the dryer skirt between the elevations of the dryer inlet and outlet elevations. The separator inlet quality can be adjusted to the desired operating point by modifying the form loss coefficient in the separator liquid discharge junction. The liquid discharge line from a dryer to the downcomer should be modeled as a separate volume or set of volumes so that the liquid removed by the dryer may be injected into the downcomer at the correct elevation below the two-phase mixture level in the downcomer. The void fraction within the dryer component can also be adjusted by use of the liquid discharge junction form loss coefficient. The separator component may represent any number of physical separators. It is required that the geometry (i.e., volume and junction flow areas) of the separator component be the volume and flow areas of all of the physical separators represented by the RELAP5 separator components and that the number of separators represented by the RELAP5 separator component be specified in Word 2 on Card ccc0002 in the separator component input data.

2.4.12 Turbine

A steam turbine is a device that converts thermal energy contained in high-pressure, high-temperature steam to mechanical work. Three different stage group types can be implemented: (a) a two-row impulse stage group, which is normally only used as the first stage of a turbine for governing purposes; (b) a general impulse-reaction stage group with a fixed reaction fraction needed as input; and (c) a constant efficiency stage group to be used for very simple modeling or as a preliminary component during the model design process. A simple efficiency formula for each of the turbine types is given in Volume I, where all the terms are defined.

The mean stage radius needed in the efficiency formulas may not be known from the actual turbine design diagrams. We recommend the mean stage radius, R , be obtained from the efficiency formulas. If the turbine model is used with a constant efficiency factor, the stage radius is not needed (except for startup) and 1.0 can be entered. If the turbine stage is a general impulse-reaction stage, then the maximum efficiency, η_o , is obtained when

$$\frac{v_t}{v} = \frac{0.5}{1-r} \quad (2.4-24)$$

Using $v_t = R\omega$ and the input values v , r , and ω at the design operating point, Equation (2.4-24) gives for R ,

$$R = \frac{0.5v}{\omega(1-r)} \quad (2.4-25)$$

This is the recommended mean stage radius that is consistent with the assumed efficiency formula. For a two-row impulse stage, the maximum efficiency occurs when

$$\frac{v_t}{v} = 0.25 \quad (2.4-26)$$

Expressing v_t as $R\omega$ gives

$$R = \frac{0.25v}{\omega} \quad (2.4-27)$$

as the mean stage radius consistent with the efficiency formula.

For a TURBINE component, the primary steam inlet junction must be input with the TURBINE component as the first junction. If a steam extraction (bleed) junction is desired, it must be input with the TURBINE component as the second junction. Thus, NJ must be either 1 or 2. Cards CCC1101 and CCC1201 represent the steam inlet junction, and Cards CC2101 and CCC2201 represent the steam extraction bleed junction (if desired). The TO connection for the steam inlet junction must refer to the inlet of the TURBINE (old format is CCC000000, and expanded format is CCC010001).

Horizontal stratification effects are not modeled in the TURBINE component. Thus, the horizontal stratification flag must be turned off ($v = 0$). If several TURBINE components are in series, the choking flag should be left on ($c = 0$) for the first component but turned off for the other components ($c = 1$). The area changes along the turbine axis are gradual, so the smooth junction option should be used at both the inlet and outlet junctions. No special modeling has been included for slip effects, nor are there any data that could be used as a guide. Thus, the inlet and outlet junctions must be input as homogeneous junctions ($h = 1$ or 2). If a steam extraction (bleed) junction is present, it should be a crossflow junction.

The standard wall friction calculation is based upon the wetted perimeter. Because of all the internal blading surfaces, the wall friction based upon the volume geometry will not give a meaningful calculation. The turbine volume must be input using the zero wall friction option.

For some off-design cases, choking can take place at the nozzle and stator throats in a turbine. The junction velocities must represent the maximum nozzle velocities if the critical flow model is to be used. Hence, the junction areas used in the TURBINE component should represent the average nozzle throat or minimum area for the stage group if proper critical flow modeling is desired.

Several of the input parameters needed may not always be easily obtainable from the limited data available to the user. In particular, the stage group nozzle throat area, A_j , and the nozzle velocity, v_j , are not always easily obtained. A steady-state turbine heat balance usually contains the representative stage group pressures, the enthalpies, and the mass flow rates. From the mass flow rate and state properties, the product $v_j A_j$ is easily obtained, but the actual value of v_j or A_j requires more information. If a geometric description of the turbine is available, then A_j is known and v_j can be calculated. This is the proper way to obtain the input data. If no geometric data are available, then the following procedure can be used to crudely estimate the needed input data. A reasonable estimate must be made for one junction area. Then, knowing $v_j A_j$ gives the corresponding v_j . The turbine momentum equation

$$v_j(v_j - v_{j-1}) = -\frac{1 - \eta}{\rho}(P_L - P_K) \quad (2.4-28)$$

along with the stage pressures, can then be used to estimate the neighboring junction velocity. The mass flow along with this new velocity gives the neighboring junction area. In this way, all the velocities and junction areas can be estimated if any one junction area A_j or junction velocity v_j is known or estimated.

Note that turbines are usually designed to run with large velocities in the nozzles. The turbine may be the component that gives the maximum Courant number in the system. For this reason, the turbine component may limit the time step size. This can be mitigated if the turbine volumes are used with an exaggerated length. This will not affect any steady-state results, but it will give slightly inaccurate storage terms during a transient. The transient storage terms are small, so this should not be a problem.

2.4.13 Accumulator

An accumulator is a *lumped* parameter component modeled by two methods. First, the component is considered to be an accumulator as long as some of the initial liquid remains in the component. In this state, the accumulator is modeled using the special formulations discussed in Volume I of this manual. However, second, when the accumulator empties of liquid, the code automatically converts the component to an equivalent single-volume with a single outlet junction and continues calculations using the normal solution algorithms. In performing this conversion, the accumulator wall heat transfer model is retained but the volume flow area (A), hydraulic diameter (D_h), and elevation change (Δz) are reset to

$$A = \frac{V}{\Delta x_{TK} + \Delta x_L} \quad (2.4-29)$$

$$D_h = \frac{4V}{\pi(D_{TK}\Delta x_{TK} + D_L\Delta x_L)} \quad (2.4-30)$$

$$\Delta z = \Delta z_{TK} + \Delta z_L \quad (2.4-31)$$

respectively. In these equations, subscript TK is for the tank, and subscript L is for the surgeline/standpipe. In addition, the accumulator mass transfer model converts to the normal mass transfer model scheme.

In setting up an accumulator component, the user must remember that at the input processing level, the code assumes that the accumulator is initially *off*; that is, flow through the accumulator junction is zero. It is further assumed that the standpipe/surgeline is initially full of liquid and that the tank liquid level is as defined by the user. These assumptions are also true for RESTART runs if the user renodalizes the accumulator. Hence, the user must be careful to define the initial accumulator pressure lower than the injection point pressure, including elevation head effects. The tank geometry may be either cylindrical (see **Figure 2.4-11**) or spherical (see **Figure 2.4-12**). In the input description (Appendix A), the standpipe/surgeline inlet refers to the end of the pipe inside the tank itself (see **Figure 2.4-13**). Also, the noncondensable (nitrogen) used in the accumulator is that defined for the entire system being modeled. Hence, the user must be sure to input the correct noncondensable name (nitrogen) on Card 110, as discussed in Appendix A.

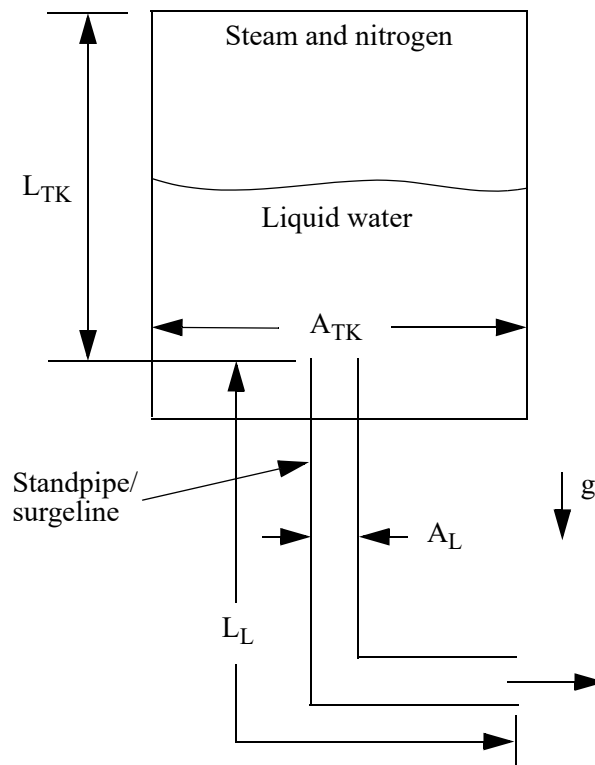


Figure 2.4-11 Schematic of a Cylindrical Accumulator

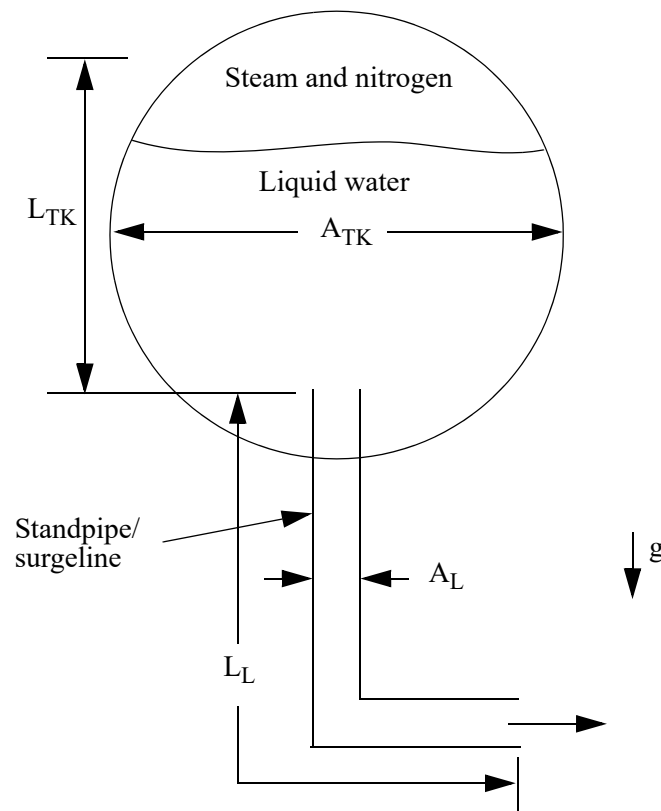


Figure 2.4-12 Schematic of a Spherical Accumulator

No other junctions (except the accumulator junction) should be connected to an accumulator volume. There are 4 possible accumulator configurations, as shown in **Figure 2.4-14**. The inclination angle (W5) and elevation change (W6) in Cards CCC0101 through CCC0109 can be either positive or negative, but both must have the same sign. The elevation drop of the surgeline and standpipe (W4) in Card CCC2200 is positive for a decrease in elevation from the standpipe/surgeline inlet to the injection point, and it is negative for an increase in elevation from the standpipe/surgeline inlet to the injection point.

2.4.14 Annulus

The annulus component is identical to a pipe component (Section 2.4.6), except the annulus component must be vertical and the annular-mist flow regime is different. If the user specifies this component, all the liquid is in the film and none is in the drops when the flow regime is annular-mist. The annulus component should be used to model a vertical annular region (i.e., reactor vessel downcomer or annular downcomer region in a U-tube steam generator).

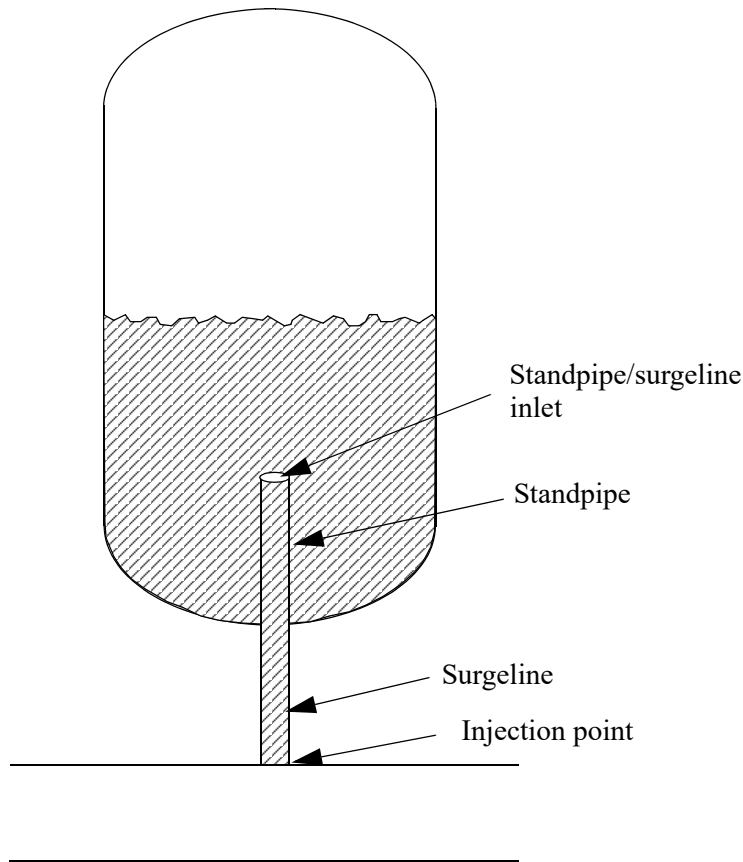


Figure 2.4-13 Schematic of an Accumulator Showing Standpipe/Surgeline Inlet

2.4.15 ECC Mixer

An ECC mixing component is a specialized branch that requires three junctions with a certain numbering order. The physical extent of the ECCMIX is a length of the cold leg pipe centered around the position of the ECC injection location. The length of this segment should be about three times the inside diameter of the cold leg pipe. Junction No. 1 is the ECC connection; junction No. 2 is the cold leg cross-section through which flow enters this component in normal reactor operation; and junction No. 3 is the one that leads to the reactor vessel. The geometrical description of the ECCMIX component is very similar to that of the JETMIXER component, except for the specification of an angle for the ECC pipe connection. The modeling details of the ECCMIX component are given in Section A7.7 of the input requirements in Appendix A of this volume. The ECCMIX component calculations are evoked only if there is subcooled ECC injection and if there is any steam to be condensed in that component. Otherwise, the ECCMIX component is treated as an ordinary BRANCH component.

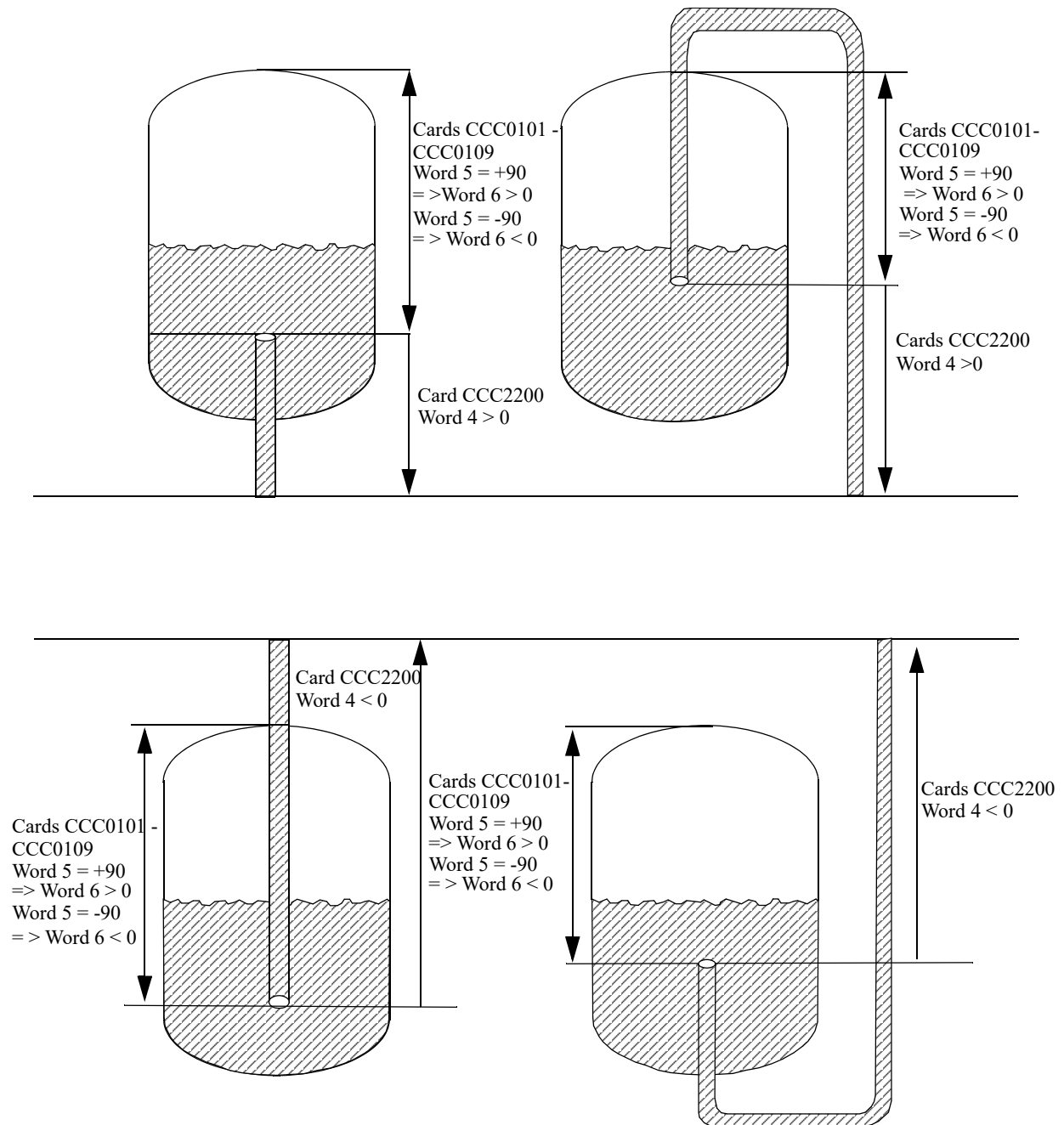


Figure 2.4-14 Possible Accumulator Configurations

2.4.16 References

- 2.4-1. E. Buckingham. "Model Experiments and the Forms of Empirical Equations." *Transactions of the ASME*. 37. 1915. pp. 263.
- 2.4-2. Aerojet Nuclear Company. *RELAP4/MOD5: A Computer Program for Transient Thermal-Hydraulic Analysis of Nuclear Reactors and Related Systems, User's Manual, Volume 1, RELAP4/MOD5 Description*. ANCR-NUREG-1335. Idaho National Engineering Laboratory. September 1976.

3 HEAT STRUCTURES

Heat structures represent the selected, solid portions of the thermal-hydrodynamic system. Being solid, there is no flow, but the total system response depends on heat transferred between the structures and the fluid, and the temperature distributions in the structures are often important requirements of the simulation. System components simulated by heat structures include fuel rods, pipe walls, core barrels, pressure vessels, and heat exchanger tubing. In simulations that do not involve core damage, heat structures can represent fuel pins, control rods, and other structural components. In core damage simulations, the SCDAP/RELAP5 code^{3.0-1} should be used. Temperatures and heat transfer rates are computed from the one-dimensional form of the transient heat conduction equation.

A heat structure is identified by a number, CCCG0NN. The subfield, CCC, is the heat structure number and is analogous to the hydrodynamic component number. Since heat structures are usually closely associated with a hydrodynamic component, it is suggested that the hydrodynamic component number and the CCC portion of the attached heat structures be the same number. Since different heat structures can be attached to the same hydrodynamic component, such as fuel pins and a core barrel attached to a core volume, the G portion can be used to distinguish the different types of heat structures. The combined field, CCCG, is the heat structure-geometry number, and input data are organized by this heat structure geometry number. Up to 99 individual heat structures may be defined using the geometry described for the heat structure geometry number. The individual heat structures are numbered consecutively starting at 01; this number is the subfield, NN, of the heat structure number. The heat structure input requirements are divided into input common to all heat structures with the heat structure geometry number, Cards 1CCCG000 through 1CCCG499, and input needed to uniquely define each heat structure, 1CCCG501 through 1CCCG999.

3.0.1 References

- 3.0-1. C. M. Allison and E. C. Johnson (eds.). *SCDAP/RELAP5/MOD3.1 Code Manuals, Volumes I, II, III*. NUREG/CR-6150, EGG-2720. Idaho National Engineering Laboratory. October 1993.

3.1 Heat Structure Geometry

Temperature distributions in heat structures are assumed to be represented adequately by a one-dimensional form of the transient heat conduction equation in rectangular, cylindrical, or spherical coordinates. The spatial dimension of the calculation is along any one of the coordinates in rectangular geometry and is along the radial coordinate in cylindrical or spherical geometry. The one-dimensional form assumes no temperature variations along the other coordinates. **Figure 3.1-1** illustrates placement of mesh points at which temperatures are computed. The mesh point spacing is taken in the positive direction from left to right. A composition is a material with associated thermal conductivity and volumetric heat capacity. Mesh points must be placed such that they lie on the two external boundaries and at any interface between different compositions. Additional mesh points may be placed at desired intervals between the interfaces or boundaries. There is no requirement for equal mesh intervals between interfaces, and compositions may vary at any mesh point.

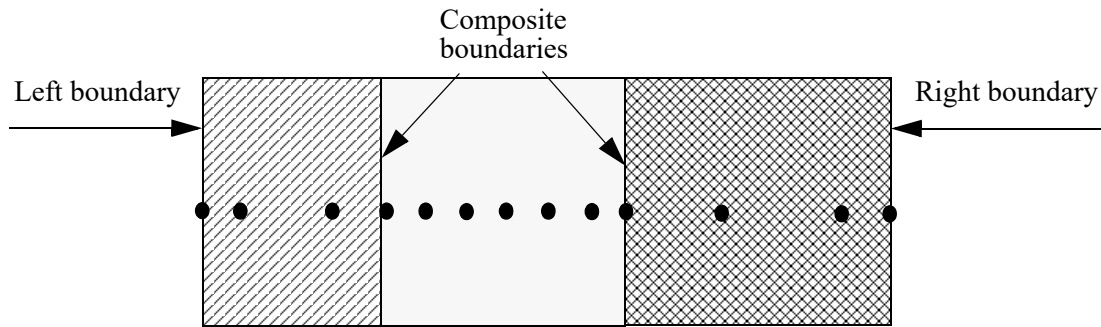


Figure 3.1-1 Mesh Point Layout

The heat structure input processing provides a convenient means to enter the mesh point spacing and composition placement. Each composition is assigned a three-digit, nonzero number (these numbers need not be consecutive). For each composition specified, corresponding thermal property data must be entered to define the thermal conductivity and volumetric heat capacity as functions of temperature. The temperature-dependence can be described by tabular data or by a set of functions. Defining thermal property data for compositions not specified in any heat structure is not considered an error but does waste storage space. Typical thermal property data for carbon steel, stainless steel, uranium dioxide, and zirconium are stored within the program. The data were entered to demonstrate the capability of the code and as a user convenience, and should not be considered recommended values. Input editing includes the thermal properties, and a list of the built-in data can be obtained by assigning the built-in materials to unused composition numbers in any input-check run. The thermal property data must span the temperature range of the problem. Problem advancement is terminated if temperatures are computed outside the range of the data.

Heat structures can have an internal volumetric heat source that can be used to represent nuclear, gamma, or electrical heating. The source $S(x,t)$ is assumed to be a separable function of space and time.

$$S(x,t) = P_f Q(x) P(t) \quad (3.1-1)$$

where

P_f = a scaling factor

$Q(x)$ = a space distribution function

$P(t)$ = power

The space function is assumed to be constant over a mesh interval but may vary from mesh interval to mesh interval. Only the relative distribution of the space function is important, and it may be scaled arbitrarily. For example, given a heat structure with two zones, the first zone having twice the internal heat generation of the second, the space distribution factors for the two zones could be 2.0 and 1.0, 200.0 and 100.0, or any numbers with the 2-to-1 ratio. Zeros can be entered for the space distribution if there is no internal heat source.

The mesh point spacings, composition placement, and source space distribution are common to all the heat structures defined with the heat structure geometry number, and only one copy of this information is stored. If a heat structure geometry has this data in common with another heat structure, input preparation and storage space can be saved by referencing the data in the other component. There are no ordering restrictions as to which heat structure geometry may reference another; one heat structure geometry may reference another, which in turn references a third, etc., as long as a defined heat structure is finally reached.

An initial temperature distribution may be entered for each heat structure geometry. This initial distribution is common to all heat structures defined with the same heat structure geometry number, but storage space for temperatures is assigned to each heat structure. Referencing initial temperature distributions in other heat structure geometries is allowed. Optionally, an initial temperature distribution may be entered for each heat structure.

The input temperature distribution can be used as the initial temperature distribution, or initial temperatures can be obtained from a steady-state heat conduction calculation using initial hydrodynamic conditions and zero-time power values. The input temperature distribution is used as the initial temperature guess for iterations on temperature-dependent thermal properties and boundary conditions. If a good temperature guess is not known, setting the temperature of any surface connected to a hydrodynamic volume equal to the volume temperature assists the convergence of the boundary conditions. The iteration process is not very sophisticated, and convergence to 0.01 K occasionally is not obtained. Input of a better initial distribution, especially surface temperatures, usually resolves the problem.

3.2 Heat Structure Boundary Conditions

Boundary condition input specifies the type of boundary condition, the possible attachment of a heat structure surface to a hydrodynamic volume, and the relating of the one-dimensional heat conduction solution to the actual three-dimensional nature of the structure. Each of the two surfaces of a heat structure may use any of the boundary conditions and may be connected to any hydrodynamic volume. Any number of heat structure surfaces may be connected to a hydrodynamic volume, but only one hydrodynamic volume may connect to a heat structure surface. When a heat structure is connected to a hydrodynamic volume, heat transferred from or to the heat structure is added to or subtracted from the internal energy content of the volume. For both left and right surfaces, a positive heat transfer rate represents heat flow out of the surface.

A symmetry or insulated boundary condition specifies no heat transfer at the surface, that is, a zero temperature gradient at the surface. This condition should be used in cylindrical or spherical coordinates when the radius of the left-most mesh point is zero, though the numerical techniques impose the condition regardless of the boundary condition specified. If a rectangular geometry is modeled with both surfaces attached to the same hydrodynamic volume, with the same boundary conditions, and having symmetry about the structure midpoint, storage space and computer time can be saved by describing only half of the structure. The symmetry boundary condition is used at one of the surfaces, and the heat surface area is doubled. This boundary condition can also be used when a surface is very well insulated.

When a heat structure is connected to a hydrodynamic volume, a set of heat transfer correlations can be used as boundary conditions. The correlations cover the various modes of heat transfer from a surface to fluid, and the reverse heat transfer from fluid to the surface. The heat transfer modes listed in the printed output are:

- Mode 0 Convection to noncondensable-water mixture.
- Mode 1 Single-phase liquid convection at supercritical pressure.
- Mode 2 Single-phase liquid convection, subcooled wall, low void fractions.
- Mode 3 Subcooled nucleate boiling.
- Mode 4 Saturated nucleate boiling.
- Mode 5 Subcooled transition boiling.
- Mode 6 Saturated transition boiling.
- Mode 7 Subcooled film boiling.
- Mode 8 Saturated film boiling.
- Mode 9 Single-phase vapor convection or supercritical pressure with the void fraction greater than zero.
- Mode 10 Condensation when the void is less than one.
- Mode 11 Condensation when the void equals one.

If the noncondensable quality is greater than 10^{-9} , 20 is added to the mode number. If the structure is a reflood structure, 40 is added. Thus, the mode number can vary from 0 to 71.

Generally, the hydrodynamic volume will not be a time-dependent volume. Caution should be used in specifying a time-dependent volume, since the elevation and length are set to zero, and the velocities in

an isolated time-dependent volume will be zero. Note that the current version of the code does not allow an isolated standard or time-dependent volume.

Users now have the option to be more specific about the type of hydraulic cell a heat slab is next to. Most of the default heat transfer coefficients originate from data taken inside vertical pipes. Users can now specify that the fluid is flowing between parallel plates, in a vertical or horizontal rod or tube bundle, or below a flat plate. When modelling a vertical bundle, the rod or tube pitch-to-diameter ratio should be input. This has the effect of increasing the convective part of heat transfer such that users can input the true hydraulic diameter and get reasonable predictions.

Other boundary condition options that can be selected are: setting the surface temperature to a hydrodynamic volume temperature, obtaining the surface temperature from a temperature-versus-time table, obtaining the heat flux from a time-dependent table, or obtaining heat transfer coefficients from either a time- or temperature-dependent table. For the last option, the sink temperature can be a hydrodynamic volume temperature or can be obtained from a temperature-versus-time table. These options are generally used to support various efforts to analyze experimental data.

A factor must be entered to relate the one-dimensional heat conduction representation to the actual heat structure. Two options are available for entry: either a heat transfer surface area or a geometry-dependent factor. For rectangular geometry, the factor is the surface area; there is no difference in the options. In cylindrical geometry, the heat structure is assumed to be a cylinder or a cylindrical shell, and the factor is the cylinder length. For a circular pipe where a hydrodynamic volume represents the flowing part of the pipe and a heat structure represents the pipe walls, the factor equals the hydrodynamic volume length. For a hydrodynamic volume representing a core volume with fuel pins or a heat exchanger volume with tubes, the factor is the product of the hydrodynamic volume length and the number of pins or tubes. In spherical geometry, the heat structure is assumed to be a sphere or a spherical shell, and the factor is the fraction of the sphere or shell. For a hemisphere, the factor would be 0.5. Except for solid cylinders or spheres where the inner surface area is zero, one surface area can be inferred from the other and the mesh point spacing information. Nevertheless, both surface areas must be entered and an input error will exist if the surfaces are not consistent. Consistency is defined to be such that the difference between the calculated left and right factors (or the input left or right factors) must be $\leq 10^{-5}$ times the sum of the calculated left and right factors (or the input left and right factors). This requirement is easily met with the second option of entering a geometry-dependent factor, since the factor is the same for the left and right boundary.

3.3 Heat Structure Sources

Volumetric heat sources for heat structures consist of the product of a scaling factor, a space-dependent function, and a time function. The space-dependent distribution has been discussed. The time function may be total reactor power, fission power, or fission product decay power from the reactor kinetics calculation; a control variable; or may be obtained from a table of power versus time. Input data provide for three factors. The first factor is applied to the power to indicate the internal heat source generated in the structure. This means that in steady-state, heat equal to the factor times the power value would be generated in the heat structure and transferred out through its left and right surfaces. If $P(t)$ is the

power in Watts and P_f is the factor, then $P_f P(t)$ is the heat generated in Watts. Within the program, this factor is divided by the integral of the space-dependent distribution to allow for the arbitrary scaling of that function. After this scaling, the internal source is in the required units of Watts/m^3 . The other two factors provide for the direct heating of the fluid in the hydrodynamic volumes attached to the surfaces. Heat equal to the factor times the power value is added to the internal energy of the fluid in the hydrodynamic volume. If $P(t)$ is the power in Watts and P_f is the factor, then $P_f P(t)$ is the heat added to the fluid. The total direct heating added to a volume is the sum of the direct heating from all structures connected to the volume. Zeros are entered where no heat source or hydrodynamic volumes exist. In a reactor problem, if a power value represents the total reactor power generated and if this power is totally accounted for in the RELAP5 model, then the sum of these three factors over all the heat structures representing that power value should equal one. The summing to one is not required, and no checks are performed by the code. In many instances, the power will not only be applied to the heat structures representing the fuel but also to the heat structures representing such items as the downcomer and pressure vessel walls.

3.4 Heat Structure Changes at Restart

At restart, heat structures may be added, deleted, or replaced. Since heat structure input data are organized with respect to a heat structure geometry, all heat structures with the heat structure geometry number are affected.

Composition and general table data can also be added, deleted, or replaced at restart. A transient or steady-state problem terminated by a heat structure temperature out of range of the thermal property data can be restarted at the restart prior to the termination by replacing the thermal property data.

3.5 Heat Structure Output and Recommended Uses

Up to six sections of heat structure output are printed at major edits. The first section prints one line of heat transfer information for each surface of each heat structure. Each line provides the heat structure number; a left or right surface indicator; the connected hydrodynamic volume or, if none, zero; surface temperature; the heat transfer rate; the heat flux; the critical heat flux; the critical heat flux multiplier; the mode of heat transfer; and the heat transfer coefficient. The first line for each heat structure also includes the heat input to the structure, the net heat loss from the structure, and the volume-average temperature for the structure. The critical heat flux multiplier is the value used to multiply the value from the CHF table.

The second section prints the mesh point temperatures for each heat structure. This section can be suppressed by an input option.

The other optional sections include output on reflood, metal-water reaction, rupture, and radiation.

For the heat structure additional boundary cards (1CCCG801 through 1CCCG899 and 1CCCG901 through 1CCCG999), it is suggested to use zero for the heat transfer hydraulic diameter (D_{he}) (i.e., the heated equivalent diameter) for a uniformly heated tube or channel. When zero is used, the heat transfer

hydraulic diameter is set the same as the hydraulic diameter (D_h) at the boundary volume, which should be determined by the user from

$$D_h = 4 \cdot \frac{\text{flow area}}{\text{wetted perimeter}} \quad (3.5-1)$$

For tube and channel tests, the same scaling method used in the hydraulic calculation should be used in the heat transfer calculation.

If the heat structure does not represent uniformly heated pipe or channel walls, the default should not be taken. The heat transfer hydraulic diameter should be determined by the user from

$$D_{he} = 4 \cdot \frac{\text{flow area}}{\text{heated perimeter}} \quad (3.5-2)$$

4 TRIPS AND CONTROLS

4.1 Trips

Extensive trip logic has been implemented in RELAP5. Each trip statement is a single logical statement, but because trip statements can refer to other trip statements, complex logical statements can be constructed.

There are two aspects to trip capability: (a) to determine when a trip has occurred, and (b) to determine what to do when a trip occurs. In the modular design of RELAP5, these two aspects have been separated. The term trip logic refers only to the first aspect and includes the input processing of the trip statements and the transient testing to set trip status. The action to be taken when a trip occurs is considered to be part of a particular model, and that aspect of trip coding is associated with the coding for the model. Examples of the second aspect of trips are the effects of trips on pump models and check valves.

Trip capability provides for variable and logical trips. Both types of trips are logical statements with a false or true result. A trip is false (that is off, not set, or has not occurred) if the result is false. A trip is true (that is on, is set, or has occurred) if the result is true. Trips can be latched or unlatched. A latched trip, once true (set), remains true (set) for the remainder of the problem execution, even if conditions change such that the logical statement is no longer true. An unlatched trip is tested at each time step, and the conditions can be switched at any step.

A TIMEOF quantity is associated with each trip. This quantity is always -1.0 for a trip with the value false. When a trip is switched to true, the time at which it switches replaces the value in TIMEOF. For a latched trip, this quantity once set to other than -1.0 always retains that value. An unlatched trip may have several TIMEOF values other than -1.0 during a simulation. Whenever an unlatched trip switches to false, TIMEOF becomes -1.0; when true again, the new time of switching to true is placed in TIMEOF. The TIMEOF quantities are used to effect delays in general tables, time-dependent volumes, time-dependent junctions, and pump speed tables, and can be referenced in the control system.

Two card formats are available for entering trip data. All trips for a problem must use the same format. At restart, the same format must be used for trip modifications unless all trips are deleted (Card 400) and desired trips are reentered. The default format uses Cards 401 through 599 for variable trips and Cards 601 through 799 for logical trips. The trip number is the same as the card number. Up to 199 variable trips and up to 199 logical trips can be defined. An alternate format is selected by entering Card 20600000. Trip data are entered on Cards 206TTTT0, where TTTT is the trip number. Trip numbers 1 through 1000 are variable trips, and trip numbers 1001 through 2000 are logical trips. The alternate format allows 1000 trips each for variable and logical trips.

As trips are input, the default initial value is false. Optionally, the TIMEOF quantity may be entered. If -1.0 is entered, the trip is false; if 0 or a positive number is entered, the trip is true, and the entered quantity is the time the trip turned true. This quantity must be less than or equal to the time of restart. For a new problem, 0.0 must be entered.

Several options are available on restart. If no trip data are entered, trips are defined at restart with the values at restart. It is possible to delete all trip definitions and enter completely new definitions. Individual trips can be deleted or redefined, and new trips can be inserted. Individual trips can be reset to false. At restart, a latched trip can be reset. Detailed discussions and examples of the use of trips are presented in Section 4 of Volume V of this code manual.

4.1.1 Variable Trips

A variable trip evaluates a comparison statement relating two variables and a constant using one of the relationships, equal (EQ), not equal (NE), greater than or equal (GE), greater than (GT), less than or equal (LE), or less than (LT). The variables currently allowed are listed in the Input Requirements (Appendix A). Most variables advanced in time are allowed, and any variable that is permanently stored can be added to the list. The only restriction on the two variables is that they have the same units. Thus, a hydrodynamic volume temperature can be compared to a heat structure temperature, but a pressure cannot be compared to a velocity. The variable trip statement is

$$\text{NUM} \quad \text{VAR1} \quad \text{OP} \quad \text{VAR2} + \text{CONSTANT} \quad \begin{bmatrix} \text{L} \\ \text{N} \end{bmatrix} \quad \text{TIMEOF} \quad (4.1-1)$$

where NUM is the card number; VAR1 and VAR2 each consist of two words that identify a variable, the first word being alphanumeric for the variable type, the second word being a number associated with the particular variable; OP is the comparison operation; CONSTANT is a signed number to be added to VAR2 before comparison; and either L or N is used to indicate a latched or unlatched trip. TIMEOF is the optional initialization value. A special form NULL,0 is used to indicate that no variable is to be used. VAR2 must be NULL,0 if VAR1 is to be compared only to the constant. Either VAR1 or VAR2 may also be TIMEOF, trip number. The trip number may refer to either a variable or a logical trip.

Three examples of variable trips are

501 P,3010000LT NULL,0 1.5+5 N

502 P,5010000GT P,3010000 2.0+5 N

510 TIME,0GE NULL,0 100.0 L

Trip 501: Is the pressure in volume 3010000 < 1.5 bar (1 bar = 10⁵ Pa)?

Trip 502: Is the pressure difference between volumes 5010000 and 3010000 > 2.0 bar?

Trip 510: Is the current advancement time ≥ 100 s?

Use of the equal (EQ) or not equal (NE) operator should be avoided because fractions expressed exactly in decimal notation may not be exact in binary notation. As an example, assume a time step of

0.01. After ten advancements, the time should be 0.10, but an equality test of time equal to 0.10 would probably fail. An analogous situation is dividing 1 by 3 on a three-digit decimal calculator, obtaining 0.333. Adding 1/3 three times should give 1.000, but 0.999 is obtained.

4.1.2 Logical Trips

A logical trip evaluates a logical statement relating two trip quantities with the operations AND, OR (inclusive), or XOR (exclusive). **Table 4.1-1** defines the logical operations where 0 indicates false, 1 indicates true. Each trip quantity may be the original value or its complement. (Complement means reversing the true and false values; that is, the complement of true is false.)

Table 4.1-1 Logical Operations

AND	OR	XOR
0 0 1 1	0 0 1 1	0 0 1 1
0 1 0 1	0 1 0 1	0 1 0 1
0 0 0 1	0 1 1 1	0 1 1 0

The logical trip statement is:

$$\text{NUM} \quad \pm \text{TRIP1} \quad \text{OP} \quad \pm \text{TRIP2} \quad \begin{bmatrix} \text{L} \\ \text{N} \end{bmatrix} \quad \text{TIMEOF} \quad (4.1-2)$$

where NUM is the card number, TRIP1 and TRIP2 are either variable or logical trip numbers, OP is the logical operator, L or N are for latched or unlatched trips, and TIMEOF is the optional initialization value. A positive trip number means the original trip value; a negative number means the complement value. Examples of logical trips are

601 501 OR 502 N

602 601 AND 510 N

620 -510 OR -510 N

Trip 602 involves a previous logical trip and illustrates the construction of a complex logical statement. With the definitions given in the examples above and using parentheses to indicate the order of logical evaluations, Trip 602 is equivalent to {(Pressure 3010000 < 1.5 bar) OR [Pressure 5010000 > (Pressure 3010000 + 2.0 bar)]} AND (Time ≥ 100 s). Trip 620 is the complement of Trip 510, and the AND operation in place of the OR operation would also give the same result. Additional examples of trips are presented in Volume V of this code manual.

4.1.3 Trip Execution

The trip printout for a new problem at time equal to 0 seconds shows trips as they were entered at input. On restarted problems, the trip printout at the restart time shows input values for new and modified trips and the values from the original problem for the unmodified trips.

Trip computations are the first calculation of a time step. Thus, trip computations use the initial values for the first time step and the results of the previous advancement for all other advancements. Because trips use old values, they are not affected by repeats of the hydrodynamic and heat structure advancements.

Trips are evaluated in order of trip numbers; thus, variable trips are evaluated first, then logical trips (refer to the discussion of trips in Volume I). Results of variable trips involving the TIMEOF quantity and logical trips involving other trips can vary, depending on their position relative to other trips. As an example, consider

6XX -650 OR -650 N

which just complements Trip 650. Also, assume Trip 650 switches to true this time step, and, thus, 650 was false and 6XX was true previous to trip evaluation. At the end of trip evaluation, 6XX is true if 6XX is < 650 and false if 6XX is > 650. If Trip 650 remains true for the following time step, Trip 6XX with 6XX < 650 becomes false one time step late. Similarly, TIMEOF quantities can be one time interval off. This can be minimized by ordering TIMEOF tests last and defining logical trips before they are used in logical statements.

4.1.4 Trip Logic Example

Techniques from Boolean algebra can assist in formulating the logical trip statements. Consider a motor-operated valve that operates such that if the valve stem is stationary, it remains stationary until a specified pressure exceeds 12 bar or drops below 8 bar. The valve starts opening when the pressure exceeds 12 bar and continues opening until the pressure drops below 11 bar. The valve starts closing when the pressure drops below 8 bar and continues closing until the pressure exceeds 9 bar. The motor valve requires two trips, one to be true when the valve should be opening, the other to be true when the valve should be closing.

The following procedure is used to derive the open trip logic. A Boolean variable has one of two possible values, false (0) or true (1). Define as Boolean variables V_0 which is to be true when the valve should be opening; V_1 as the current value of the valve motion; P_1 true when the pressure is > 11 bar; and P_2 true when the pressure is > 12 bar. **Table 4.1-2** is a truth table that has been constructed by listing all possible combinations of the three input variables, V_1 , P_2 , and P_1 , and the desired output, V_0 . The number in the rightmost column is the number resulting from assuming the input values form a binary number; this is done to ensure that all combinations are listed. From the truth table, the following expression can be written,

$$V_0 = (\bar{V}_1 \otimes P_2 \otimes P_1) \oplus (V_1 \otimes \bar{P}_2 \otimes P_1) \oplus (V_1 \otimes P_2 \otimes P_1) \quad (4.1-3)$$

where $\otimes$ indicates AND, $\oplus$ indicates OR, and the bar indicates the complement. The expression is derived by combining (with OR operations) terms from each line having a true value in the output column. Each term consists of the combining of each input variable with AND operations, using the direct variable if the value is true and the complement if the value is false. **Table 4.1-2** shows that two of the combinations are impossible. This is because if P_2 is true, P_1 must also be true; that is, if the pressure is > 12 bar, it is also > 11 bar. Because of the relationship between P_2 and P_1 ,

$$P_2 \otimes P_1 = P_2 \quad P_2 \oplus P_1 = P_1 \quad (4.1-4)$$

Using the Boolean identities from **Table 4.1-3**, the logical expression can be reduced to

$$V_0 = (\bar{V}_1 \otimes P_2) \oplus [(V_1 \otimes P_1) \otimes (\bar{P}_2 \oplus P_2)] = (\bar{V}_1 \otimes P_2) \oplus (V_1 \otimes P_1) \quad (4.1-5)$$

Table 4.1-2 Truth Table Examples

Output	Input			
V_0	V_1	P_2	P_1	Num
0	0	0	0	0
0	0	0	1	1
impossible	0	1	0	2
1	0	1	1	3
0	1	0	0	4
1	1	0	1	5
impossible	1	1	0	6
1	1	1	1	7

Table 4.1-3 Boolean Algebra^a Identities

$A \otimes A = A$	$A \oplus A = A$	$A \otimes 0 = 0$	$A \oplus 0 = A$
$A \otimes \bar{A} = 0$	$A \oplus \bar{A} = 1$	$A \otimes 1 = A$	$A \oplus 1 = \bar{A}$
$A \otimes B = B \otimes A$		$A \oplus B = B \oplus A$	

Table 4.1-3 Boolean Algebra^a Identities (Continued)

$A \otimes (B \oplus C) = (A \otimes B) \oplus (A \otimes C)$			
$A \oplus (B \otimes C) = (A \oplus B) \otimes (A \oplus C)$			

a. $\otimes$ denotes AND; $\oplus$ denotes OR; bar above quantity denotes complement.

The following trip input implements the logic. Trips 601 through 603 implement the rightmost expression in Equation (4.1-5). Trip 603 specifies the open trip in a motor valve. The trip logic is written as follows:

501	P,1010000GT	NULL,0	11.0+5	N	(P1)
502	P,1010000GT	NULL,0	12.0+5	N	(P2)
601	-603 AND 502	N	(FIRST TERM OF EQ)		
602	603 AND 501	N	(SECOND TERM OF EQ)		
603	601 OR 602	N	(OPEN TRIP)		

The close trip logic can be written similarly.

4.2 Control Components

The control system provides the capability to evaluate simultaneous algebraic and ordinary differential equations. The capability is primarily intended to simulate control systems typically used in hydrodynamic systems, but it can also model other phenomena described by algebraic and ordinary differential equations. Another use is to define auxiliary output quantities (such as differential pressures) so they can be printed in major and minor edits and be plotted.

4.2.1 Basic Control Components

The control system capability consists of several types of control components, each type of component defining a control variable as a specific function of time-advanced quantities. The time-advanced quantities include hydrodynamic volume, junction, pump, valve, heat structure, reactor kinetics, and trip quantities, and the control variables themselves, including the control variable being defined. Permitting control variables to be input to control components allows complex expressions to be developed from components that perform simple, basic operations. The basic control components are listed below, followed by a brief review of the evaluation procedure. Familiarity with the control system numerical techniques documented in Section 6 of Volume I is recommended. In the definitions that follow, Y_i is the control variable defined by the i -th control component; A_j , R , and S are real constants input the by user; I is

an integer constant input by the user; V_j is a quantity advanced in time by RELAP5 and can include Y_i ; t is time; and s is the Laplace transform variable. Superscripts involving the index n denote time levels. Some components include a definition in Laplace transform notation. The name in parentheses is the name used in the input data to select the type of component.

4.2.1.1 Constant (CONSTANT)

$$Y_i = S \quad (4.2-1)$$

4.2.1.2 Addition-Subtraction (SUM)

$$Y_i = S(A_0 + A_1 V_1 + A_2 V_2 + \dots) \quad (4.2-2)$$

4.2.1.3 Multiplication (MULT)

$$Y_i = S V_1 V_2 \quad (4.2-3)$$

4.2.1.4 Division (DIV)

$$Y_i = \frac{S}{V_1} \text{ or } S_1 \frac{V_2}{V_1} \quad (4.2-4)$$

4.2.1.5 Integer Exponentiation (POWERI)

$$Y_i = S V_1^I \quad (4.2-5)$$

4.2.1.6 Real Exponentiation (POWERR)

$$Y_i = S V_1^R \quad (4.2-6)$$

4.2.1.7 Variable Exponentiation (POWERX)

$$Y_i = S V_1^{V_2} \quad (4.2-7)$$

4.2.1.8 Table Lookup Function (FUNCTION)

$$Y_i = SF(V_1) \quad (4.2-8)$$

where F is a function defined by table lookup and linear interpolation.

4.2.1.9 Standard Functions (STDFNCTN)

$$Y_i = SF(V_1, V_2, V_3, \dots) \quad (4.2-9)$$

where F can be $|V_1|$, $\exp(V_1)$, $\ln(V_1)$, $\sin(V_1)$, $\cos(V_1)$, $\tan(V_1)$, $\tan^{-1}(V_1)$, $(V_1)^{1/2}$, $\text{MAX}(V_1, V_2, \dots)$, and $\text{MIN}(V_1, V_2, \dots)$. Only MAX and MIN may have multiple arguments.

4.2.1.10 Delay (DELAY)

The delay component is defined by

$$Y_i = SV_1(t - t_d) \quad (4.2-10)$$

where t_d is the delay time. A user-input h determines the number of time-function pairs in the table used to store past values of V_1 . The maximum number of time-function pairs is $h + 2$. The delay table time increment is $\frac{t_d}{h}$. The delayed function is obtained by linear interpolation using the stored past history. As time is advanced, new time values are added to the table. Once the table is filled, new values replace values that are older than the delay time.

4.2.1.11 Unit Trip (TRIPUNIT)

$$Y_i = SU(\pm t_r) \quad (4.2-11)$$

4.2.1.12 Trip Delay (TRIPDLAY)

$$Y_i = ST_r(t_r) \quad (4.2-12)$$

In the two definitions above, t_r is a trip number and, if negative, indicates that the complement of the trip is to be used; and U is 0.0 or 1.0, depending on trip t_r (or its complement if t_r is negative) being false or true. T_r is -1.0 if the trip is false, and the time the trip was last set true if the trip is true. The trip delay result is -S if the trip is false and can be values between 0 and St (t is time) if the trip is true. The trip delay can be

limited to values between 0 and St (instead of $-S$ and St) by use of the optional minimum value for the component.

4.2.1.13 Integration (INTEGRAL)

$$Y_i = S \int_0^t V_1 dt : \text{or } Y_i(s) = \frac{SV_1(s)}{s} \quad (4.2-13)$$

4.2.1.14 Differentiation (DIFFERNI or DIFFERND)

$$Y = S \frac{dV_1}{dt} : \text{or } Y_i(s) = sV_1(s) \quad (4.2-14)$$

Use of DIFFERNI is *not* recommended, and, if possible, any differentiation should be avoided. See the discussion in Volume I of this manual.

4.2.1.15 Proportional-Integral (PROP-INT)

$$Y_i = S \left(A_1 V_1 + A_2 \int_0^t V_1 dt \right) \text{ or } Y_i(s) = S \left(A_1 + \frac{A_2}{s} \right) V_1(s) \quad (4.2-15)$$

4.2.1.16 Lag (LAG)

$$Y_i = \int_0^t \frac{(SV_1 - Y_i)}{A_1} dt : \text{or } Y_i(s) = S \left(\frac{1}{1 + A_1 s} \right) V_1(s) \quad (4.2-16)$$

4.2.1.17 Lead-Lag (LEAD-LAG)

$$Y_i = \frac{A_1 SV_1}{A_2} + \int_0^t \frac{(SV_1 - Y_i)}{A_2} dt : \text{or } Y_i(s) = s \left(\frac{1 + A_1 s}{1 + A_2 s} \right) V_1(s) \quad (4.2-17)$$

Each control component generates an equation, and, together, the components generate a system of nonlinear simultaneous equations. The solution of the simultaneous equations is approximated by simply evaluating the equation for each component in order of increasing component numbers and using the currently available information. Evaluation of algebraic control components uses only currently defined

values, but evaluation of components involving integration and differentiation use both old (V^n) and new (V^{n+1}) values. For time-advanced variables other than control variables, both the old and new quantities are available. If a control variable is defined (by appearing on the left side of an equation) before it appears on the right side, the correct old and new variables are available. If a control variable appears on the right side before it is defined, or if it appears in the defining equation, the new and old values are off by a time step. That is, V^{m+1} uses V^m and V^m uses V^{m-1} . For good results, the user should try to define a control variable before using it. This is not always possible, as shown in the second example in Section 4.2.2.

Except for a CONSTANT component, each control component may optionally specify a minimum, a maximum, or both. After the component is evaluated by its defining equation, the value is limited by the minimum and maximum values if they are specified.

The control system input provides for an initial value and a flag to indicate that the initial value is to be computed during the initialization phase of input processing. The initialization of all other systems, such as trips, hydrodynamics, heat structures, and reactor kinetics, precedes that for control systems. If one of those systems needs an initial value of a control system variable, the input value is used. Thus, the control variable value used in servo valve initialization, initialization of time-dependent volumes and junctions if control variables are specified as search arguments, initialization of heat structures when a control variable is specified as a heat source, and computation of bias reactivity when control variables contribute to reactivity use input values. However, the input edit and first major edit after introduction of a control variable show the value after initialization.

Except for the SHAFT component, RELAP5 treats control system variables as dimensionless quantities. No unit conversion of the input scaling factors or multiplier constants is done when British input units are specified, and no unit conversion is done on output when British output units are specified. All dimensioned variables are stored within the program in SI units, and the units for variables that can be used in control components are stated in the input description. The user may assume any desired unit for each control variable. It is the user's responsibility to enter appropriate scale factors and multiplier constants to achieve the desired units and to maintain unit consistency.

Two card formats are provided for input of control system data, but only one format may be used in a problem. The default format uses Card 205CCCNN, where CCC is the control component number and NN is a card sequence number. The card format limits the number of control components to 999. The alternate format using Card 205CCCCN can be selected by entering Card 20500000. With the alternate format, only one digit is used for card sequencing, and up to 9999 control components can be used with the four digit CCCC. Control variables are printed in major edits, can be specified for minor edits, and can be plotted.

4.2.2 Control System Examples

Two examples of control system use are given. (See Section A.14 of Appendix A for input format descriptions.) Input for the examples are shown except that symbols enclosed in parentheses are sometimes

used where the actual input would need a number. Also, all examples use control component numbers beginning with one.

The first example is the computation of total flow rate in a volume from

$$W = (\alpha_f \rho_f v_f + \alpha_g \rho_g v_g) A \quad (4.2-18)$$

where

α = void fraction

ρ = density

v = velocity

A = flow area

g = vapor

f = liquid

Two multiplication components and one addition-subtraction component are used. The time-advanced quantities, α , ρ , and v , are specified as V_1 , V_2 , and V_3 , respectively, in the two multiplication components, one for each phase. The area A is entered as the scaling factor. An addition-subtraction component adds the results from the multiplication components with $A_0 = 0$, $A_1 = A_2 = S = 1.0$, and V_1 and V_2 are the control variables defined by the multiplication components. For the present numerical scheme, the products should be defined first. This control system is assumed to generate a quantity for plotting only, so initial values are entered as zeros and initialization is selected. For volume number 123010000, input data using the default format would be the following:

```
20500100 FFLOW MULT (A) 0.0 1
20500101 VOIDF,123010000 RHOF,123010000
20500102 VELF,123010000
20500200 GFLOW MULT (A) 0.0 1
20500201 VOIDG,123010000 RHOG,123010000
20500202 VELG,123010000
20500300 TFLOW SUM 1.0 0.0 1
20500301 0.0 1.0,CNTRLVAR,1 1.0,CNTRLVAR,2
```

The second example is to solve

$$A_2\ddot{x} + A_1\dot{x} + A_{10}\dot{x} + A_0x + B\int_0^t x dt = C \quad (4.2-19)$$

Assignment of control variables, Y_i , are made to derivative, integral, and product terms, as listed below. In addition, each line shows equivalent expressions derived from algebraic manipulation and definition of an integral.

$$Y_1 = \dot{x} = Y_3 Y_4 \quad (4.2-20)$$

$$Y_2 = \ddot{x} = \frac{1}{A_2}(C - A_1 Y_3 - A_{10} Y_1 - A_0 Y_4 - B Y_5) \quad (4.2-21)$$

$$Y_3 = \dot{x} = \int_0^t \ddot{x} dt = \int_0^t Y_2 dt \quad (4.2-22)$$

$$Y_4 = x = \int_0^t \dot{x} dt = \int_0^t Y_3 dt \quad (4.2-23)$$

$$Y_5 = \int_0^t x dt = \int_0^{tm} Y_4 dt \quad (4.2-24)$$

The control components are defined by the rightmost expression. Thus, the third-order, nonlinear equation is defined by a multiplication, an addition-subtraction, and three integration components. Note that the above expressions cannot be rearranged so that all control variables are defined on the left before being used as operands on the right. The above order is recommended for the current numerical scheme. Assuming zero as the initial value for all the quantities, no initialization, and that the integral should be limited between zero and one (no reason except to demonstrate the input), input cards in the alternate format would be the following.

```

20500010 XD1*X MULT 1.0 0.0 0
20500011 CNTRLVAR,3 CNTRLVAR,4
20500020 XD2 SUM (1.0/A2) 0.0 0
20500021 (C) (-A1), CNTRLVAR,3 (-A10), CNTRLVAR,1
20500022 (-A0), CNTRLVAR,4 (-B), CNTRLVAR,5
20500030 XD1 INTEGRAL 1.0 0.0 0

```

```

20500031 CNTRLVAR,2
20500040 X INTEGRAL 1.0 0.0 0
20500041 CNTRLVAR,3
20500050 "INT OF X" INTEGRAL 1.0 0.0 0.3,0.0,1.0
20500051 CNTRLVAR,4

```

4.2.3 Shaft Control Component

The shaft component is a specialized control component that computationally couples motor, turbine, pump, and generator components analogously to a shaft mechanically coupling these devices. The primary purpose for the shaft component is to couple multiple turbine hydrodynamic components to represent a multi-stage turbine with steam extraction and liquid drain lines and to allow the turbines to drive a pump or generator. Computations associated with the shaft are advanced in time in the same manner as other control components. The shaft component evaluates the rotational velocity equation as

$$\sum_i I_i \frac{d\omega}{dt} = \sum_i \tau_i - \sum_i f_i \omega + \tau_c \quad (4.2-25)$$

where I_i is the moment of inertia from component i , τ_i is torque from component i , f_i is friction from component i , and τ_c is an optional torque from a control component. The summations are over the pump, generator, motor, or turbine components that might be connected to the shaft and the shaft itself. The rotational velocity is considered positive when rotating in the normal operating direction. A torque is positive when it would accelerate the shaft in the positive direction. In their normal operating modes, motors and turbines would generate positive torque, and pumps and generators would have negative torque.

Each component contains its own model, data, and storage for inertia, friction, and torque and has storage for its rotational velocity. For example, the pump model allows cubic expressions for inertia and friction. The friction expression shown in Equation (4.2-25) is used for the shaft itself and the generator component. Each component also has a disconnect trip number. If zero (no trip), the component is always connected to the shaft. If a trip is specified, the component is connected when false and disconnected when true. Any disconnected component is advanced separately and, thus, can have a different rotational velocity than the shaft. All connected components have the same rotational velocity.

The shaft equation is advanced explicitly by

$$\sum_i I_i^n \frac{(\omega^{n+1} - \omega^n)}{\Delta t} = \sum_i \tau_i^n - \sum_i f_i^n \omega^n + \tau_c \quad (4.2-26)$$

where superscripts indicate time levels. Inertias, torques, and friction are evaluated using old time information. The torque from the control system, τ_c , would be in terms of new time values for quantities other than control variables and would use new or old time values for control variables, depending on their component numbers relative to the shaft component number. Except when a generator component is involved, the shaft component calculations consist of solving Equation (4.2-26) for ω^{n+1} separately for each component disconnected from the shaft (if any) and for the shaft and the connected components as one system. For separated components, the new rotational velocity is stored with the component data, and the summations are only over terms within the component. (Each component has only one term, except the pump/motor component, which has two terms.) For the shaft and the connected components, the new rotational velocity is stored as the rotational velocity of the shaft and each connected component.

The following sections discuss the components that can be connected to a shaft.

4.2.3.1 Motor Component

No separate motor component exists in RELAP5. A motor capability is an optional feature of a pump component, and input describing the motor features are entered as part of the pump input. Specifying a pump as being connected to a shaft includes the motor if it is described in the pump input.

A motor model can also be described through the control system and its torque applied to the shaft through a control variable [τ_c in Equation (4.2-26)].

4.2.3.2 Pump Component

A pump need not be connected to a shaft, since the pump component optionally includes a model for advancing the angular velocity equation. A review of the pump when not associated with a shaft follows, so that the pump with a shaft can be described by their differences.

4.2.3.2.1 Pump Not Associated with Shaft

A pump rotational velocity table and associated trip may be entered. If a rotational velocity table is entered, its use depends on the optional trip. If the trip is not entered, the table is always used; if the trip is entered, the table is used when the trip is true and not used when the trip is false. The dependent variable of the table is rotational velocity. The search variable may be time or any other variable allowed in minor edits, including control variables. This allows a model for pump velocity to be computed by the control system. A motor is implied by the table, since a torque is needed to match the friction and hydrodynamic torque and to accelerate the pump velocity from the previous time-step value. The torque from this implied motor is labeled by MTR.TORQUE in the pump output of major edits.

When the pump speed table is not being used or is not entered, the pump rotational velocity equation is used,

$$I \frac{d\omega}{dt} = \tau_m + \tau_h \quad (4.2-27)$$

where I is the moment of inertia of the pump, ω is the rotational velocity, τ_m is the pump motor torque, and τ_h is the sum of the frictional and hydrodynamic torques. An operational pump trip may be specified. If not specified (trip number is zero) or if specified and false, electric power is supplied to the pump motor. If the trip is true, the pump breaker has tripped. (This is the origin of the name trips for the Boolean logic in RELAP4, and the name has been continued in RELAP5.) No electric power is supplied to a tripped pump, and, thus, the motor torque, τ_m , is zero.

A pump motor is directly specified when a table of pump motor torque versus rotational velocity is entered. An induction motor can be modeled by entering a function similar to that shown in Volume I. The key features of an induction pump are the negative slope of the torque with respect to velocity near the synchronous velocity, and the fact that the torque is zero at the synchronous velocity. In steady-state, the velocity is slightly less than the synchronous speed such that a positive torque balances the negative torque imposed by the pump. Pump transients such as pump startups from rest to operating speeds can be modeled. A simple AC or DC motor could also be modeled by a table that would have only positive torque values and negative slope. The motor torque table is not searched when the pump trip is true, since the motor torque is always zero when the pump is tripped. The motor torque is labeled by MTR.TORQUE in the pump output in major edits.

If a motor torque table is not entered, a pump motor is implied. When the pump trip is true, the torque from the implied pump motor is zero. If the trip is not entered or is false, a motor torque is assumed that is equal to the sum of frictional and hydrodynamic torques, resulting in no change to the rotational velocity over the time step. In this mode, the field labeled MTR.TORQUE has the same magnitude as the pump torque but has opposite sign.

The implied pump motor is normally used in cases where the pump is initially operating at normal velocity and, if tripped, is never restarted. Note that with the implied motor, if the pump trip is set true (pump tripped), the pump is free to change velocity. If the pump trip is reset to false (pump trip reset), the rotational velocity remains at the previous time step velocity; it is not reset to the initial velocity. To return to the initial velocity, the pump rotational velocity table can be used.

Optional input can prevent reverse rotation and stop the pump based on elapsed time and exceeding a maximum rotational speed in either direction.

4.2.3.2.2 Pump Associated With Shaft

Optional pump component input can be entered to associate the pump component with a shaft component. When a pump is associated with a shaft component, the rotational velocity is computed by the shaft component logic and not by the pump logic. The following describes the differences in pump logic when the pump is associated with a shaft.

The pump speed table cannot be entered. The options to prevent reverse velocity and to stop the pump based on time or velocity also cannot be used.

With one exception, the motor torque computation using either the motor torque table or the implied motor with a shaft component is identical to that without a shaft. If no components other than the pump are attached to the shaft, the moment of inertia of the pump-shaft combination is equal to that of the pump alone. Identical results can be obtained with or without using the shaft. The shaft must have a nonzero moment of inertia; to have the inertia of a pump alone equal that of the pump-shaft combination, some of the pump inertia must be apportioned to the shaft.

The one exception noted above is that with an implied pump motor (no motor torque table entered) and no pump trip entered (trip number is zero), the implied motor torque is always zero. This same situation without the shaft generates motor torque sufficient to maintain constant velocity. This option with the shaft forces the pump motor torque always to be zero. It would be used when a turbine is attached to the shaft, or the torque is computed by the control system.

The pump and shaft components offer several options; and, in some cases, the same model can be specified in more than one manner. Some general application recommendations follow. For motor-driven pumps that are either on or off (untripped or tripped), use the pump component without a shaft. For a variable-speed pump where the speed is computed by the control system, use a pump component with a *one-to-one* velocity table. The one-to-one table is a stratagem for forcing pump velocity to be equal to a control variable. Specify the search variable to be the control variable containing the velocity and enter a two-point velocity table. The independent and dependent variable for each point are the same. The first point is for the minimum possible velocity; the second point is for the highest velocity expected. The output from the table lookup and interpolation is just the input search argument. For a motor-driven, variable-speed pump where the torque is computed by the control system, use the shaft component. For a turbine-driven pump, use a shaft with the pump and turbine stages attached.

4.2.3.3 Turbine Component

A turbine component is a hydrodynamic component consisting of one volume and has additional modeling to compute torque based on volume conditions and rotational velocity. One junction may connect to the turbine volume inlet to represent the steam line. Multiple junctions may connect to the outlet to represent steam exit, extraction steam for regenerative heating of feedwater, and drain lines to remove liquid. A small turbine, such as might be used to drive a pump, is usually modeled by one turbine component. The turbine used to drive the electrical generator typically has steam extraction points and drain lines and, thus, is usually modeled by two or more turbine components. The shaft component is the only mechanism for providing the rotational velocity common to each turbine component and summing the torque developed in each turbine component. The shaft is also the only mechanism to couple the turbine to a pump or generator.

4.2.3.4 Generator Component

The generator component consists of the minimum model to load a turbine. Because of the simple model and its small input data requirements, it has been made an option of the shaft component.

The generator model allows two operating modes. One mode is having the generator connected to a large electrical grid; the generator, the shaft, and other connected components are forced to the synchronous speed. The other mode is tripped, and the rotational velocity then responds to the torques applied to the shaft. When the generator is connected to the grid, the torque necessary to maintain synchronous velocity is computed and the generator power is that torque times the synchronous velocity. If the torque is negative, the generator is in its normal mode of generating electricity. If the torque is positive, the generator is acting as a synchronous motor and power is being drawn from the grid to maintain the synchronous velocity. When the generator is tripped, the generator torque is zero.

A generator can be connected to a pump through the shaft component. This allows a synchronous motor-pump combination, which is yet another pump-motor option that can yield results identical to the pump without a shaft using an implied motor.

4.2.3.5 Pump, Generator, and Shaft Sample Problem

Table 4.2-1 shows input data for a sample problem to test pump, generator, and shaft components. The test problem consists of two identical but separate loops. Each loop has a pump and a pipe connecting the pump discharge to the pump suction. The normal wall friction model is used, and an orifice is included for additional dissipation. The loops are filled with subcooled water at zero velocity. The two pumps are driven differently. The first pump uses an implied pump motor operating at normal speed. The water is accelerated to near steady-state velocity within a few seconds. A true steady-state is not possible, since there is no provision for removing dissipation heat. The pump is then tripped, the pump coasts down, and flow velocities diminish. The second pump uses a pump motor torque table representing an induction motor with the rotational velocity initially zero. The pump accelerates to near the synchronous velocity, and, in turn, the water velocity is accelerated similarly to the first loop. The second loop is tripped similarly to the first loop.

In the second problem, the pumps are driven identically but using a different mechanism. The first pump uses a shaft and a generator acting as a motor. The second pump uses a shaft and control system to develop the torque. A general table duplicates the motor torque table, and a unit trip applies the trip action. Identical results are obtained in the two cases.

Table 4.2-1 Input Data for a Sample Problem to Test Pump, Generator, and Shaft

=two loops with pumps
*

* This problem has two loops, each with friction, an orifice, and a pump. Built-in pump data are used.

* The first loop is similar to the pump problem. The second loop uses pump motor torque data to

Table 4.2-1 Input Data for a Sample Problem to Test Pump, Generator, and Shaft (Continued)

* represent an induction motor. The pump is initially at rest. The pump accelerates to near synchronous
 * speed, and fluid is accelerated. Reaching near steady-state, pump trip and decreasing pump speed and
 * fluid velocity are similar to pump. The second problem is identical to the first except that shaft and
 * generator (acting as a motor) components are used.
 *

100 new transnt

102 british british

104 none

201 1.0 1.0-6 0.010 15001 1 20 1000

202 40.0 1.0-6 0.200 15001 1 20 1000

301 p 1010000

302 p 1040000

303 p 1060000

304 p 1070000

305 p 1100000

306 p 1150000

307 p 1180000

308 p 2010000

309 velfj 1010000

310 velfj 1070000

311 velfj 1180000

312 velfj 2010000

313 velfj 2020000

314 pmpvel 002

315 pmphead 002

316 pmptrq 002

351 p 3010000

352 p 3040000

353 p 3060000

354 p 3070000

355 p 3110000

356 p 3150000

357 p 3180000

358 p 4010000

359 velfj 3010000

Table 4.2-1 Input Data for a Sample Problem to Test Pump, Generator, and Shaft (Continued)

360	velfj	3070000
361	velfj	3180000
362	velfj	4010000
363	velfj	4020000
364	pmpvel	004
365	pmphead	004
366	pmptrq	004
501	time 2 q null	0 20.2 1
10000	loop	pipe
10001		19
10101		0.0376, 19
10201		0.0376, 6 0.01, 7 0.0376, 18
10301		2.0, 19
10601		0.0, 4 90.0, 9 0.0, 14 -90.0, 19
10801		0, 0, 19
11001		0, 19
11101		0, 6 100, 7 0, 18
11201		3, 2244.780, 540.0, 0, 0, 0, 19
11301		0, 0, 0, 18
20000	loop	pump
20101		0.0468 0 0.1660 0 0 0 0
20108		1010000 0.0376 0 0 0
20109		1000000 0.0376 0 0 0
20200		3 2264.78 540.0 0
20201		0 0 0 0
20202		0 0 0 0
20301		-1 0 -2 -1 -1 501 -1
20302		3560.0 0.66573 180.0 192.0 34.8 38.3 62.3 0 6.7 0 0 0
23000		0
23001		0.0, 0.0 0.1 0.0 0.15, 0.05 0.24, 0.8 0.3, 0.96 0.4, 0.98
23002		0.6 0.97 0.8, 0.9 0.9 0.8 0.96, 0.5 1.0, 1.0
23100		0
23101		0.0, -.17 0.0001, -.017 0.0006, 0.0 0.1, 0.0 0.15, 0.05
23102		0.24, 0.56 0.8, 0.56 0.96, 0.45 1.0, 0.0
30000	loop2	pipe

Table 4.2-1 Input Data for a Sample Problem to Test Pump, Generator, and Shaft (Continued)

```

30001 19
30101 0.0376, 19
30201 0.0376, 6 0.01, 7 0.0376, 18
30301 2.0, 19
30601 0.0, 4 90.0, 9 0.0, 14 -90.0, 19
30801 0, 0, 19
31001 0, 19
31101 0, 6 100, 7 0, 18
31202 3, 2265.780, 540.0, 0, 0, 0, 19
31301 0, 0, 0, 18
40000 loop2 pump
40101 0.0468 0 0.1600 0 0 0 0
40108 3010000 0.0376 0 0 0
40109 3000000 0.0376 0 0 0
40200 3 2244.78 540.0 0
40201 0 0 0 0
40202 0 0 0 0
40301 2 2 2 0 -1 501 1
40302 3560.0 0.0 180.0 192.0 34.8 38.3 62.3 35.0 6.7 0 0 0
46001 1440., 1.00 2160., 1.10 2880., 1.50
46002 3526., 2.80 3672., -2.70 4320., -1.90
46003 5040., -1.20 5760., -1.05 6480., -1.00
46004 7200., -0.98
.end of first case
=two loops with pumps using shaft component
20301 -1 0 -2 -1 -1 0 1
20302 3560.0 0.66573 180.0 192.0 34.8 37.0 62.3 0 6.7 0 0 0
20309 20
40301 2 2 2 -1 -1 0 0
40302 3560.0 0.0 180.0 192.0 34.8 38.0 62.3 0 6.7 0 0 0
40309 10
46001
46002
46003
46004

```

Table 4.2-1 Input Data for a Sample Problem to Test Pump, Generator, and Shaft (Continued)

```

20500100 mtr.trq function 47.4536282 0 0
20500101 cntrlvar, 10 10
20500200 trip tripunit 1.0 1.0 0
20500201 -501
20500300 torque mult 0.7375621495 0 0
20500401 cntrlvar, 3
20501000 shaft4 shaft 1.0 0.0 0
20501001 3 0.3 0.0 pump, 4
20201000 reac-t 0 0.10471975512 1.0
20201001 1440.0, 1.00 2160.0, 1.10 2880.0, 1.50
20201002 3528.0, 2.80 3672.0, -2.70 4120.0, -1.90
20201003 5040.0, -1.20 5760.0, -1.05 6480.0, -1.00
20201004 7200.0, -.098
20502200 shaft2 shaft 1.0 2370.0 1
20502001 0 1.0 0.0 pump, 2 generatr, 20
20502006 1800.0 2370.0 0.3 0.0 501 0
.end of job

```

4.3 Conditional Execution of Trips and Control Variables

Control variables and trips may be used to model specific behavior of the system components. Control variables are calculated at the end of the converged time step, in the increasing order of a control variable number. Trips are processed at the beginning of a time step, after the time step size and the final time for the actual step has been determined. All variable trips are processed before logical trips. This should be taken into account if the sequence of trip processing is important in the logical scheme utilized by the user.

By default, all control variables and trips are processed unconditionally for every time step (latched trips are skipped in the evaluation once they are set to true). In this case, modeling complex control systems can be difficult. Therefore, conditional execution of trips and control variables has been added to RELAP5. For variable trips, this logic can be activated via W9 on Cards 401-599 or 20600010-20610000 or 20600001-20610000. For logical trips, this logic can be activated via W6 on Cards 601-799 or 20610010-20620000 or 20610001-20620000. For control variables, this logic can be activated via W9 on Card 205CCC00 or 205CCCC0.

5 REACTOR KINETICS

The reactor kinetics capability can be used to compute the power behavior in a nuclear reactor. The power is computed using the space-independent or point kinetics approximation, which assumes that power can be separated into the product of space and time functions. This approximation is adequate for those cases in which the space distribution remains nearly constant.

Reactor kinetics data may be entered for new or restart problems. In restart problems, reactor kinetics data completely replaces previous reactor kinetics data if present; thus, all needed data must be entered even if they duplicate existing data.

5.1 Power Computation Options

Data for the six generally accepted delayed neutron groups are built into the code. Optionally, yield ratios and decay constants for up to 50 groups may be entered.

The total reactor power is the sum of immediate fission power and the power from decay of fission fragments. The immediate power is that released at the time of fission and includes power from fission fragment kinetic energy, prompt gammas, and neutron moderation. Decay power is generated as the fission products undergo radioactive decay. The user can specify one of three options for computing reactor power: fission power only; fission and fission product decay product power; or fission, fission product decay, and actinide decay power. Actinide decay power is the power resulting from production of ^{239}U by neutron absorption in ^{238}U and subsequent two-stage beta decay to ^{239}Pu .

Two sets of fission product decay data are built into the code. The default set is the eleven-group ANS standard proposed in 1973.^{5.1-1} The other set of data is from the 1979 ANS Standard for Decay Heat Power in Light Water Reactors.^{5.1-2} The 1979 standard specifies data for three isotopes, ^{235}U , ^{238}U , and ^{239}Pu , using 23 groups for each isotope. To use the three-isotope data, the user must furnish the fraction of power produced by each isotope. An option exists to use only the ^{235}U isotope data from the 1979 standard. Actinide data are from the 1979 standard. An input fraction is applied to both the fission product and actinide yield data. For fission products, the factor is usually 1.0 for best-estimate calculations; and 1.2 has been used for conservative calculations with the 1973 data. For actinide data, the factor is the ratio of ^{238}U atoms consumed per ^{235}U atoms fissioned; but additional conservative factors can be applied. Fission product and actinide data can also be entered by the user. Earlier versions of the code did not include the G factor, which is part of the 1979 standard. This factor can now be optionally included and should be for long-term transients.

The built-in data for delayed neutrons, fission products, and actinides are recommended and are listed in the reactor kinetic input edit when used. Use of the fission power plus fission product decay power is recommended, as is actinide decay power if an appreciable amount of ^{238}U is present. The new standard is recommended because it is an approved standard and the variance between 1979 data and experimental

data is much less than for the 1973 data. The three-isotope option is recommended unless the power fractions for each isotope are not available.

The reactor kinetics output lists total reactor power, fission power, decay power, reactivity, and reciprocal period. Either the total power, fission power, or decay power can be specified as the time-varying part of the heat source in heat structures.

5.1.1 References

- 5.1-1. American Nuclear Society Proposed Standard. ANS 5.1. *Decay Energy Release Rates Following Shutdown of Uranium-Fueled Thermal Reactors*. October 1971, revised October 1973.
- 5.1-2. American National Standard for Decay Heat Power in Light Water Reactors. ANSI/ANS-5.1-1979.

5.2 Reactivity Feedback Options

Five reactivity feedback options are provided. One assumes separability of feedback effects; the others use three- or four-dimensional table lookup and linear interpolation. The defining equations are given in Volume 1 of this manual. Note that the sign of the feedback terms is positive. Negative quantities must be entered where negative feedback is desired. All options include an input reactivity, r_o , a bias reactivity, r_B , and sums over scram curves and control variables.

The quantity r_o is an input quantity and is the reactivity corresponding to the assumed steady-state reactor power at time equal to zero. This quantity must be ≤ 0 . A nonzero quantity indicates that a neutron source is present. For most applications, $r_o = 0$ is acceptable.

The bias reactivity, r_B , is calculated during input processing such that $r(0) = r_o$. The purpose of the bias reactivity is to ensure that the initial reactivity is still equal to the input reactivity after including the feedback effects. Without this quantity, the user would have to manually adjust a scram curve or control variable to obtain the input value of initial reactivity or have a step input of reactivity as the transient starts. The bias reactivity, r_B , is printed out during input processing.

The scram curves are obtained from general tables defining reactivity as a function of time. Each table can have an associated trip number. If the trip number is not entered or zero, time is the search argument. If the trip number is nonzero, the search argument is -1.0 if the trip is false. If the trip is true, the search argument is time minus the time at which the trip last turned true. These tables can be used to describe reactivity changes from rod motion.

Control variables can be defined to represent power control systems or to implement alternate feedback models. However, reactor kinetics advancement precedes control system evaluation; thus, feedback from control variables is delayed one time step.

The separable option uses two tables, one defining reactivity as a function of volume density and the other defining reactivity as a function of volumetric average fuel temperatures. The tables allow nonlinear feedback owing to moderator density and fuel temperature changes. A constant temperature coefficient allows for linear moderator temperature feedback, and an additional linear fuel temperature feedback is provided. The separable option is so named because of the assumption that each feedback mechanism is independent and the total reactivity is the sum of the individual effects. The separable option does not directly allow boron feedback, but boron effects can be modeled through the control system.

Data for the separable option can be obtained from reactor operating data, reactor physics calculations, or a combination of the two. The required moderator temperature coefficient is not the usually quoted quantity. Assume the moderator feedback is a function of density and temperature, $r(\rho, T)$, and density is a function of temperature, $\rho(T)$. The usual temperature coefficient is the total derivative, $\frac{dr}{dT}$.

The input requires partial derivatives. The moderator density feedback is $\left. \frac{\partial r}{\partial \rho} \right|_T$; the temperature coefficient is $\left. \frac{\partial r}{\partial T} \right|_\rho$.

The three- and four-dimensional table lookup and interpolation option uses three or four quantities as the independent variables. The four-dimensional table includes the effects of boron; the three-dimensional table does not include boron effects. Two suboptions allow a choice of independent variables. One choice is reactivity as a function of moderator density void weighted moderator temperature, volumetric average fuel temperature, and boron density. This option uses the same variables as the separable option plus boron effects if four variables are used. The other option uses void fraction, liquid moderator temperature, volume averaged fuel temperature, and boron concentration as independent variables. Feedback effects in light water power reactors are usually expressed in terms of these quantities. The multidimensional interpolation allows nonlinearities and interaction of feedback effects but burdens the user with obtaining a larger amount of reactivity data. As with the separable option, required data can be obtained from plant data or reactor physics calculations. As discussed in Volume 1, a data point must be entered for each combination of coordinate values. Accurate reactivity data need only be entered for points near zero reactivity. Once the shutdown reactivity decreases below -2.0 dollars, little change in fission energy release occurs with further decreases in reactivity. Thus, in sections of the multidimensional table where reactivity is known to be very much shut down, data can be determined from extrapolation and need not be accurate. Similarly, some parts of the table may contain large values of reactivity. The user does not expect the transient to use this portion of the table, but the code input requires all tabular points to be entered. Again, accurate data need not be entered; if the transient should enter this area, the large power rises will be evident and the user can investigate the modeling difficulty. In some instances, a coordinate value is introduced to ensure accuracy in one section of a table, but the detail is not needed in other parts of the table. Where the detail is not needed, data could be obtained at a more coarse mesh, and the user can interpolate to meet the input requirements of the code.

Usually, several hydrodynamic volumes are used to represent the coolant channels in a reactor core and several heat structures represent the fuel pins. Weighting factors are input to specify the reactivity contribution of each hydrodynamic volume and heat structure to the total. Reactivity feedback is usually defined such that the weights for volumes and heat structures each should sum to one. The code does not check that the weights sum to one.

The use of the weights is different between the separable and table options. In the separable option, a reactivity effect is first computed for a volume or for a heat structure. Then its contribution to the total reactivity is obtained by multiplying the effect by the weighting factor. This order is reversed for the table option.

Weighted-averaged independent variables for table lookup and interpolation are obtained by using volume or heat structure values and the weighting factors. Table evaluation for total feedback uses the averaged values. It is possible to define a table equivalent to the separable data. However, slightly different transient results would be obtained using the equivalent data owing to the difference in application of the weighting factors.

In steady-state problems, the user usually wishes to specify reactor power. If reactivity feedback data are entered, reactor power will vary as the reactor system moves toward a steady-state condition. To prevent this, a control system could be defined to adjust reactivity to maintain constant power. A simpler alternative is to omit reactivity feedback in steady-state, and the reactor power will remain constant at the input value. At the restart to start the transient, the original reactor kinetics data plus feedback data can be entered.

6 GENERAL TABLES AND COMPONENT TABLES

General tables provide data for several models, including heat structures, valves, reactor kinetics, and control systems. The general table input provides for the following tables: power versus time, temperature versus time, heat flux versus time, heat transfer coefficient versus time, heat transfer coefficient versus temperature, reactivity versus time, and normalized valve area versus normalized stem position. An input item identifies each table so that proper unit conversion and input checking can be done. For example, specifying a temperature table when a power table is required is detected as an error. Because these tables are often experimental data, or scaling may be needed for parametric studies, the input provides for conversion and/or scaling factors for these tables. Input editing of these tables includes both the original and scaled data. General tables can be entered, deleted, or replaced at restart. The tables are linearly interpolated between table values, and the end-point values are used when the search arguments are beyond the range of entered data.

6.1 General Table

Figure 6.1-1 shows input data for a power-type general table, and the graph shows its time history. The first entry of the first line indicates that it is a power versus time table, the second entry, indicates that the trip number for the table is trip number 605, the third entry indicates that the time values are to be used as entered, i.e., multiplied by 1.0, the last entry indicates that the power values that are input should be scaled by $50.0 + 6$, i.e., 50 MW. The search logic is determined by the trip input. A nonzero trip number specifies the following logic: when the trip is false, the table is interpolated using a search argument of -1.0 , resulting in a power of zero up to the trip time t_r ; when the trip is true, the table is interpolated with search argument $t - t_r$, effectively shifting the origin of the table to time t_r . This is analytically equivalent to the application of a unit step function and delay. If a zero trip number is specified, current time is always the search argument. The remaining input defines the time history. The data are input as x-y pairs of time and power. The tabular data show two data points having the same time value, zero, but having different power values. This allows entry of step changes, as shown on the graph. The graph also illustrates that when search arguments are beyond the range of entered data, endpoint values are used rather than extrapolation.

Entry of a nonzero trip number in general tables is valid only when time is the independent variable.

Time-dependent volumes, time-dependent junctions, and pump angular velocity tables are examples of component tables. These tables provide for entry of a trip number and in the default mode use time as the independent variable. In this mode, the use of the trip time is identical to that described for time-dependent general tables. But the time-dependent volumes, junctions, and pump velocity tables also permit any time-advanced quantity to be specified as the independent variable. If a trip is specified and is false, the table is interpolated with -1.0×10^{75} as the search argument. If no trip is specified, or the trip is true, the specified time-advanced quantity is the search argument.

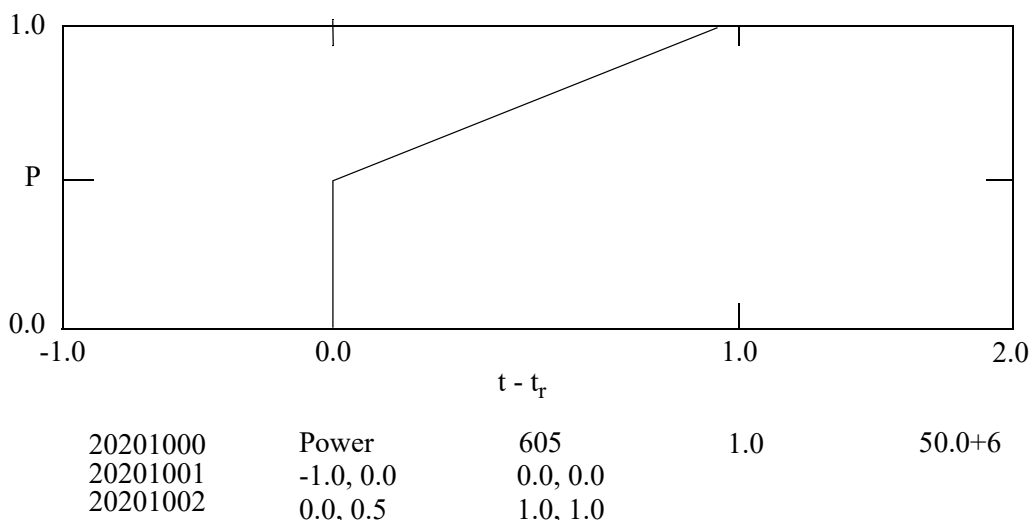


Figure 6.1-1 Input Data for a Power-Type General Table and Graph

A typical use of tables using a quantity other than time as the independent variable is the modeling of a high- or low-pressure reactor safety injection system. Rather than model the valve, pump, and motor for the system, a time-dependent junction is used to approximate the injection system. The pressure at the injection point is specified as the independent variable, and flow rate is the dependent variable. The table would define zero flow for the first zero pressure value, then appropriate flow rates for the second zero pressure and following pressure values. The last pressure value would be the cutoff pressure of the pump and have a corresponding zero flow. In normal reactor operation, the trip would be false, and the table interpolation would return zero flow. When the safety system is actuated, flow may still be zero if the reactor pressure exceeds the cutoff pressure. As the reactor pressure drops, flow would start; and the table could indicate increasing flow with decreasing pressure, possibly up to a maximum flow rate. The source of injection water is usually a time-dependent volume. This technique would not add pump work to the injected fluid. Some approximation of the pump work could be made by also specifying the injection point pressure as the independent variable of the time-dependent volume and entering appropriate thermodynamic conditions as dependent variables.

6.2 FREETAB Table

The following sections provide example FREETAB input for interpolation types "linear" and "polynom".

6.2.1 Interpolation Type "linear"

The example provided here is for interpolation type "linear", with the number on knot points in the independent vectors set to: $n_x=3$, $n_y=3$, and $n_z=2$. The x-, y-, and z-coordinates are as follows:

$x=\{0.0, 1.0, 2.0\}$

$y=\{0.0, 0.5, 1.0\}$

$z=\{1.0, 1.5\}$

For a table component ID of CCC=153, and assuming a trip ID of 101, the FREETAB input for this example would be as shown below:

```

20215300 freetab 101
20215301 linear 3 3 2 1.0      * nx, ny, nz, default value
20215302 0.0, 1.0, 2.0,      * x-coordinates
20215303 0.0, 0.5, 1.0      * y-coordinates
20215304 1.0 1.5            * z-coordinates
20215311 3.00000e+00 1.05000e+01 3.90000e+01 * x={:}, y=0.0, z=1.0
20215312 4.12500e+00 1.44375e+01 5.36250e+01 * x={:}, y=0.5, z=1.0
20215313 3.00000e+00 1.05000e+01 3.90000e+01 * x={:}, y=1.0, z=1.0
20215314 5.12500e+00 1.79375e+01 6.66250e+01 * x={:}, y=0.0, z=1.5
20215315 7.04688e+00 2.46641e+01 9.16094e+01 * x={:}, y=0.5, z=1.5
20215316 5.12500e+00 1.79375e+01 6.66250e+01 * x={:}, y=1.0, z=1.5

```

6.2.2 Interpolation Type "polynom"

The example provided here is for interpolation type "polynom", and assumes the following polynomial:

$$\begin{aligned}
 F(x, y, z) = & (((a_x \cdot x + b_x) \cdot x + c_x) \cdot x + d_x) \\
 & \cdot (((a_y \cdot y + b_y) \cdot y + c_y) \cdot y + d_y) \cdot \\
 & \cdot (((a_z \cdot z + b_z) \cdot z + c_z) \cdot z + d_z)
 \end{aligned}
 \tag{6.2-1}$$

where

$$\begin{aligned}
 (a_x = 1.0) \quad (b_x = 0.5) \quad (c_x = 1.0) \quad (d_x = 1.0) \\
 (a_y = 0.0) \quad (b_y = -1.5) \quad (c_y = 1.5) \quad (d_y = 1.0) \cdot \\
 (a_z = 0.0) \quad (b_z = 1.5) \quad (c_z = 0.5) \quad (d_z = 1.0)
 \end{aligned}
 \tag{6.2-2}$$

Based on the above, the dimensions of the polynomial in the $\{x,y,z\}$ coordinates would be: $k_x=4$, $k_y=3$, and $k_z=3$. For this example, the limits on the $\{x,y,z\}$ coordinates are: $x=\{0.0,3.0\}$, $y=\{0.0,1.0\}$, and $z=\{1.0,1.5\}$. Thus, for a table component ID of CCC=153, and assuming a trip ID of 101, the FREETAB input for this example would be as shown below:

```

20215300 freetab 101
20215301 polynom 4 3 3 1.0    * kx, ky, kz, default value
20215302 0.0 3.0             * x-range
20215303 0.0 1.0             * y-range

```

FREETAB Table

20215304	1.0	1.5						* z-range
20215311	1.0	0.5	1.0	1.0				
20215312	-1.5	1.5,	1.0	1.5	0.5	1.0		

7 INITIAL AND BOUNDARY CONDITIONS

All transient analysis problems require initial conditions from which to begin the transient simulation. Usually, the initial conditions will correspond to a steady-state, with the transient initiated from a change of some boundary condition. In general, the initial conditions required are a determinate set of the dependent variables of the problem. The hydrodynamic model requires four thermodynamic state variables in each volume and the velocities at each junction. Heat structures require the initial temperature at each node, control systems require the initial value of all control variables, and kinetics calculations require initial power and reactivity. All of these parameters are established through the code input and initialization process for a new problem. For a restart problem, the values are established from the previous calculation. For restart with renodalization or problem changes, the initialization will result from a combination of the two processes, and care must be exercised to ensure that the input values are compatible with those from the restart, especially if an initial steady-state is to be simulated.

Boundary conditions may be required for hydrodynamic models, heat structures, or control components if these parameters are governed by conditions outside of the problem boundaries. Examples of these could be mass and energy inflows or an externally specified control parameter.

Obtaining a desired simulation is very dependent upon proper specification of initial and boundary conditions. The purpose of this section is to summarize recommended approaches for these specifications.

7.1 Initial Conditions

All variables of the problem that are established by integration require initial values in order to begin a calculation or simulation. Problem variables related to the integration variables through quasi-steady relationships do not require initial conditions, since they can be established from the initial values required for the integration variables. An example is the pump head, which is related to the pump flow and speed, both of which are obtained by integration. Thus, the initial conditions for pump flow and speed must be specified.

7.1.1 Input Initial Values

Input initial values are required in order to begin a new problem regardless of whether a steady-state or a transient run is specified. These initial values are supplied by the user through input for each component. (Heat structures are an exception and can be initialized either by input or by steady-state initialization using the heat structure boundary conditions at time zero.)

The hydrodynamic volume components have seven options for specifying the volume initial conditions (see Section 2.4.1 for more detail). Four options are provided for pure steam/water systems, and the remaining three options allow noncondensables. Boron concentration can be specified with all seven options by adding 10 to the control word, Word W1(I). Regardless of what option is used, the initialization computes initial values for all primary and secondary dependent variables. The primary variables are

pressure, void fraction, two-phasic energies, noncondensable quality, and boron concentration. Secondary variables are quality, density, temperature, and so on.

The most common specification will be an equilibrium condition for the steam/water system. The options 1-3 [control word W1(I) on Card CCC0200, in Appendix A] are equilibrium specifications using temperature and quality, pressure and quality, and pressure and temperature. The first two conditions are valid combinations for single- (at the saturation point) or two-phase conditions. The third combination is valid only for single-phase nonsaturated conditions. When air is specified, it is best to use conditions of humid or saturated air at the initial pressure and temperature of the system. The specification of dry air can cause numerical difficulties when mixing with water or water vapor occurs. If air is the only system component and mixing with water or water vapor does not occur, the specification of pure air will cause no problems.

Heat structure initial temperatures must be input. Depending upon the initialization option selected, these temperatures are either used as the initial temperatures or as the initial guess for an iterative solution for a steady-state temperature profile. The iteration solution will attempt to satisfy the boundary conditions and heat sources/sinks that have been specified through input. Some care is needed, since an indeterminate solution can result from specification of some boundary conditions (e.g., a two-sided conductor with specified heat fluxes). If the initial temperature of a heat structure is unknown, it is generally safer to use the steady-state option and supply as a first guess a uniform temperature distribution equal to the temperature of a hydrodynamic volume to which it is connected. In the case of a two-sided structure, either side may be selected. The steady-state solution algorithm will rapidly converge to a steady-state temperature distribution.

Initial conditions must be specified for each control component used, even if the option to compute the initial condition is selected. As stated above, only the integral functions should require initial conditions. However, since control components are initialized using a sequential single-pass solution scheme and since some control variables may be specified as arguments for other control variables, it is possible for some to be initially undefined. Hence, the initial value for all control variables must be specified. Also, the code does not check whether initial values are needed nor whether they are reasonable; thus, the user should always supply an accurate initial value.

The reactor kinetics model requires specification of an initial power and reactivity. Previous power history data may also be entered.

7.1.2 Steady-State Initialization

RELAP5 contains an option to perform steady-state calculations. This option uses the transient hydrodynamic, kinetics, and control system algorithms and a modified heat structure thermal transient algorithm to converge to a steady-state. The differences between the steady-state and transient options are that a lowered heat structure thermal inertia is used to accelerate the response of the thermal transient, and a testing scheme is used to check if steady-state has been achieved. When steady-state is achieved, the run is terminated, thus saving computer time. The results of the steady-state calculation are saved so that a

restart can be made in the transient mode. In this case, all initial conditions for the transient are supplied from the steady-state calculation. It is also possible to restart in either the transient or steady-state mode from either a prior transient or steady-state run.

The user should be aware that use of the steady-state option provides a more optimum solution than simply running the problem as a transient and monitoring the results. This occurs because the code monitors results for the entire system, including the effects of calculational precision. Also, thermal inertia for the heat structures is generally quite large, so that for the transient option, the heat structure temperature distribution will not achieve steady-state in the time that a hydrodynamic steady-state can be achieved. Hence, use of the steady-state option will provide the user with a precise steady-state, including a precise heat structure steady-state.

It is still necessary to supply input specifying initial conditions for a steady-state run. However, the accuracy of the input data is less critical, since they are simply used as a starting point for convergence to a steady-state. The values used should be reasonable, however, since the closer they are to the actual steady-state, the shorter the calculation will be to achieve steady-state.

Once an initial steady-state is calculated, the user can save the RESTART-PLOT file and perform subsequent new steady-state runs using the previous steady-state results. This results in reduced calculational times for the subsequent runs and at the same time maintaining a complete set of steady-state initializations.

The steady-state initialization calculation is an open-loop calculation unless control functions are defined such that active control systems are used to obtain desired operating points. Active control is achieved using controlled variables such as pressure, flow rate, etc. The user must design and implement such control functions, and only a limited number of system parameters can be controlled independently. In this regard, the model behaves exactly as a real system, and, if a resistance to flow must be varied to achieve the desired steady-state, then a valve must be used with a controller. The use of a controller to achieve a desired steady-state can save considerable time compared to the process of open-loop control, in which a resistance or other parameter is varied from run to run until the desired steady-state is achieved.

In providing control systems and trips to drive the solution to steady-state, two rules of thumb must be considered, both of which revolve around the basic purpose of the steady-state run. The first rule is that if the run is to simulate the real behavior of a plant in achieving steady-state, then control systems and trips simulating real plant controls or control procedures should be designed. However, the second rule of thumb is that if the run is simply to achieve a steady-state initialization of the system model, then controls not representative of the actual system may be designed that will drive the solution to steady-state in the fastest manner possible. The only restriction is that stability of the calculations must be maintained.

There is also an option, Word 4, digits tt, on the time step control cards, 201 through 299, which allows the user to select part of the steady-state calculation to be used. The thermal inertia of the heat structures is lowered, but the testing scheme to check the derivatives of variables to determine a steady-state is not used. This gives a user the advantage of using the artificially accelerated thermal steady-state in

the heat structures while allowing use of either a set end time or else the user's own choice for a variable to monitor for a steady-state through a simple control system.

7.2 Boundary Conditions

Boundary conditions are required in most transient calculations. In reality, boundary conditions take the form of the containment atmosphere, operator actions, or mass and energy sources that are not explicitly modeled as part of the system. Such boundary conditions are simulated by means of time-dependent volumes for specified sources or sinks of mass, time-dependent junctions for specified flows, or specified heat structure surface heat fluxes and energy sources. Specified variation of parameters in control components to simulate an operator action may also be used. The time variation of the boundary conditions is specified by input tables that can also be varied dynamically by using trips.

7.2.1 Mass Sources or Sinks

Hydrodynamic mass sources or sinks are simulated by the use of a time-dependent volume with a time-dependent junction. The thermodynamic state of the fluid is specified as a function of time by input or by a control variable. The thermodynamic state is needed for inflow because the densities and energies are needed in the donored flux terms in the density and energy equations. The time-dependent junction flows or velocities are also specified. This approach can be used to model either an inflow or an outflow condition; however, care is required in modeling outflows. A time-dependent junction is analogous to a positive displacement pump in that the flow is independent of the system pressure. In the case of outflow, it is possible to specify a greater outflow than inflow to a volume or even outflow that will exhaust the volume. In this case, a numerical failure will result when the equivalent of a negative density is calculated. For this reason, modeling outflows using a time-dependent junction is not recommended.

It should be noted that for the case of a TMDPVOL (time-dependent volume) and TMDPJUN (time-dependent junction), the pressure upstream of the TMDPJUN in the TMDPVOL is neither changed nor checked by the code to be consistent with that derived from the pressure drop obtained in the momentum equation if the velocity option is used. The pressure upstream is the value that the user enters for the TMDPVOL. This is because this type of boundary condition is analogous to a positive displacement pump where the inflow rate is independent of the system pressure (see Volume II, Section 2 of the manual). An additional fact that should be considered when using a time-dependent junction as a boundary is that pump work is required for system inflow if the system pressure is greater than the time-dependent volume pressure. In particular, any energy dissipation associated with a real pumping process is not simulated.

7.2.2 Pressure Boundary

A pressure boundary condition is modeled using a time-dependent volume in which the pressure and thermodynamic state variables are specified as a function of time through input by tables or by a control variable. The time-dependent volume is connected to the system through a normal junction; thus, inflow or outflow will result, depending upon the pressure difference. Several precautions are needed when specifying a pressure boundary, since flow invariably accompanies such a boundary. First, the time-

dependent volume conditions must represent the state of fluid that would normally enter the system for an inflow condition. Second, there are implied boundary conditions for a time-dependent volume in addition to the specified values. Third, only the static energy of an incoming flow is fixed by a time-dependent volume. The total energy will include the inflow kinetic energy that increases with increasing velocity.

The additional boundary conditions represented by a time-dependent volume concern the virtual viscosity terms inherent in the numerical formulation of the momentum equation (see Section 3 in Volume I for a detailed discussion). For this purpose, the derivative of velocity across the time-dependent volume is zero, and the length and volume are assumed to be zero (regardless of the specified input). The fact that the energy of inflow increases with velocity can lead to a nonphysical result, since the stagnation pressure also increases, and for a fixed system pressure, an unmitigated increase in inflow velocity can result. This effect can be avoided by making the cross-sectional area of the time-dependent volume large compared to the junction so that the volume velocity of the time-dependent volume is small, and thus the total energy of the inflow is constant. When a large area ratio exists between the time-dependent volume and the junction connecting it to the system, a reservoir or plenum is simulated. As a general rule, all pressure boundary conditions having either inflow or outflow should be modeled as plenums for stability and realism. In particular, when an outflow is choked, the critical flow model more closely approximates the conditions at a large expansion (i.e., little or no diffusion occurs). Thus, this assumption is consistent with the choked flow model and is, therefore, recommended.

Volume I of the manual indicates that linear interpolation between neighboring volume properties is used for junction properties (e.g., densities) that are not donored. These junction properties are used in the momentum equations. For the case of a regular junction (i.e., non-time-dependent junction) next to a time-dependent volume (i.e., the back pressure case), the momentum equations are modified to set the junction properties equal to the properties from the volume connected to the junction that is not the time-dependent volume. This results in the correct situation where changing the outlet time-dependent volume temperature will not change the calculation results by changing the momentum equations junction properties. Thus, the code does not allow overspecification of the boundary conditions for this case.

8 PROBLEM CONTROL

8.1 Problem Types and Options

RELAP5 provides for four problem types--NEW, RESTART, PLOT, and STRIP. The first two are concerned with simulating hydrodynamic systems; NEW starts a simulation from input data describing the entire system; RESTART restarts a previously executed NEW or RESTART problem. PLOT is no longer operational. STRIP is an output-type runs using the restart-plot file written by NEW or RESTART problems. NEW and RESTART problems require an additional option to be selected, STDY-ST or TRANSNT.

A RESTART problem may restart from any restart record. A note indicating the restart number and record number is printed at the end of the major edit whenever a restart record is written. The restart number is equal to the number of attempted advancements and is the number to be used on Card 103 to identify the desired restart record. The record number is simply a count of the number of restart records written, with the restart record at time equal zero having record number zero. Quantities written in the restart-plot records by default are noted in the input data description. User-specified input can add additional quantities to the restart-plot records.

PLOT and STRIP are output-type runs. PLOT generates plots from data stored on the restart-plot file. The PLOT capability is not now operational but is still documented. The PLOT capability may be dropped from the code since NPA^{8.1-1} and XMGR5,^{8.1-2} an extension of XMGR,^{8.1-3} and ACGRACE allow very general and high quality plots of RELAP5 results and associated information. STRIP writes selected information from a restart-plot file onto a new file. The new file consists of records containing time and the user-selected variables in the order selected by the user. Data to be plotted or stripped are limited to that written in the plot records on the restart-plot file.

8.1.1 References

- 8.1-1. D. M. Snider, K. L. Wagner, W. H. Grush, and K. R. Jones. *Nuclear Plant Analyzer, Volumes 1-4*. NUREG/CR-6291, INEL-94/0123. Idaho National Engineering Laboratory. December 1994.
- 8.1-2. K. R. Jones and J. E. Fisher. *XMGR5 Users Manual*. INEL/EXT-97-00346. Idaho National Engineering Laboratory. March 1997.
- 8.1-3. P. J. Turner. *ACE/gr User's Manual*. SDS3, 91-3. Beaverton, OR. 1992.

8.2 Time Step Control

Input data for time step control consist of one or more cards containing a time limit, minimum time step, requested (maximum) time step, control option, minor edit plot/frequency, major edit frequency, and restart frequency. The time limit must increase with increasing card numbers. The information on the first card is used until the problem time exceeds the card limit; then, the next card is used, and so on. In restart

problems, these cards may remain or may be totally replaced. Cards are skipped if necessary until the problem time at restart is properly positioned with regard to the time limit values.

The control option is a packed five digit (ssdt) word containing a major edit select option (ss), a debug output option (d), and the time step control (tt). The major edit select option (ss) allows sections of major edits for the hydrodynamic volumes and junctions, heat structures, and statistics to be skipped. The debug output option (d) forces any combination of plot, minor edits, or major edit output to be written at each successful advancement rather than at just the completion of advancement over a requested time step. The time step control option (tt) allows the user to change the time step control logic. All options can be changed with each time step control card.

Specifically, digit tt allows the user to select several time step control options. This time step control option is represented by a number between 0 and 31 that can be thought of as a five-bit number. Entering zero (no bits set) attempts to advance both the hydrodynamic and heat conduction advancements at the requested time step. However, the hydrodynamic time step will be reduced if necessary such that the Courant limit is satisfied. If out-of-range thermodynamic property conditions are encountered, the hydrodynamic advancement will be retried with reduced time steps. The problem will be terminated if the time step must be reduced beyond the minimum time step. Each time step reduction halves the previously attempted time step. At the beginning of an advancement for a requested time step, a step counter is set to one. Whenever a reduction occurs, the step counter is doubled. When a successful advancement occurs, the step counter is reduced by one. When the step counter is decremented to 0, the problem has been advanced over one requested time step. Doubling of the time step is allowed only when the step counter is even, and the step counter is halved when the time step is doubled. With no bits set, the time step is doubled whenever possible. At the completion of advancements over a requested time step, the next requested advancement is obtained and may be different from the previous requested time step if data from the next time step control card are used. If necessary, the new requested time step is reduced by halving until the new actual time step is < 1.5 times the last successful time step.

Setting bit one (entering 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, or 31) includes the features described for entering zero and, in addition, uses the halving and doubling procedures to maintain an estimate of hydrodynamic truncation error within program-defined limits. The estimate is based on the mass error computed by comparing densities derived from the mass conservation equations and the equations of state. If an acceptable error is not reached and the next reduction would lead to a time step below the minimum time step, the advancement is accepted. The first 100 such occurrences are noted in the output.

If the second bit is set (entering 2, 3, 6, 7, 10, 11, 14, 15, 18, 19, 22, 23, 26, 27, 30, or 31), the heat structure time step will be the same as the hydrodynamic time step. The time step control for the hydrodynamics is determined by the status of the first bit as described above, and both the heat conduction and hydrodynamic advancements are repeated when a time step reduction occurs.

If the third bit is set (entering 4, 5, 6, 7, 12, 13, 14, 15, 20, 21, 22, 23, 28, 29, 30, or 31), the heat conduction-transfer and the hydrodynamics are advanced implicitly. When the third bit is set indicating

implicit coupling of heat conduction-transfer, the second bit indicating that the two advancements use the same time step must also be set. (Input checking does not now enforce this, but a future code change will include this checking.) If the third bit is not set, the heat conduction-transfer and hydrodynamic advancements are serially coupled. That is, the heat conduction transfer is advanced first using old hydrodynamic information, and the hydrodynamics is then advanced using new heat transfer information. The time step control for hydrodynamics is determined by the status of the first bit, as described above.

If the fourth bit is set (entering 8, 9, 10, 11, 12, 13, 14, 15, 24, 25, 26, 27, 28, 29, 30, or 31), the hydrodynamics will use the nearly-implicit hydrodynamic numerical scheme. The time step can be as large as 20 times the Courant limit for the TRANSNT option and 40 times the Courant limit for the STDY-ST option. The time step control for hydrodynamics is determined by the status of the first bit, as described above.

If the fifth bit is set (entering 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, or 31), control of termination of the steady-state advancement is used. At the end of each advancement in steady-state problems, an algorithm measures the approach to steady-state. If this bit is set, advancement will not be terminated by the algorithm; if the bit is not set, steady-state can be terminated by the algorithm when it detects steady-state has been reached. This control can allow the user to ensure that a steady-state run always uses a defined minimum advancement time, then can allow another period of advancement time for the algorithm to determine steady-state, and, finally, manually terminate the run if necessary by trip control or exceeding the end time of the last time-step control card.

Note that combinations of the effects of setting of the individual bits are achieved by setting bits in combination. For example, entering three (setting bits two and one) results in the combined effects described above for bits two and one. Older versions of RELAP5 would convert 2 to 3 to maintain compatibility; this is no longer done.

Entering zero is not recommended except for special program testing situations. If bit one is set but not set for 2 and 3, care must be taken in selection of the requested time step. Individually, the hydrodynamic and heat conduction advancements are stable; the hydrodynamic time step is controlled to ensure stability, the heat conduction solution with constant thermal properties is stable for all time steps, and the change of thermal properties with temperature has not been a problem. The serial coupling of the hydrodynamic volumes and heat structures through heat structure boundary conditions can be unstable, and excessive truncation error with large time steps can occur. This has been observed in test problems. Entering three usually eliminates the problem, and nearly all assessment of the code has been done using option three. Using option seven, which includes the implicit coupling of heat conduction and hydrodynamics should lead to an improved advancement. Users are encouraged to use option seven, but with the caution that this is a recent addition to RELAP5 and is still under assessment. Several improvements have been made to the nearly-implicit advancement, but use of that option is still under development and assessment. Use of option 15 is recommended for steady-state runs and slow transients; and users are encouraged to use option 19, but with the same caution as option 7.

The minor edit, major edit, and restart frequencies are based on the requested time-step size. A frequency n means that the action is taken when a period of time equal to n requested time steps has elapsed. The edits and the restart record are written at time zero and at the specified frequencies up to the time limit on the time step control card. The maximum time step is reduced if needed, and the edits and restart record are forced at the time limit value. Actions at the possibly new specified frequencies begin with the first advancement with a new time step control card. A restart forces a major and minor edit to be written, and a major edit forces a minor edit to be written. Plot information is written to the internal plot and restart-plot files whenever a minor edit is written. Note that minor edits are produced only if minor edit requests are entered; an internal plot file is written only if internal plot requests are entered; and plot and restart data are written on the restart-plot file only if the file is requested.

An option (d) used for program testing can force a plot print, minor edit, major edit, or combinations of these to be written at each advancement. Care should be used, since considerable output can be generated.

Major edits forced by the program testing option or the last major edit of the problem terminated by approach to the job CPU limit may not coincide with the requested time step. When this occurs, a warning message is printed that states that not all quantities are advanced to the same time points.

8.3 Printed Output

A program version identification is printed at the beginning of printed output and the first page following the list of input data.

8.3.1 Input Editing

Printed output for a problem begins with a listing of the input; each line of input is preceded by a sequence number. The sequence number is not the same as the card number. Notification messages are listed when data card replacement or deletion occurs. Punctuation errors, such as an alphabetic character in numeric fields, multiple signs, periods, etc., are noted by an error message, and a ^ (caret) is printed under the card image indicating the column position of the error.

Input processing consists of three phases. The first phase simply reads and stores all the input data for a problem such that the data can later be retrieved by card number. Error checking is limited to punctuation checking, and erroneous data flagged during this phase nearly always causes additional diagnostics in later phases. The second phase does the initial processing of data. Input data are moved and expanded into dynamic arrays sized for the problem being solved, and default options are applied. Processing and error checking is local to the data being processed. That is, when processing a single-junction component, no checking is performed regarding the existence of connected volumes. Similarly, hydrodynamic volumes connected to heat structure surfaces are not checked during processing of heat structure boundary data. At the end of this phase, all data cards should have been used. Unused cards are considered errors and are listed. Asterisks following the card number indicate that the card number was bad, that an error was noted in the card image listing, and that the number is the sequence number rather than the card number. The

third phase completes input processing and performs requested initialization. Once the second phase has been completed, data specifying linkages between various blocks of data can now be processed and checked. Examples of error checking are junction connections made to nonexistent volumes, heat structure surfaces connected to nonexistent hydrodynamic volumes, specified thermal properties, and power data not entered. Solution of steady-state heat conduction for initial temperature distribution in heat structures is an example of initialization.

The flow map used for a particular volume is printed out during the input editing of the hydrodynamic volumes. **Table 8.3-1** shows the flow map and the corresponding number printed out under the label Flow Map.

Table 8.3-1 Flow Map Identifiers

Flow map	Number (input edit)
Vertical	1
Horizontal	2
Annular	3
Pump	4
ECC Mixer	5

Depending on the type of data, input is edited in only one of the last two edits or in both of them. Error diagnostics can be issued during either phase, even if no editing for the erroneous data is done in a phase. When an error is detected, possible corrective actions are disregarding the data, which usually leads to other diagnostics, inserting benign data, or marking data as being entered but useless for further processing. These actions are taken so that input processing continues despite severe errors (other than on problem type and options). Regardless of errors, all data are given preliminary checking. Severe errors can limit cross-checking. Correcting input errors diagnosed in a submittal may lead to other diagnostics in a subsequent submittal, as elimination of errors allow more detailed checking. Except for exceeding requested computer time, disk limits, and printed output limits, any abnormal termination is considered a programming error and even exceeding computer time limits is prevented during transient execution. The final message of input processing indicates successful input processing or that the problem is being terminated as the result of input errors.

8.3.2 Major Edits

Major edits are an editing of most of the key quantities being advanced in time. The amount of output depends on the input deck and output options chosen by the code user. Output includes a time-step summary, trip information, reactor kinetics information, one to four sections of hydrodynamic volume information, hydrodynamic volume time-step control information, one or two sections of hydrodynamic junction information, metal-water reaction information, heat structure/heat transfer information, heat

structure temperatures, reflood information, reflood surface temperatures, cladding rupture information, surface radiation information, control variable information, and generator information. Major edits are quite lengthy, and care should be used in selecting print frequencies. Some sections of major edits can be bypassed through input data on time-step control cards. An example of a major edit is shown in **Figure 8.3-1**.

Each section of information is discussed below in the order that each appears in a major edit. In particular, what the abbreviated labels stand for as well as how they relate to variables used in Volume 1 of this manual are indicated.

8.3.2.1 Time Step Summary

As shown in **Figure 8.3-1**, the first section of a major edit prints the problem time and statistics concerning time step control. ATTEMPTED ADV. is the total number of successful and repeated advancements. REPEATED ADV. is the number of advancements that were not accepted and were retried with a halved time step. SUCCESSFUL ADV. is the number of accepted advancements. REQUESTED ADV. is the number of advancements with the specified requested maximum time step. These are presented in two columns. The TOT. column is over the entire problem; the EDIT column contains the number since the previous major edit. MIN. DT, MAX. DT, and AVG. DT are the minimum, maximum, and average time step used since the last major edit. REQ. DT is the requested maximum time step used since the last major edit. This quantity may not be the requested time step entered on the card if the major edit is for the final time value on the card. LAST. DT is the time step used in the last advancement. CRNT. DT is the time step limit based on the Courant stability criterion for the last advancement. ERR. EST is the estimate of the truncation mass error fraction at the last advancement. This is the maximum of the two types of computed mass error (ϵ_m and ϵ_{rms}) discussed in Volume I, Section 8. Entering 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, or 31 for the time step control option will reduce or double the time step to keep this quantity between the limits of 8.0×10^{-4} and 8.0×10^{-3} if the mass error criterion is controlling the time step. In **Figure 8.3-1**, the problem is running at the requested (maximum) time-step, and the ERR. EST is below the lower limit. CPU is the CPU time for the entire problem up to the time of the major edit. TOT. MS is the total mass currently contained in the hydrodynamic systems, and MS. ERR is an estimate of the cumulative error in the total mass owing to truncation error. M. RATO is the ratio of the cumulative mass error to the total mass at the start of the transient; M. RATN is the ratio of the cumulative mass error to the current total mass. The output lists the ratio with the largest denominator, thus the smaller of the two ratios. TIME is the simulated time for the entire problem up to the time of the major edits.

8.3.2.2 Trip Information

At major edits, each defined trip number and the current TIMEOF quantity are printed. The TIMEOF quantity is -1.0 when the trip is false, and when ≥ 0 , indicates that the trip is true and is the time the trip last switched to true. **Figure 8.3-1** includes an example of a trip edit.

REACTOR LOSS OR COOLANT ANALYSIS PROGRAM														
edward's pipe problem base case with extras														
MAJOR EDIT !!time= 0.100000 sec														
attempted adv: tot.=	109	edit=	20	min.dt=	1.000000E-03	sec	last dt=	1.000000E-03	sec	ms.red=	-1.787187E-03	kg		
repeated adv: tot.=	3	edit=	0	max.dt=	1.000000E-03	sec	crnt.dt=	7.025645E-03	sec	tot.ms=	11.1019	kg		
successful adv: tot.=	106	edit=	20	avg.dt=	1.000000E-03	sec	err.est=	3.891584E-06		m.ratio=	-1.149045E-04			
requested adv: tot.=	100	edit=	20	req.dt=	1.000000E-03	sec	cpu=	2.00080	sec	time=	0.100000	sec		
Trip number, trip time (sec)														
501	-1.000000		502	-1.000000		505	6.000000E-03							
601	-1.000000		602	-1.000000		603	0.000000E+00							
Total power fission power gamma power reactivity (dollars)														
6.46074E+10	6.45997E+10	7.75141E+06	1.5008											
System 1 *none* mass= 11.102 kg mass error= -1.78719E-03 kg err.est.= 3.89158E-06														
Vol.no.	pressure	voidf	voidg	voidgo	tempf	tempg	sat. temp.	uf	ug	vol-flag				
	(Pa)				(k)	(K)	(K)	(J/kg)	(J/kg)	tlpvbf				
edward's pipe														
3-010000	2.58633E+06	0.87323	0.12677	0.12564	500.483	498.867	498.905	9.74638E+05	2.60139E+06	0.00000				
3-020000	2.58627E+06	0.88929	0.11071	0.10962	500.538	498.872	498.904	9.74892E+05	2.60140E+06	0.00000				
3-030000	2.58627E+06	0.86369	0.13631	0.13531	500.353	498.870	498.904	9.74035E+05	2.60140E+06	0.00000				
3-040000	2.58638E+06	0.87074	0.12926	0.12806	500.378	498.894	498.910	9.74150E+05	2.60144E+06	0.00000				
3-050000	2.58749E+06	0.89085	0.10915	0.10795	500.524	498.939	498.929	9.74826E+05	2.60151E+06	0.00000				
3-060000	2.58914E+06	0.90717	9.28307E-02	9.18278E-02	500.647	498.997	498.963	9.75398E+05	2.60158E+06	0.00000				
3-070000	2.59131E+06	0.91570	8.43030E-02	8.34841E-02	500.717	499.055	499.008	9.75723E+05	2.60161E+06	0.00000				
3-080000	2.59326E+06	0.91516	8.48351E-02	8.41059E-02	500.714	499.091	499.048	9.75708E+05	2.60161E+06	0.00000				
3-090000	2.59310E+06	0.91185	8.81460E-02	8.72750E-02	500.694	499.033	499.045	9.75614E+05	2.60148E+06	0.00000				
3-100000	2.58992E+06	0.89705	0.10295	0.10177	500.584	498.885	498.979	9.75102E+05	2.60126E+06	0.00000				
3-110000	2.58203E+06	0.88244	0.11756	0.11629	500.472	498.674	498.816	9.74589E+05	2.60111E+06	0.00000				
3-120000	2.56699E+06	0.73794	0.26206	0.23511	498.981	498.471	498.503	9.67675E+05	2.60133E+06	0.00000				
3-130000	2.46936E+06	0.56045	0.43955	0.43627	496.660	496.433	496.438	9.57014E+05	2.60099E+06	0.00000				
3-140000	2.40135E+06	0.49097	0.50903	0.50478	495.333	494.953	494.962	9.50972E+05	2.60059E+06	0.00000				
3-150000	2.36217E+06	0.46318	0.53682	0.53349	494.703	494.063	494.097	9.48107E+05	2.60031E+06	0.00000				
3-160000	2.33684E+06	0.44284	0.55716	0.55561	494.183	493.443	493.532	9.45740E+05	2.60003E+06	0.00000				
3-170000	2.31703E+06	0.42559	0.57441	0.57370	493.704	492.949	493.087	9.43550E+05	2.59979E+06	0.00000				
3-180000	2.30117E+06	0.41320	0.58680	0.58622	493.310	492.561	492.728	9.41752E+05	2.59962E+06	0.00000				
3-190000	2.28849E+06	0.39956	0.60044	0.59981	492.991	492.248	492.440	9.40295E+05	2.59949E+06	0.00000				
3-200000	2.28174E+06	0.43684	0.56316	0.56283	492.797	492.163	492.286	9.39410E+05	2.59962E+06	0.00000				
rhtbody tmdpvcl														
5-010000	1.00000E+05	0.00000E+00	1.00000	1.00000	372.782	372.782	372.782	4.17407E+05	2.50606E+06	0.11000				
Vol.no. rhof rhog rho-mix rho-boron vel-liquid vel-vapor sounde														
	(kg/m3)	(kg/m3)	(kg/m3)	(kg/m3)	(m/sec)	(m/sec)	(m/sec)	mix-cup	quality	quality	quality	static	non-cond.	
3-010000	830.80	12.945	727.12	0.00000E+00	0.25810	0.12238	26.817	5.163E-03	2.257E-03	0.000E+00				
3-020000	830.73	12.945	740.19	0.00000E+00	0.37923	0.24891	26.638	5.538E-03	1.936E-03	0.000E+00				
3-030000	830.98	12.945	719.47	0.00000E+00	0.61811	0.47123	26.627	5.361E-03	2.452E-03	0.000E+00				
3-040000	830.94	12.945	725.21	0.00000E+00	0.87499	0.68065	26.526	5.582E-03	2.307E-03	0.000E+00				
3-050000	830.75	12.949	741.49	0.00000E+00	1.1313	0.98186	26.488	5.874E-03	1.906E-03	0.000E+00				
3-060000	830.58	12.956	754.68	0.00000E+00	1.3522	1.2668	26.433	5.912E-03	1.594E-03	0.000E+00				

Figure 8.3-1 Example of Major Edit

3-070000	830.49	12.966	761.57	0.00000E+00	1.5330	1.4546	26.351	5.776E-03	1.435E-03	0.000E+00
3-080000	830.49	12.976	761.14	0.00000E+00	1.6950	1.5898	26.239	5.583E-03	1.446E-03	0.000E+00
3-090000	830.52	12.978	758.46	0.00000E+00	1.8733	1.7781	26.252	5.590E-03	1.508E-03	0.000E+00
3-100000	830.67	12.966	746.48	0.00000E+00	2.1180	2.1369	26.401	5.734E-03	1.788E-03	0.000E+00
3-110000	830.82	12.929	734.67	0.00000E+00	2.4275	2.6664	26.828	6.314E-03	2.069E-03	0.000E+00
3-120000	832.82	12.849	617.94	0.00000E+00	5.9491	9.0256	26.775	6.677E-03	5.449E-03	0.000E+00
3-130000	835.85	12.363	473.88	0.00000E+00	11.926	13.998	32.083	1.000E-02	1.147E-02	0.000E+00
3-140000	837.56	12.027	417.33	0.00000E+00	16.202	17.195	35.657	1.467E-02	1.467E-02	0.000E+00
3-150000	838.36	11.834	394.66	0.00000E+00	19.321	20.204	37.688	1.754E-02	1.610E-02	0.000E+00
3-160000	839.02	11.711	378.08	0.00000E+00	21.396	22.623	38.938	1.919E-02	1.726E-02	0.000E+00
3-170000	839.64	11.615	364.01	0.00000E+00	22.839	24.531	39.881	2.057E-02	1.833E-02	0.000E+00
3-180000	840.14	11.538	353.92	0.00000E+00	23.920	26.099	40.554	2.180E-02	1.913E-02	0.000E+00
3-190000	840.55	11.476	342.74	0.00000E+00	24.935	27.307	41.438	2.276E-02	2.010E-02	0.000E+00
3-200000	840.79	11.440	373.74	0.00000E+00	24.500	29.176	38.238	2.346E-02	1.724E-02	0.000E+00
Vol.no.	rhof	rhog	rho-mix	rho-boron	vel-liquid	vel-vapor	sonde	quality	quality	quality
	(kg/m3)	(kg/m3)	(kg/m3)	(kg/m3)	(m/sec)	(m/sec)	(m/sec)	mix-cup	static	non-cond.
5-010000	958.39	0.59041	0.59041	0.00000E+00	23.455	30.878	473.21	2.257E-02	1.00	0.000E+00
Vol.no.	Q.wall.tot	Qwg.wall.gas	Vapor.gen	gamma.boil	Hif.liq.int	Hig.vap.int	Mass-flux	Reynolds	Reynolds	Flow
	(Watts)	(Watts)	(kg/sec-m3)	(kg/sec-m3)	(Watts/m3-k)	(Watts/m3-k)	(kg/sec-m2)	liquid	vapor	regi
3-010000	834.80	0.00000E+00	15.228	0.48854	1.70679E+07	1.44099E+06	93.722	60278.	462.43	hst
3-020000	606.63	0.00000E+00	15.638	0.35504	1.70830E+07	1.58281E+06	278.02	1.78759E+05	1761.8	hst
3-030000	595.04	0.00000E+00	14.506	0.34830	1.78500E+07	1.52113E+06	450.88	2.89812E+05	3469.4	hst
3-040000	621.34	0.00000E+00	17.690	0.36378	2.15460E+07	1.82284E+06	631.71	4.05983E+05	5387.5	hst
3-050000	633.11	0.00000E+00	16.963	0.37075	1.89565E+07	2.04745E+06	829.37	5.33072E+05	6977.4	hst
3-060000	635.68	0.00000E+00	13.633	0.37233	1.43152E+07	1.84202E+06	1011.4	6.50298E+05	7628.4	hst
3-070000	644.67	0.00000E+00	10.825	0.37763	1.10888E+07	1.81699E+06	1162.2	7.47516E+05	7686.1	hst
3-080000	671.42	0.00000E+00	10.210	0.39332	1.06929E+07	1.71363E+06	1290.4	8.30236E+05	8027.0	hst
3-090000	704.52	0.00000E+00	11.828	0.41277	1.26306E+07	2.08088E+06	1423.2	9.15610E+05	9187.5	hst
3-100000	791.54	0.00000E+00	17.081	0.46389	1.90636E+07	2.96775E+06	1593.7	1.02475E+06	12194.	hst
3-110000	901.27	0.00000E+00	17.188	0.52830	1.89090E+07	6.49353E+06	1798.1	1.15480E+06	17530.	hst
3-120000	2867.0	0.00000E+00	484.66	1.6981	1.81826E+09	9.10588E+07	4030.9	2.57650E+06	1.01646E+05	bby
3-130000	7513.2	0.00000E+00	265.72	4.4462	2.16831E+09	3.92132E+08	6519.7	4.11257E+06	2.85180E+05	bby
3-140000	9875.7	0.00000E+00	194.69	5.8180	9.46904E+08	3.34815E+08	7225.3	4.50771E+06	4.61445E+05	slg
3-150000	10613.	0.00000E+00	136.36	6.2235	4.02689E+08	9.68110E+07	7850.0	4.86687E+06	5.86370E+05	slg
3-160000	10994.	0.00000E+00	98.479	6.4244	2.65440E+08	2.74888E+07	8274.8	5.10903E+06	6.78599E+05	slg
3-170000	11173.	0.00000E+00	77.516	6.5122	2.16145E+08	1.22905E+07	8485.4	5.22052E+06	7.54779E+05	slg
3-180000	11144.	0.00000E+00	64.022	6.4808	1.85848E+08	7.84995E+06	8601.2	5.27597E+06	8.18613E+05	slg
3-190000	10976.	0.00000E+00	54.341	6.3708	1.63814E+08	5.89453E+06	8701.7	5.32495E+06	8.70984E+05	slg
3-200000	10621.	0.00000E+00	40.610	6.1564	1.27093E+08	7.48087E+06	8783.6	5.36694E+06	9.11898E+05	slg
5-010000	0.00000E+00	0.00000E+00	0.00000E+00	0.00000E+00	1.00000E+12	0.00000E+00	4406.9	1.17198E+06	6.30557E+05	mpr
vol.no.	voidla	voidlb	vollev	vlev						
	(m)	(m)	(m/sec)	(m/sec)						
Vol.no.	lrst.mass err.	reduce-quality	reduce-extrap.	reduce-mass	reduce-propty.	min.courant	reduce-courant			
	edit total	edit total	edit total	edit total	edit total	edit total	edit total			
3-010000	0	6	0	0	0	0	0	0	0	0
3-020000	0	0	0	0	0	0	0	0	0	0

Figure 8.3-1 Example of Major Edit (Continued)

Jun.no.	*none*	to vol.	liq.j.vel. (m/sec)	vap.j.vel. (m/sec)	mass flow (kg/sec)	jun.area (m2)	throat ratio	jun-flag jefvcahs	flow no. regi last	adv. edit	choked total
3-030000	0	0	0	0	0	0	0	0	0	0	0
3-040000	0	0	0	0	0	0	0	0	0	0	0
3-050000	0	0	0	0	0	0	0	0	0	0	0
3-060000	0	0	0	0	0	0	0	0	0	0	0
3-070000	0	0	0	0	0	0	0	0	0	0	0
3-080000	0	0	0	0	0	0	0	0	0	0	0
3-090000	1	1	0	0	0	0	0	0	0	0	0
3-100000	2	4	0	0	0	0	0	0	0	0	0
3-110000	5	6	0	0	0	0	0	0	0	0	0
3-120000	9	9	0	0	0	0	0	0	0	0	0
3-130000	3	13	0	0	0	0	0	0	0	0	0
3-140000	0	11	0	0	0	0	0	0	0	0	0
3-150000	0	9	0	0	0	0	0	0	0	0	0
3-160000	0	7	0	0	0	0	0	0	0	0	0
3-170000	0	8	0	0	0	0	0	0	0	0	0
3-180000	0	11	0	0	0	0	0	0	0	0	0
3-190000	0	8	0	0	0	0	0	0	0	0	0
3-200000	0	13	0	0	2	0	0	2	105	0	0
5-010000	0	0	0	0	0	0	0	0	0	0	0
System 1											
edward's pipe											
3-010000	3-010002	3-020001	0.25810	0.12238	0.85590	4.56037E-03	1.0000	0000000	hst	0	0
3-020000	3-020002	3-030001	0.49819	0.39379	1.6830	4.56037E-03	1.0000	0000000	hst	0	0
3-030000	3-030002	3-040001	0.74154	0.53413	2.4341	4.56037E-03	1.0000	0000000	hst	0	0
3-040000	3-040002	3-050001	1.0074	0.83515	3.3347	4.56037E-03	1.0000	0000000	hst	0	0
3-050000	3-050002	3-060001	1.2524	1.1555	4.2397	4.56037E-03	1.0000	0000000	hst	0	0
3-060000	3-060002	3-070001	1.4502	1.3975	4.9960	4.56037E-03	1.0000	0000000	hst	0	0
3-070000	3-070002	3-080001	1.6152	1.5175	5.6139	4.56037E-03	1.0000	0000000	hst	0	0
3-080000	3-080002	3-090001	1.7750	1.6616	6.1653	4.56037E-03	1.0000	0000000	hst	0	0
3-090000	3-090002	3-100001	1.9719	1.8903	6.8265	4.56037E-03	1.0000	0000000	hst	0	0
3-100000	3-100002	3-110001	2.2664	2.3482	7.7258	4.56037E-03	1.0000	0000000	hst	0	0
3-110000	3-110002	3-120001	2.5913	2.9460	8.6965	4.56037E-03	1.0000	0000000	hst	0	0
3-120000	3-120002	3-130001	9.9548	11.770	29.068	4.56037E-03	1.0000	0000000	bby	0	0
3-130000	3-130002	3-140001	14.513	15.378	31.561	4.56037E-03	1.0000	0000000	bby	0	0
3-140000	3-140002	3-150001	18.126	18.809	34.802	4.56037E-03	1.0000	0000000	slg	0	0
3-150000	3-150002	3-160001	20.588	21.549	37.336	4.56037E-03	1.0000	0000000	slg	0	0
3-160000	3-160002	3-170001	22.242	23.669	38.519	4.56037E-03	1.0000	0000000	slg	0	0
3-170000	3-170002	3-180001	23.460	25.374	39.064	4.56037E-03	1.0000	0000000	slg	0	0
3-180000	3-180002	3-190001	24.394	26.814	39.499	4.56037E-03	1.0000	0000000	slg	0	0
3-190000	3-190002	3-200001	25.494	27.791	39.979	4.56037E-03	1.0000	0000000	slg	0	0
rhtbdy	sngljun										
4-000000	3-200002	5-010001	26.960	35.493	40.222	3.96752E-03	1.0000	0000000	slg	1	20
											98

Figure 8.3-1 Example of Major Edit (Continued)

Jun.no.	voidfj	voidgj	fij (N-s2/ms)	fwalfj	fwalgj	fjunf	fjunr	formfj	formgj	no. advs. last edit	ccfl total			
3-010000	0.87323	0.12677	12.181	6.057E-02	0.498	0.000E+00	0.000E+00	0.000E+00	0.000E+00	0	0			
3-020000	0.88929	0.11071	44.597	4.627E-02	0.254	0.000E+00	0.000E+00	0.000E+00	0.000E+00	0	0			
3-030000	0.86369	0.13631	18.655	4.403E-02	0.254	0.000E+00	0.000E+00	0.000E+00	0.000E+00	0	0			
3-040000	0.87074	0.12926	31.701	4.077E-02	0.228	0.000E+00	0.000E+00	0.000E+00	0.000E+00	0	0			
3-050000	0.89085	0.10915	59.239	3.862E-02	0.233	0.000E+00	0.000E+00	0.000E+00	0.000E+00	0	0			
3-060000	0.90717	9.28307E-02	115.43	3.725E-02	0.243	0.000E+00	0.000E+00	0.000E+00	0.000E+00	0	0			
3-070000	0.91570	8.43030E-02	71.569	3.624E-02	0.249	0.000E+00	0.000E+00	0.000E+00	0.000E+00	0	0			
3-080000	0.91516	8.48351E-02	6.0437	3.544E-02	0.236	0.000E+00	0.000E+00	0.000E+00	0.000E+00	0	0			
3-090000	0.91185	8.81460E-02	3.5683	3.483E-02	0.212	0.000E+00	0.000E+00	0.000E+00	0.000E+00	0	0			
3-100000	0.89705	0.10295	4.7712	3.378E-02	0.177	0.000E+00	0.000E+00	0.000E+00	0.000E+00	0	0			
3-110000	0.88244	0.11756	18.112	4.765E-02	0.114	0.000E+00	0.000E+00	0.000E+00	0.000E+00	0	0			
3-120000	0.73794	0.26206	51326.	1.966E-02	2.760E-02	0.000E+00	0.000E+00	0.000E+00	0.000E+00	0	0			
3-130000	0.56045	0.43955	2.00582E+05	1.720E-02	2.013E-02	0.000E+00	0.000E+00	0.000E+00	0.000E+00	0	0			
3-140000	0.49097	0.50903	2.00261E+05	1.658E-02	9.475E-03	0.000E+00	0.000E+00	0.000E+00	0.000E+00	0	0			
3-150000	0.46318	0.53682	67008.	1.631E-02	2.725E-03	0.000E+00	0.000E+00	0.000E+00	0.000E+00	0	0			
3-160000	0.44284	0.55716	23980.	1.602E-02	1.223E-03	0.000E+00	0.000E+00	0.000E+00	0.000E+00	0	0			
3-170000	0.42559	0.57441	9935.6	1.581E-02	6.896E-04	0.000E+00	0.000E+00	0.000E+00	0.000E+00	0	0			
3-180000	0.41320	0.58680	4688.7	1.569E-02	4.393E-04	0.000E+00	0.000E+00	0.000E+00	0.000E+00	0	0			
3-190000	0.39956	0.60044	2620.7	1.555E-02	8.236E-04	0.000E+00	0.000E+00	0.000E+00	0.000E+00	0	0			
4-000000	0.43684	0.56316	2944.0	6.521E-03	4.580E-04	0.000E+00	0.000E+00	0.000E+00	0.000E+00	0	0			
RELAP5/3.1#2#2														
Reactor Loss Of Coolant Analysis Program														
edward's pipe problem base case with extras														
HEAT STRUCTURE OUTPUT +++time= 0.100000 sec														
str.no.	side	bdry.vol.	surface	temp. (K)	number	heat-trf. convection (Watt)	heat-flux convection (Watt/m2)	critical heat-flux (Watt/m2)	CHF mul	ht mode	heat-trf. coef.conv (Watt/m2-k)	int.-heat source (Watt)	conv+rad -source (Watt)	vol.ave. temp. (K)
30-001	left	3-010000	501.543	834.80	17027.	3.72688E+06	0.72	4	7212.4	0.00000E+00	834.80	502.05		
	right	0-000000	502.183	0.00000E+00	0.00000E+00	0.00000E+00	0.00	0	0.00000E+00					
30-002	left	3-020000	501.803	606.63	12373.	5.06775E+06	0.73	4	5916.0	0.00000E+00	606.63	502.11		
	right	0-000000	502.183	0.00000E+00	0.00000E+00	0.00000E+00	0.00	0	0.00000E+00					
30-003	left	3-030000	501.849	595.04	12137.	5.20362E+06	0.71	4	6233.7	0.00000E+00	595.04	502.13		
	right	0-000000	502.183	0.00000E+00	0.00000E+00	0.00000E+00	0.00	0	0.00000E+00					
30-004	left	3-040000	501.853	621.34	12673.	5.36164E+06	0.72	4	7114.5	0.00000E+00	621.34	502.13		
	right	0-000000	502.183	0.00000E+00	0.00000E+00	0.00000E+00	0.00	0	0.00000E+00					
30-005	left	3-050000	501.857	633.11	12913.	5.55411E+06	0.74	4	8225.1	0.00000E+00	633.11	502.13		
	right	0-000000	502.183	0.00000E+00	0.00000E+00	0.00000E+00	0.00	0	0.00000E+00					
30-006	left	3-060000	501.856	635.68	12966.	5.70648E+06	0.75	4	9259.3	0.00000E+00	635.68	502.13		
	right	0-000000	502.183	0.00000E+00	0.00000E+00	0.00000E+00	0.00	0	0.00000E+00					
30-007	left	3-070000	501.847	644.67	13149.	5.66860E+06	0.76	4	10149.	0.00000E+00	644.67	502.13		
	right	0-000000	502.183	0.00000E+00	0.00000E+00	0.00000E+00	0.00	0	0.00000E+00					
30-008	left	3-080000	501.829	671.42	13695.	5.59057E+06	0.76	4	10879.	0.00000E+00	671.42	502.13		
	right	0-000000	502.183	0.00000E+00	0.00000E+00	0.00000E+00	0.00	0	0.00000E+00					

Figure 8.3-1 Example of Major Edit (Continued)

30-009	left	3-090000	501.810	704.52	14370.	5.49912E+06	0.75	4	11586.	0.00000E+00	704.52	502.13
	right	0-000000	502.183	0.00000E+00	0.00000E+00	0.00000E+00	0.00	0	0.00000E+00			
30-010	left	3-100000	501.775	791.54	16145.	5.45335E+06	0.74	4	12478.	0.00000E+00	791.54	502.12
	right	0-000000	502.183	0.00000E+00	0.00000E+00	0.00000E+00	0.00	0	0.00000E+00			
30-011	left	3-110000	501.737	901.27	18383.	5.59273E+06	0.73	4	13538.	0.00000E+00	901.27	502.12
	right	0-000000	502.183	0.00000E+00	0.00000E+00	0.00000E+00	0.00	0	0.00000E+00			
30-012	left	3-120000	501.611	2867.0	58477.	5.44584E+06	0.83	4	24834.	0.00000E+00	2867.0	502.11
	right	0-000000	502.183	0.00000E+00	0.00000E+00	0.00000E+00	0.00	0	0.00000E+00			
30-013	left	3-130000	500.913	7513.2	1.53245E+05	4.27520E+06	0.84	4	36433.	0.00000E+00	7513.2	502.06
	right	0-000000	502.183	0.00000E+00	0.00000E+00	0.00000E+00	0.00	0	0.00000E+00			
30-014	left	3-140000	500.003	9875.7	2.01433E+05	4.15672E+06	0.85	4	43779.	0.00000E+00	9875.7	501.98
	right	0-000000	502.183	0.00000E+00	0.00000E+00	0.00000E+00	0.00	0	0.00000E+00			
30-015	left	3-150000	499.165	10613.	2.16480E+05	4.10480E+06	0.85	4	49141.	0.00000E+00	10613.	501.88
	right	0-000000	502.183	0.00000E+00	0.00000E+00	0.00000E+00	0.00	0	0.00000E+00			
30-016	left	3-160000	498.469	10994.	2.24237E+05	4.10462E+06	0.86	4	52609.	0.00000E+00	10994.	501.78
	right	0-000000	502.183	0.00000E+00	0.00000E+00	0.00000E+00	0.00	0	0.00000E+00			
30-017	left	3-170000	497.862	11173.	2.27894E+05	4.10481E+06	0.86	4	54902.	0.00000E+00	11173.	501.67
	right	0-000000	502.183	0.00000E+00	0.00000E+00	0.00000E+00	0.00	0	0.00000E+00			
30-018	left	3-180000	497.329	11144.	2.27297E+05	4.10517E+06	0.86	4	56630.	0.00000E+00	11144.	501.55
	right	0-000000	502.183	0.00000E+00	0.00000E+00	0.00000E+00	0.00	0	0.00000E+00			
30-019	left	3-190000	496.851	10976.	2.23867E+05	4.10547E+06	0.86	4	58069.	0.00000E+00	10976.	501.43
	right	0-000000	502.183	0.00000E+00	0.00000E+00	0.00000E+00	0.00	0	0.00000E+00			
30-020	left	3-200000	496.460	10621.	2.16638E+05	4.10587E+06	0.87	4	59203.	0.00000E+00	10621.	501.31
	right	0-000000	502.182	0.00000E+00	0.00000E+00	0.00000E+00	0.00	0	0.00000E+00			
200-001	left	0-000000	9.220	0.00000E+00	0.00000E+00	0.00000E+00	0.00	0	0.00000E+00	0.00000E+00	13510.	5.86
	right	0-000000	0.000	13510.	1.10741E+05	0.00000E+00	0.00	0	0.00000E+00			
Sum of int.-heat-source= 0.00000E+00 (watt)												
At time= 0.100000 sec mesh point temperatures (K)												
30-001	501.54	501.74	501.90	502.02	502.09	502.14	502.17	502.18	502.18	502.18	502.18	502.18
	502.18											
30-002	501.80	501.94	502.04	502.10	502.14	502.16	502.18	502.18	502.18	502.18	502.18	502.18
	502.18											
30-003	501.85	501.98	502.07	502.12	502.15	502.17	502.18	502.18	502.18	502.18	502.18	502.18
	502.18											
30-004	501.85	501.99	502.07	502.13	502.16	502.17	502.18	502.18	502.18	502.18	502.18	502.18
	502.18											
30-005	501.86	501.99	502.08	502.13	502.16	502.17	502.18	502.18	502.18	502.18	502.18	502.18
	502.18											
30-006	501.86	501.99	502.08	502.13	502.16	502.17	502.18	502.18	502.18	502.18	502.18	502.18
	502.18											
30-007	501.85	501.99	502.08	502.13	502.16	502.17	502.18	502.18	502.18	502.18	502.18	502.18
	502.18											
30-008	501.83	501.98	502.08	502.13	502.16	502.17	502.18	502.18	502.18	502.18	502.18	502.18
	502.18											
30-009	501.81	501.97	502.07	502.13	502.16	502.17	502.18	502.18	502.18	502.18	502.18	502.18
	502.18											
30-010	501.78	501.95	502.06	502.13	502.16	502.17	502.18	502.18	502.18	502.18	502.18	502.18

Figure 8.3-1 Example of Major Edit (Continued)

30-011	502.18	501.74	501.93	502.05	502.12	502.16	502.17	502.18	502.18	502.18	502.18
	501.74	502.18									
30-012	501.61	501.91	502.05	502.12	502.16	502.17	502.18	502.18	502.18	502.18	502.18
	502.18										
30-013	500.91	501.79	502.03	502.12	502.16	502.17	502.18	502.18	502.18	502.18	502.18
	502.18										
30-014	500.00	501.47	501.95	502.10	502.15	502.17	502.18	502.18	502.18	502.18	502.18
	502.18										
30-015	499.17	501.04	501.79	502.05	502.14	502.17	502.18	502.18	502.18	502.18	502.18
	502.18										
30-016	498.47	500.61	501.59	501.98	502.12	502.16	502.18	502.18	502.18	502.18	502.18
	502.18										
30-017	497.86	500.16	501.35	501.88	502.08	502.15	502.17	502.18	502.18	502.18	502.18
	502.18										
30-018	497.33	499.72	501.08	501.74	502.02	502.13	502.17	502.18	502.18	502.18	502.18
	502.18										
30-019	496.85	499.29	500.79	501.58	501.95	502.10	502.16	502.17	502.18	502.18	502.18
	502.18										
30-020	496.46	498.88	500.48	501.39	501.85	502.05	502.14	502.17	502.18	502.18	502.18
	502.18										
200-001	9.2198	9.1063	8.7686	8.2149	7.4590	6.5194	5.4193	4.1857	2.8491	1.4423	
	0.00000E+00										

	Control variable edit; at time= 0.100000										
	1	ctl1	sum	mult	2.586302E+06	4	ctl4	sec	mult		
5	ctl5		mult		228.657	10	ctl10		div	1.140872E+06	
11	ctl10		div		1.00296	12	ctl12		diffreni	4.382614E-07	
13	ctl13		integral		5.000000E-03	14	ctl14		integral	1.00000	
15	ctl15		diffreni		1.000000E-01	99	ctl99		diffrend	0.100000	
201	ctl201		function		3.00000	202	ctl202		stdfnctn	9.75990	
203	ctl203		tripunit		0.000000E+00	204	ctl204		tripdelay	0.199667	
205	ctl205		poweri		0.200000	206	ctl206		power	-2.00000	
207	ctl207		powerx		0.300000	300	ctl300		delay	5.000000E-03	
301	ctl301		prop-int		2.15000	302	ctl302		lag	0.000000E+00	
303	ctl303		lead-lag		0.683938	304	ctl304		constant	0.367876	
401	con1		constant		0.000000E+00	402	con2		constant	0.387000	
403	pumpctl		pumpctl		-1.000000E-03	404	steamctl		constant	0.100000	
405	feedctl		feedctl		-2.500000E-04				steamctl	2.00167	

---Restart no. 109 written, block no. 2---

Figure 8.3-1 Example of Major Edit (Continued)

8.3.2.3 Reactor Kinetics Information

At major edits, the total reactor power (labeled TOTAL POWER), fission power (labeled FISSION POWER), decay power (labeled GAMMA POWER), reactivity (labeled REACTIVITY), and reciprocal period (labeled REC. PERIOD) are printed. Either the total power, fission power, or decay power can be specified as the time-varying part of the heat source in heat structures. **Figure 8.3-1** illustrates a reactor kinetics edit; however, it is not intended to be physically realistic.

8.3.2.4 Hydrodynamic Volume Information--First Section

Systems are labeled SYSTEM, followed to the right by the system number (1, 2, 3, etc.) and the name of the system (optional; *none* if no name is input on Cards 120 through 129). To the right of this are the labels MASS, MASS ERROR, and ERR. EST. for this system, followed immediately by the actual value and unit. These three quantities correspond to the TOT. MS, MS. ERR, and ERR. EST listed in the Time Step Summary, except that these are only for the particular system whereas the Time Step Summary quantities are the sum for all the systems. The overall system mass error (ϵ_{ms} in Volume I, Section 8) is included in ERR. EST for the Time Step Summary but not for the individual systems. As **Figure 8.3-1** illustrates, quantities are grouped by component within each system. Each component is first labeled with the component name (supplied by the user) and the component type. Underneath this are the values for each volume within the component.

The first items printed in this section are the abbreviated labels and units for the quantities to be printed out. The first label is VOL. NO., which is the component number (CCC) and the six-digit volume subfield number (XXYYZZ) within the component. These numbers are separated by a hyphen (-). Next is PRESSURE, which is the pressure (P_L^{n+1}) used in the hydrodynamic equation of Volume 1 of this manual. Next are VOIDF, VOIDG, and VOIDGO, which are the new liquid and vapor void fraction and the previous time step vapor void fraction ($\alpha_{f,L}^{n+1}$, $\alpha_{g,L}^{n+1}$, and $\alpha_{g,L}^n$) used in the equations. The previous time step void fraction is significant because it helped determine the wall and interfacial terms on the current edit. Next are TEMPF, TEMPG, and SAT. TEMP., which are the liquid temperature ($T_{f,L}^{n+1}$), the vapor temperature ($T_{g,L}^{n+1}$), and the saturation temperature [$T_L^{s,n+1}(P_s)$] based on the steam partial pressure used in the equations. For single-phase, the temperature of the missing phase is set to the saturation temperature. After this are UF and UG, which are the liquid specific internal energy ($U_{f,L}^{n+1}$) and the vapor specific internal energy ($U_{g,L}^{n+1}$) used in the equations. Finally, the label VOL. FLAG is listed, which is the volume control flag (tlpvbfe) input by the user for hydrodynamic volume components. Following the labels, the title supplied by the user and type of component are given, followed by the actual values of the quantities for each volume.

Additional information is printed in the first hydrodynamic volume section that is unique to certain components. In **Figure 8.3-2**, additional information for a pump, turbine, and accumulator are given. For a pump, five additional quantities are printed. In the normal operating mode, these are the rotational velocity

(RPM), pump head (HEAD), torque exerted by the fluid (TORQUE), pump octant number (OCTANT), and torque generated from the pump motor (MTR. TORQUE). These terms are discussed in Volume 1. For an accumulator, four additional quantities are printed. These are the volume of liquid in the tank-standpipe-surge line (LIQ. VOLUME), the mass of liquid in the tank-standpipe-surge line (MASS), the liquid level of water contained in the tank-standpipe-surge line (LEVEL), and the mean tank wall metal temperature (WALL TEMP). These terms are discussed in Volume 1 and in Section 2 of this volume. For a turbine, four additional quantities are printed. In the normal operating mode, these are the power extracted from the turbine (POWER), the torque extracted from the turbine (TORQUE), the turbine rotational speed (SPEED), and the efficiency factor used to represent nonideal internal processes (EFFICIENCY). These terms are also discussed in Volume 1 and in Section 2 of this volume.

```

ipump    pump
rpm = 125.22    (rad/sec)    head = 0.59115E+06 (pa)    torque = -0.10090E+06    (n-m)
octant = 2                                mtr. torque = 0.10090E+06    (n-m)

snglaccm accum
liq. volume = 24.404    m3, mass = 24167    kg, level = 19.028    m, wall temp = 322.18 k

stage3 turbine
power = 1.75174E+08 (watt)    torque = 0.30571E+06 (n-m)    speed = 573.00 (rad/sec)
                                efficiency = 0.62945

```

Figure 8.3-2 Example of Additional Output for Pumps, Turbines, and Accumulators

8.3.2.5 Hydrodynamic Volume Information--Second Section

This information appears in every major edit if noncondensable species were specified in the input. In this section, no system or component label information is printed. The volume number (labeled VOL. NO.) and five to nine other quantities are printed on each line. These are printed out in numerical order within each system. The quantities are PART. PRESS., the partial pressure of steam ($P_{s,L}^{n+1}$); DIRECT HTC, the direct heat transfer coefficient ($H_{gf,L}^n$); SATT-TOT, the saturation temperature [$T_L^{s,n+1}(P)$] based on the total pressure; NONCOND. VAPOR MASS, noncondensable mass ($M_{n,L}^{n+1}$); and the mass fraction of each of the noncondensable species ($X_{ni,L}^{n+1}$), labeled by the element name and NCOND. QUAL. The noncondensable qualities X_{ni} sum to 1.0 in each volume.

8.3.2.6 Hydrodynamic Volume Information--Third Section

This section of output is optional and can be skipped by setting bit three in the ss digits of Word 4 (W4) on the time step control cards (Cards 201 through 299). This section is printed in **Figure 8.3-1**. In this section, no system information and no component label information is printed. Furthermore, no additional component quantities are printed out. Instead, just the volume number (VOL. NO.) and ten other

quantities are printed out on each line. These are printed out in numerical order within each system. The quantities are RHO_F, liquid density ($\rho_{f,L}^{n+1}$); RHO_G, vapor density ($\rho_{g,L}^{n+1}$); RHO-MIX, void-averaged mixture density (ρ_L^{n+1}); RHO-BORON, spatial boron density ($\rho_{B,L}^{n+1}$) [this is the volume liquid fraction (α_f) times the liquid density (ρ_f) times the boron concentration (C_b). Boron concentration is used for hydrodynamic input, and boron density is used for major edits.]; VEL-LIQUID, liquid volume-average velocity ($v_{f,L}^{n+1}$); VEL-VAPOR, vapor volume-average velocity ($v_{g,L}^{n+1}$); SOUNDE, isentropic sonic velocity for single-phase or homogeneous equilibrium isentropic sonic velocity for two-phase ($a_{HE,L}^{n+1}$); QUALITY MIX-CUP, mixing cup or equilibrium quality used in wall heat transfer ($X_{e,L}^{n+1}$); QUALITY STATIC, static quality (X_L^{n+1}); and QUALITY NON-COND., noncondensable quality ($X_{n,L}^{n+1}$).

8.3.2.7 Hydrodynamic Volume Information--Fourth Section

This section prints whenever the third section prints. This section is printed in **Figure 8.3-1**. Following the volume number is TOT.HT.INP., the total wall heat transfer rate to the liquid and vapor ($Q_L^n \bullet V_L$); VAP.HT.INP., wall heat transfer rate to the vapor ($Q_{wg,L}^n \bullet V_L$); VAPOR-GEN., total mass transfer rate per unit volume at the vapor/liquid interface in the bulk fluid for vapor generation/condensation and in the boundary layer near the wall for vapor generation/condensation ($\Gamma_{g,L}^{n+1}$); WALL FLASHING, for explicit coupling of heat conduction/transfer and hydrodynamics, the mass transfer rate at the vapor/liquid interface in the boundary layer near the wall for vapor generation/condensation; for implicit coupling of heat conduction/transfer and hydrodynamics, the mass transfer rate per unit volume at the vapor/liquid interface in the boundary layer near the wall for vapor generation; LIQ.INT.HTC, liquid-to-saturation interfacial heat transfer coefficient times area per unit volume ($H_{if,L}^n$); VAP.INT.HTC, vapor-to-saturation interfacial heat transfer coefficient times area per unit volume ($H_{ig,L}^n$); MASS-FLUX, volume average mass flux (G); REYNOLDS LIQUID, liquid Reynolds number; REYNOLDS VAPOR, vapor Reynolds number; and finally, FLOW REGI, flow regime. See Section 2.1 of this volume for the meaning of the flow regime label.

8.3.2.8 Hydrodynamic Volume Time Step Control Information

This section is also optional and can be skipped by setting bit four in the ss digits of Word 4 (W4) on the time step control cards (Cards 201 through 299). This section is printed in **Figure 8.3-1**. As with the previous section, no system or component label information is printed, no additional component quantities are printed, and all quantities are printed in volume numerical order within each system. All quantities are presented in two columns. The EDIT column contains the number since the previous major edit; the TOTAL column is over the entire problem.

The numbers under LRGST. MASS ERR give the number of times a volume had the largest mass error. The numbers under MIN. COURANT give the number of times a volume had the smallest time step based on the Courant stability limit. One volume under each of the headings is incremented by one for each

successful advancement. The columns under REDUCE indicate volumes that have caused time step reductions. The MASS and PROPTY columns are for reductions resulting from mass error and out-of-range thermodynamic properties. The MASS column is for time step size reduction resulting from local mass error; it does not include reductions resulting from overall (global) mass error. The QUALITY column is for reductions resulting from problems with void fraction (α_g), noncondensable quality (X_n), and mixture density from the phasic continuity equations (ρ_m). Advancements that result in α_g and X_n being slightly less than 0.0 or slightly greater than 1.0 are allowed, and the variable is reset to 0.0 or 1.0. Advancements that result in values much less than 0.0 or much greater than 1.0 are considered an error, and the time step is repeated. The cutoff points are based on a functional relationship. This relation is tied to the mass error upper limit (8×10^{-3}). Advancements that result in ρ_m being ≤ 0 are also counted in the QUALITY column. The final cause of a QUALITY column reduction relates to the one-phase to two-phase (appearance) case discussed in Volume 1 of this manual. If too much of one phase appears (more than a typical thermal boundary layer thickness), an error is assumed to have occurred, the time step is halved and repeated, and the QUALITY column counter is incremented. The EXTRAP column is for reductions when extrapolation into a metastable thermodynamic state causes problems (see Section 8 of Volume I for a discussion of metastable thermodynamic conditions). These problems are vapor density ($\rho_g \leq 0.0$), vapor temperature ($T_g \leq 274$ K), liquid density ($\rho_l \leq 0.0$), liquid temperature ($T_l \geq$ saturation temperature (T^s) + 50 K, and vapor temperature ($T_g \leq$ saturation temperature (T^s) - 50 K. The COURANT column is for reductions resulting from the material Courant limit check. When the semi-implicit numerical scheme is used, the time step is reduced to the material Courant limit. When the nearly-implicit numerical scheme is used, the time step is reduced to 20 times the material Courant limit for the TRANSNT option and to 40 times the material Courant limit for the STDY-ST option.

Columns under the first four REDUCE headings are incremented only after a successful advancement following one or more successive reductions. Quantities are incremented only for those volumes that caused the last reduction. More than one column and row quantity can be incremented in a time step. Because of this characteristic, quantities in the first four REDUCE headings do not necessarily equal the REPEATED ADV quantity in the Time Step Summary at the top of a major edit. Since the REDUCE-COURANT column is for a reduction that occurs before the advancement takes place, it does not cause the time step to be repeated and, thus, does not increase the REPEATED ADV quantity.

New items have been recently added to this section that are not shown in **Figure 8.3-1**. Columns under the REPEAT headings are incremented if an advancement is repeated for several different reasons. If noncondensable gas first appears in a volume during an advancement, the quantity under the REPEAT-AIR APP is incremented and the advancement repeated with the same time step size. If water packing is detected in a volume, the quantity under the heading REPEAT-PACKING is incremented and the time step is repeated with the same time step size. In either of these situations, the time advancement and the time step is repeated with the same time step size. In either of these situations, the time advancement algorithm is modified to accommodate the appearance of noncondensable gas or water packing. The modifications to the hydrodynamic advancement algorithm are described in Volume I in the section entitled Special Techniques. In addition, if in the time step in which noncondensable gas first appears in a volume, the

pressure change in that cell is too large, the time step is repeated with a smaller time step size to reduce the pressure change in that cell. The same advancement may be repeated several times with smaller and smaller time steps until the pressure change criterion is satisfied. Like the quantities under the REDUCE headings, the quantity under the REPEAT - DEL PRES heading is incremented only after the pressure change criterion is satisfied.

8.3.2.9 Hydrodynamic Junction Information--First Section

This section of output is not optional and always appears in a major edit. This section is printed in **Figure 8.3-1**. As with the first section of the hydrodynamic volume information, quantities are grouped by system. For each system, the label SYSTEM, the system number (1, 2, 3, etc.), and the system name (optional) are printed on the first line. The first printed quantity for each junction is the junction number. [Labeled JUN. NO., it denotes the component number (CCC) and the six-digit junction subfield number (XXYYZZ) within the component.] These numbers are separated by a hyphen (-). The next two quantities are the volume numbers for the *from* and *to* volumes associated with the junction (labeled FROM VOL. and TO VOL.). A minus sign will be printed in front of the *from* volume number if it is not the outlet end of the volume. Similarly, a minus sign will be printed in front of the *to* volume number if it is not the inlet end of the volume. Next are the liquid junction velocity and vapor junction velocity, LIQ.J.VEL. and VAP.J.VEL ($v_{f,j}^{n+1}$ and $v_{g,j}^{n+1}$). These velocities correspond to the junction area A_j , which is described in this section. In single-phase, the velocities are equal. This is followed by MASS FLOW, the mass flow rate ($\dot{\alpha}_{f,j}^n \dot{\rho}_{f,j}^n v_{f,j}^{n+1} + \dot{\alpha}_{g,j}^n \dot{\rho}_{g,j}^n v_{g,j}^{n+1}$) A_j . The next two quantities are JUN. AREA, junction area (A_j) and THROAT RATIO, throat ratio $\frac{A_T}{A_j}$, where A_T is the physical junction area at the throat. For the smooth area option, A_j is the physical junction area (full open area if a valve). For the abrupt area option, A_j is the minimum area of the two connecting volumes. The throat ratio is the ratio of the physical junction area to the defined junction area A_j . This quantity may be less than one for orifices and valves. The velocities are based on the junction area A_j . The next quantity is the junction control flag (JUNCTION FLAGS), which is the eight-digit packed number jefvcahs that the user inputs for each junction. The next quantity is the junction flow regime, FLOW REGI; see Section 2.1 of this volume of the manual for the meaning of the flow regime label. The last three columns are a choking summary (NO.ADVS.CHOKED). The subheading LAST indicates whether the choking model was applied on the last time step (set to 1 if it was, set to 0 if it was not). The subheading EDIT lists the number of times the choking model was applied since the last major edit; the subheading TOTAL lists the number of times the choking model was applied for the entire problem. As with the first section of the hydrodynamic volume information, quantities within each system are grouped by component, with the component name and type printed above the quantities.

8.3.2.10 Hydrodynamic Junction Information--Second Section

This section of output is optional and can be skipped by setting bit two in the ss digits of Word 4 (W4) on the time step control cards (Cards 201 through 299). This section is printed in **Figure 8.3-1**. As with the second section of the hydrodynamic volume information, no system information is printed, no

component label information is printed, no additional component quantities are printed, and all quantities are printed in numerical order within each system. The junction number (JUN. NO.) and twelve other quantities are next printed out on each line. These are printed out in numerical order within each system. The quantities are VOIDFJ, liquid junction void fraction ($\dot{\alpha}_{f,j}^{n+1}$); VOIDGJ, vapor junction void fraction ($\dot{\alpha}_{g,j}^{n+1}$); FIJ, interphase drag coefficient C_i^n ; FWALFJ and FWALGJ, dimensionless liquid and vapor wall friction $\left[2 \cdot FWF_f^n \cdot \frac{\Delta x_i}{|v_{f,j}^n|} \right]$ and $\left(2 \cdot FWG_g^n \cdot \frac{\Delta x_i}{|v_{g,j}^n|} \right)$ in most cases]; user-specified dimensionless forward and reverse flow energy loss coefficients [FJUNF and FJUNR, corresponds to $\left(2 \cdot \frac{HLOSSF_i^n}{|v_{f,j}^n|} \right)$ and $\left(2 \cdot \frac{HLOSSG_i^n}{v_{g,j}^n} \right)$, where FJUNF is used in HLOSSF and HLOSSG for forward flow and FJUNR is used in HLOSSF and HLOSSG for reverse flow]; and the dimensionless abrupt area change liquid and vapor loss coefficients, FORMFJ and FORMGJ $\left[\left(2 \cdot \frac{HLOSSF_i^n}{|v_{f,j}^n|} \right) \right]$ and $\left(2 \cdot \frac{HLOSSG_i^n}{v_{g,j}^n} \right)$ in most cases, where FORMFJ is used in HLOSSF and FORMGJ is used in HLOSSG]. The previous six quantities were all made dimensionless so that the relative importance of each in the momentum equations could be determined from the major edits. The last three quantities are a countercurrent flow limitation (CCFL) model summary (NO. ADVS. CCFL). The subheading LAST indicates whether the CCFL model was applied on the last time step (set to 1 if it was or set to 0 if it was not); the subheading EDIT lists the number of times the CCFL model was applied since the last major edit; the subheading TOTAL lists the number of times the CCFL model was applied for the entire problem.

8.3.2.11 Heat Structure-Heat Transfer Information

This section of output is not optional and always appears in a major edit when heat structures are present. Quantities in this section are printed in numerical order. The first printed quantity for each heat structure is the individual heat structure number STR. NO., denoting the heat structure-geometry number (CCCG) and the three-digit individual heat structure subfield number (0NN). These numbers are separated by a hyphen (-). Following this, nine quantities are printed out for both sides of the heat structure. First, the surface indicator is printed for both sides (SIDE, printed as either LEFT or RIGHT). Next, the volume number for the hydrodynamic volume connected on each side is printed (BDRY. VOL. NUMBER, 0-000000 is printed if no volume is present). Then the surface temperature is printed for both sides (SURFACE TEMP.). After this is the heat transfer rate out of the structure for both sides (HEAT-TRF. CONVECTION). This is followed by two fluxes for both sides, the heat flux and the critical heat flux (HEAT-FLUX CONVECTION and CRITICAL HEAT-FLUX). After these, the critical heat flux multiplier, the mode of heat transfer, and the heat transfer coefficient are printed for both sides (CHF-MUL, HT MODE, and HEAT-TRF. COEF.CONV.). The multiplier is the coefficient multiplied times the critical heat flux found in the CHF table to obtain the final value printed here. Section 3.2 describes the meaning of the modes. The heat transfer coefficient is discussed in Volume IV. Finally, three quantities are printed for the individual heat structure. These are the heat generated within the structure (INT.-HEAT SOURCE), the net heat transfer rate out of the structure, i.e., convection plus radiation minus generation

(CONV+RAD-SOURCE), and the volume-average temperature for the structure (VOL. AVE. TEMP.). **Figure 8.3-1** shows an example of this section of the major edit. Following this section, the sum of the sources is given.

8.3.2.12 Heat Structure Temperature

This section of output is optional and can be skipped by setting bit one in the ss digits of Word 4 (W4) on the time step control cards (Cards 201 through 299). As in the first heat structure section, the individual heat structure number (STR. NO) is printed in the first column. Then, all the mesh point temperatures (MESH POINT TEMPERATURES) for the individual heat structure are printed, starting with the left side and proceeding toward the right side (read from left to right across the page). In **Figure 8.3-1**, 11 mesh point temperatures are printed out.

8.3.2.13 Reflood Information

This section of output is not optional and always appears in a major edit when heat structures are present and the reflood model is turned on. Once the model is turned on, it stays on, and this section continues to be printed out. **Figure 8.3-3** shows an example of this section preceded by the normal heat structure printouts. The section begins with the label REFLOOD EDIT and the time. The first quantity printed is the heat structure-geometry number (CCCG, labeled GEOM. NO.). Following this are two columns providing information about the number of axial nodes (AXIAL NODES NUMBER). The first of these columns is the assigned maximum number of axial nodes (MAXIMUM). This number is computed at input time, and it is the theoretical maximum [(number of heat structures with this geometry) • (maximum number of axial intervals) + 1] when the user requests 2, 4, or 8 maximum number of axial intervals. Owing to storage limitations, this number is calculated by a formula that reduces the number below the theoretical maximum for 16, 32, 64, or 128 maximum number of axial intervals. For the example in **Figure 8.3-3**, the user requested 16, so the theoretical maximum is 321, which is larger than the assigned maximum of 153. The next column is the actual number of axial nodes used for the last time advancement (EDIT), and, in this case, it is 59. If the EDIT column is ever larger than the MAXIMUM column, the code will abort. The next four quantities are used in deciding on the number of nodes needed to define the boiling curve. The first three are the wall temperature at incipience of boiling, INC. BOIL. TEMP. (T_{IB}); the wall temperature at critical heat flux, CRITICAL TEMP. (T_{CHF}); and the wall rewetting or quench temperature, REWETTING TEMP. (T_Q). These numbers are set to 5 degrees below and 40 and 250 degrees above the saturation temperature, respectively. The final number is the location of the critical temperature, CRIT. TEMP. POSITION. This location is the distance from the start of the first heat structure. This last output does not appear to be working, but the quench position can be found by examining the next two sections. Next is the axial position of all 59 nodes, followed by the left- and right-side surface temperatures at these axial positions.

This axial section of output is optional, and it is skipped when the heat structure temperatures are skipped. As with the previous section on reflood information, this section is not printed until the reflood model is turned on, and then it continues to be printed out. An example of this section is also shown in

Figure 8.3-3. The temperatures are printed from left to right, beginning with the first heat structure. In this example of 20 heat structures, 59 axial mesh point surface temperatures are printed.

8.3.2.14 Cladding Oxidation and Rupture Information

If the user has activated the metal-water reaction model by using a 1cccg003 card, the cladding inside and outside oxide penetration depth is printed prior to the heat structure output. **Figure 8.3-4** gives an example where there are two stacks of eight heat structures. The second stack (31) at elevation 5 shows some inside cladding oxidation. This is because this elevation has ruptured, as can be seen in the next section in **Figure 8.3-4**. The pressure shown is the pressure inside the gap.

8.3.2.15 Surface Radiation Model Output

Figure 8.3-5 shows an example of output from the radiation model. This radiation enclosure of six heat structures was tripped on at 0.25 seconds and was never tripped off. The heat flux is out of five structures and into number six. In the energy exchange calculation, -0.28 W is unaccounted for.

8.3.2.16 Control Variable Information

This section of output is not optional and always appears in a major edit when control systems are present. **Figure 8.3-1** shows examples of such printout, which begins with the label CONTROL VARIABLE EDIT. Four items are printed for each variable, with two sets of information printed per line. The four items are the control variable number (NNN), the alphanumeric name of the control variable, the control component type, and the value of the control variable at the end of the last advancement.

8.3.2.17 Generator Information

This section of output is not optional and always appears when a generator control component is present. As discussed in Volume 1, the generator component is an optional feature of the shaft component. As a result, the first column under the GENERATOR label in the major edit is the control variable number (NNN) of the corresponding shaft component. To the right of this, under normal operating conditions, is the torque exerted by the generator (TORQUE). Under normal conditions, the torque will be negative, since it is required to turn the generator. The next quantity printed, under normal conditions, is the power applied by the generator (INPUT POWER). Again, under normal conditions, the power will be negative.

8.3.3 Minor Edits

Minor edits are condensed edits of user-specified quantities. The frequency of minor edits is user-specified and may be different from the major edit frequency. **Figure 8.3-6** shows one page of minor edits. The selected quantities are held until 50 time values are stored. The minor edit information is then printed, 50 time values on a page, nine of the selected quantities per page, with time printed in the leftmost column on each page. Minor edits can print selected quantities at frequent intervals using much less paper than major edits. Section 4 of Appendix A of this volume indicates how to request minor edits and what the user-specified quantities represent.

geom.no.	Reflood edit at time= 6.556816E-02 sec				crit. temp.			
	axial nodes number	inc. boil.	critical rewetting	temp.	position	temp.	position	
	maximum	edit	temp.	(K)	(m)	(K)	(m)	
61	153	59	510.14	555.14	765.14	0.00000E+00		
Present position of axial mesh (0.0 is bottom) (m)								
	0.00000E+00	1.14300E-02	2.28600E-02	3.42900E-02	4.57200E-02	5.71500E-02	6.85800E-02	8.00100E-02 9.14400E-02 0.10287
	0.11430	0.12573	0.13716	0.14859	0.16002	0.17145	0.18288	0.27432 0.36576 0.45720
	0.54864	0.64008	0.73152	0.82296	0.91440	1.0058	1.0973	1.1887 1.2802 1.3716
	1.4630	1.5545	1.6459	1.7374	1.8288	1.9202	2.0117	2.1031 2.1946 2.2860
	2.3774	2.4689	2.5603	2.6518	2.7432	2.8346	2.9261	2.9718 3.0175 3.0632
	3.1090	3.1547	3.2004	3.2461	3.2918	3.3833	3.4747	3.5662 3.6576
geom.no. surface axial mesh point temperatures (K)								
and side								
61 left	469.29	473.46	477.65	481.84	486.02	490.21	494.40	498.59 502.78 506.97
	511.16	515.34	519.53	523.72	527.91	532.10	536.29	569.80 613.17 656.55
	694.73	732.92	784.23	835.55	883.84	932.11	946.97	961.83 984.99 1008.2
	1033.0	1057.8	1082.7	1107.6	1108.3	1109.0	1097.2	1085.4 1079.5 1073.6
	1032.9	992.32	952.23	912.13	871.17	830.20	780.18	755.16 730.15 713.58
	697.00	680.43	663.85	650.65	637.46	611.06	601.23	591.40 581.57 508.97
61 right	471.45	475.61	479.78	483.95	488.12	492.29	496.46	500.63 504.80 508.97
	513.14	517.31	521.48	525.65	529.82	533.99	538.05	571.30 614.73 658.07
	696.02	733.91	784.68	835.36	882.87	930.31	944.58	958.83 981.16 1003.5
	1027.2	1050.9	1074.5	1098.1	1098.1	1098.0	1085.5	1073.0 1065.7 1058.4
	1016.3	974.46	942.90	910.13	861.79	815.16	768.24	745.17 721.40 705.53
	690.41	674.96	659.10	646.32	633.89	608.50	598.81	589.07 579.50

Figure 8.3-3 Example of Reflood Major Edit

Printed Output

Str.no.	side	penetration depth (m)
21-001	inside	0.00000E+00
21-001	outside	3.85866E-09
21-002	inside	0.00000E+00
21-002	outside	3.65601E-08
21-003	inside	0.00000E+00
21-003	outside	1.52705E-07
21-004	inside	0.00000E+00
21-004	outside	3.07263E-07
21-005	inside	0.00000E+00
21-005	outside	3.36703E-07
21-006	inside	0.00000E+00
21-006	outside	2.10397E-07
21-007	inside	0.00000E+00
21-007	outside	7.34253E-08
21-008	inside	0.00000E+00
21-008	outside	1.35699E-08
31-001	inside	0.00000E+00
31-001	outside	4.44638E-09
31-002	inside	0.00000E+00
31-002	outside	5.52415E-08
31-003	inside	0.00000E+00
31-003	outside	2.15746E-07
31-004	inside	0.00000E+00
31-004	outside	3.91791E-07
31-005	inside	1.83263E-07
31-005	outside	2.52239E-07
31-006	inside	0.00000E+00
31-006	outside	2.89031E-07
31-007	inside	0.00000E+00
31-007	outside	9.74868E-08
31-008	inside	0.00000E+00
31-008	outside	1.49986E-08

Total hydrogen generated 3.82445E-04 (kg)

Str.no.	gas gap m,	clad radius m,	ruptured	pressure (Pa)
21-001	1.07668E-04	6.15165E-03	no	1.25221E+07
21-002	1.04248E-04	6.15778E-03	no	
21-003	1.02303E-04	6.16311E-03	no	
21-004	1.01429E-04	6.16637E-03	no	
21-005	1.01504E-04	6.16685E-03	no	
21-006	1.02673E-04	6.16458E-03	no	
21-007	1.05300E-04	6.16027E-03	no	
21-008	1.09669E-04	6.15496E-03	no	
31-001	1.04044E-04	6.14859E-03	no	4.13800E+06
31-002	1.00966E-04	6.15537E-03	no	
31-003	9.81420E-05	6.16033E-03	no	
31-004	9.61598E-05	6.16288E-03	no	
31-005	7.39275E-04	6.80334E-03	yes	
31-006	9.92277E-05	6.16150E-03	no	
31-007	1.01248E-04	6.15719E-03	no	
31-008	1.05442E-04	6.15135E-03	no	

Figure 8.3-4 Example of Cladding Oxidation and Rupture Major Edit


```

Radiation set 1. last time when radiation calculation became active was    0.25000
                        last time when radiation calculation became inactive was    0.00000E+00
Num   str.no. side      radiation
                        heat flux      energy
                        (Watt/m2)      (Watt)
1     2111- 1 right      8159.0        1507.2
1     2222- 1 right      8534.2        2627.5
1     2333- 1 right      7788.8        1918.4
1     2444- 1 right      6320.7        1264.9
1     2555- 1 right      677.39        31.282
1     2666- 1 left      -34278.        -7349.6
The sum of the radiation energy=    -0.28

```

Figure 8.3-5 Example of Radiation Major Edit

8.3.4 Diagnostic Edit

During a transient (TRANSNT on Card 100) or steady-state (STDY-ST on Card 100) problem, additional tables of variables can be printed out by inputting Words 4 and 5 on Card 105, or the tables often will be printed out when a failure occurs. These tables will be discussed in this section. This printout contains key variables from the hydrodynamic and heat transfer subroutines. The main variable in the code that activates this output is the variable HELP. Normally, HELP = 0, and no diagnostic printout occurs. The various ways that this diagnostic edit can occur will be presented, along with the value of the variable HELP. Some examples of the type of printout that occurs in the diagnostic edit will also be presented.

One way a diagnostic edit occurs is when it is forced out for more than one time step. This can be done by inputting Words 4 and 5 on Card 105 which sets HELP = 3, which will force out the hydrodynamic diagnostic edit. This, in turn, will set IWRITE = 1 in the heat transfer subroutines, forcing out the heat transfer diagnostic edit. The diagnostic edit will continue to appear for successive time steps until the count number reaches W5. Then, the calculation will stop. This method is often used by the development staff in debugging the code. An example of a diagnostic edit for one time step when HELP = 3 is presented in Appendix B.

Another way a diagnostic edit can occur is to set HELP = 2 with a debugger in any of the hydrodynamic subroutines. This will force out the diagnostic edit for the remainder of the hydrodynamic subroutines in this time step. Then, the time step will be repeated with HELP set to -2 and IWRITE set to 1 in the heat transfer subroutines. As a result, the entire time step will be repeated with the diagnostic edit obtained for the hydrodynamic and heat transfer subroutines. After this, the code continues the calculation with HELP reset to 0, resulting in no further diagnostic edits.

The final way a diagnostic edit can occur is when a code failure occurs. This does not occur for every code failure, but it does occur for a large number of them. When this occurs, HELP will be set to 1 in most cases. When it is set to 1, the diagnostic edit will be forced out for the remainder of the time step. Then, the time step will be repeated with HELP set to -1 and IWRITE set to 1 in the heat transfer subroutines. As with the previous case, the entire time step will be repeated with diagnostic edit obtained for the hydrodynamic and heat transfer subroutines. For this case, however, the calculation terminates and a final major edit plus a minor edit are printed out.

time (sec)	P (Pa)	P (Pa)	P (Pa)	P (Pa)	P (Pa)	P (Pa)	P (Pa)	3100000 (Pa)	3150000 (Pa)	3200000 (Pa)	quals 3010000	quals 3200000	velfj (m/sec)	3010000 (m/sec)	velgj (m/sec)
0.000000E+00	7.000000E+06	7.000000E+06	7.000000E+06	7.000000E+06	7.000000E+06	7.000000E+06	7.000000E+06	7.000000E+06	7.000000E+06	7.000000E+06	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00	0.000000E+00
0.000000E-03	6.09595E+06	5.59016E+06	3.90506E+06	3.90506E+06	2.63640E+06	2.63640E+06	2.63640E+06	2.63640E+06	2.63640E+06	2.63640E+06	3.15487E-03	3.15487E-03	0.21781	0.21781	0.21781
4.000000E-03	1.37441E+06	1.49439E+06	2.04011E+06	2.04011E+06	2.53500E+06	2.53500E+06	2.53500E+06	2.53500E+06	2.53500E+06	2.53500E+06	9.50141E-03	9.50141E-03	0.35402	0.35402	0.35402
6.000000E-03	1.96118E+06	2.38019E+06	2.55980E+06	2.55980E+06	2.64681E+06	2.64681E+06	2.64681E+06	2.64681E+06	2.64681E+06	2.64681E+06	1.24858E-02	1.24858E-02	0.54599	-0.61598	-0.61598
8.000000E-03	2.61603E+06	2.61606E+06	2.63588E+06	2.63588E+06	2.65547E+06	2.65547E+06	2.65547E+06	2.65547E+06	2.65547E+06	2.65547E+06	1.67083E-04	1.21614E-02	0.12627	-1.7106	-1.7106
1.000000E-02	2.67931E+06	2.65608E+06	2.65518E+06	2.65518E+06	2.65501E+06	2.65501E+06	2.65501E+06	2.65501E+06	2.65501E+06	2.65501E+06	2.04469E-04	1.06812E-02	0.21780	-1.2075	-1.2075
1.200000E-02	2.67327E+06	2.65938E+06	2.65889E+06	2.65889E+06	2.65107E+06	2.65107E+06	2.65107E+06	2.65107E+06	2.65107E+06	2.65107E+06	2.45838E-04	9.28386E-03	0.26593	-0.80086	-0.80086
1.400000E-02	2.66856E+06	2.66000E+06	2.65810E+06	2.65810E+06	2.64618E+06	2.64618E+06	2.64618E+06	2.64618E+06	2.64618E+06	2.64618E+06	2.92285E-04	8.53122E-03	0.29181	-0.49368	-0.49368
1.600000E-02	2.66440E+06	2.65952E+06	2.65666E+06	2.65666E+06	2.64487E+06	2.64487E+06	2.64487E+06	2.64487E+06	2.64487E+06	2.64487E+06	3.41520E-04	8.47477E-03	0.30592	-0.28538	-0.28538
1.800000E-02	2.66036E+06	2.65806E+06	2.65475E+06	2.65475E+06	2.64555E+06	2.64555E+06	2.64555E+06	2.64555E+06	2.64555E+06	2.64555E+06	3.92138E-04	8.98618E-03	0.31254	-0.15440	-0.15440
2.000000E-02	2.65624E+06	2.65578E+06	2.65247E+06	2.65247E+06	2.64469E+06	2.64469E+06	2.64469E+06	2.64469E+06	2.64469E+06	2.64469E+06	4.43197E-04	9.91746E-03	0.31422	-7.47591E-02	-7.47591E-02
3.000000E-02	2.63579E+06	2.63869E+06	2.63721E+06	2.63721E+06	2.62933E+06	2.62933E+06	2.62933E+06	2.62933E+06	2.62933E+06	2.62933E+06	6.90530E-04	1.53013E-02	0.29283	3.46642E-02	3.46642E-02
4.000000E-02	2.61627E+06	2.61820E+06	2.61488E+06	2.61488E+06	2.60294E+06	2.60294E+06	2.60294E+06	2.60294E+06	2.60294E+06	2.60294E+06	9.15933E-04	1.56999E-02	0.26775	5.00349E-02	5.00349E-02
5.000000E-02	2.59669E+06	2.59811E+06	2.60207E+06	2.60207E+06	2.59739E+06	2.59739E+06	2.59739E+06	2.59739E+06	2.59739E+06	2.59739E+06	1.12553E-03	1.61790E-02	0.25628	7.19093E-02	7.19093E-02
6.000000E-02	2.58577E+06	2.58423E+06	2.58937E+06	2.58937E+06	2.60217E+06	2.60217E+06	2.60217E+06	2.60217E+06	2.60217E+06	2.60217E+06	1.33803E-03	1.63810E-02	0.26613	0.11720	0.11720
7.000000E-02	2.58946E+06	2.58786E+06	2.57898E+06	2.57898E+06	2.49350E+06	2.49350E+06	2.49350E+06	2.49350E+06	2.49350E+06	2.49350E+06	1.57807E-03	1.67368E-02	0.27204	0.14126	0.14126
8.000000E-02	2.59205E+06	2.59227E+06	2.58084E+06	2.58084E+06	2.45390E+06	2.45390E+06	2.45390E+06	2.45390E+06	2.45390E+06	2.45390E+06	1.80915E-03	1.68859E-02	0.25649	0.10747	0.10747
9.000000E-02	2.58958E+06	2.59177E+06	2.58565E+06	2.58565E+06	2.39619E+06	2.39619E+06	2.39619E+06	2.39619E+06	2.39619E+06	2.39619E+06	2.03421E-03	1.70636E-02	0.25519	0.10989	0.10989
0.100000	2.58633E+06	2.58749E+06	2.58992E+06	2.58992E+06	2.36217E+06	2.36217E+06	2.36217E+06	2.36217E+06	2.36217E+06	2.36217E+06	2.25691E-03	1.72378E-02	0.25810	0.12238	0.12238
0.110000	2.58230E+06	2.58331E+06	2.59420E+06	2.59420E+06	2.34174E+06	2.34174E+06	2.34174E+06	2.34174E+06	2.34174E+06	2.34174E+06	2.48113E-03	1.73292E-02	0.25970	0.13130	0.13130
0.120000	2.59000E+06	2.58786E+06	2.60043E+06	2.60043E+06	2.32430E+06	2.32430E+06	2.32430E+06	2.32430E+06	2.32430E+06	2.32430E+06	7.70341E-03	1.75719E-02	0.23041	6.28296E-02	6.28296E-02
0.130000	2.58640E+06	2.58175E+06	2.48120E+06	2.48120E+06	2.31092E+06	2.31092E+06	2.31092E+06	2.31092E+06	2.31092E+06	2.31092E+06	2.91332E-03	1.78266E-02	0.24140	9.40326E-02	9.40326E-02
0.140000	2.58288E+06	2.57279E+06	2.43458E+06	2.43458E+06	2.30155E+06	2.30155E+06	2.30155E+06	2.30155E+06	2.30155E+06	2.30155E+06	3.12678E-03	1.80562E-02	0.23671	8.66769E-02	8.66769E-02
0.150000	2.57992E+06	2.56282E+06	2.39006E+06	2.39006E+06	2.29433E+06	2.29433E+06	2.29433E+06	2.29433E+06	2.29433E+06	2.29433E+06	3.34363E-03	1.82649E-02	0.22555	6.39216E-02	6.39216E-02
0.160000	2.57310E+06	2.55244E+06	2.35333E+06	2.35333E+06	2.28703E+06	2.28703E+06	2.28703E+06	2.28703E+06	2.28703E+06	2.28703E+06	3.56586E-03	1.84566E-02	0.22539	6.82506E-02	6.82506E-02
0.170000	2.56001E+06	2.53912E+06	2.33132E+06	2.33132E+06	2.27954E+06	2.27954E+06	2.27954E+06	2.27954E+06	2.27954E+06	2.27954E+06	3.80216E-03	1.86416E-02	0.22808	8.04724E-02	8.04724E-02
0.180000	2.55186E+06	2.52416E+06	2.31369E+06	2.31369E+06	2.27185E+06	2.27185E+06	2.27185E+06	2.27185E+06	2.27185E+06	2.27185E+06	4.05832E-03	1.88338E-02	0.22644	8.31920E-02	8.31920E-02
0.190000	2.54663E+06	2.43665E+06	2.29849E+06	2.29849E+06	2.26364E+06	2.26364E+06	2.26364E+06	2.26364E+06	2.26364E+06	2.26364E+06	4.33389E-03	1.90561E-02	0.25844	0.16503	0.16503
0.200000	2.53246E+06	2.37423E+06	2.28615E+06	2.28615E+06	2.25508E+06	2.25508E+06	2.25508E+06	2.25508E+06	2.25508E+06	2.25508E+06	4.62236E-03	1.93277E-02	0.42554	0.55780	0.55780
0.210000	2.51779E+06	2.33807E+06	2.27584E+06	2.27584E+06	2.24647E+06	2.24647E+06	2.24647E+06	2.24647E+06	2.24647E+06	2.24647E+06	5.11058E-03	1.96530E-02	0.83404	1.5376	1.5376
0.220000	2.50015E+06	2.31672E+06	2.26673E+06	2.26673E+06	2.23758E+06	2.23758E+06	2.23758E+06	2.23758E+06	2.23758E+06	2.23758E+06	8.93717E-03	2.00231E-02	6.1628	11.401	11.401
0.230000	2.30867E+06	2.29389E+06	2.25941E+06	2.25941E+06	2.22924E+06	2.22924E+06	2.22924E+06	2.22924E+06	2.22924E+06	2.22924E+06	1.81870E-02	2.04248E-02	7.1194	7.4004	7.4004
0.240000	2.16806E+06	2.22330E+06	2.24440E+06	2.24440E+06	2.22077E+06	2.22077E+06	2.22077E+06	2.22077E+06	2.22077E+06	2.22077E+06	2.65854E-02	2.08391E-02	4.2868	3.3204	3.3204
0.250000	2.09789E+06	2.14246E+06	2.20520E+06	2.20520E+06	2.20649E+06	2.20649E+06	2.20649E+06	2.20649E+06	2.20649E+06	2.20649E+06	3.23567E-02	2.12484E-02	2.5818	1.3673	1.3673
0.260000	2.06772E+06	2.08623E+06	2.14102E+06	2.14102E+06	2.17666E+06	2.17666E+06	2.17666E+06	2.17666E+06	2.17666E+06	2.17666E+06	3.65658E-02	2.17062E-02	1.7801	0.48090	0.48090
0.270000	2.04889E+06	2.05279E+06	2.08302E+06	2.08302E+06	2.12231E+06	2.12231E+06	2.12231E+06	2.12231E+06	2.12231E+06	2.12231E+06	4.00966E-02	2.24343E-02	1.5093	0.35386	0.35386
0.280000	2.02845E+06	2.02863E+06	2.04013E+06	2.04013E+06	2.05998E+06	2.05998E+06	2.05998E+06	2.05998E+06	2.05998E+06	2.05998E+06	4.34078E-02	2.38038E-02	1.4108	0.40779	0.40779
0.290000	2.00588E+06	2.00501E+06	2.00481E+06	2.00481E+06	1.99889E+06	1.99889E+06	1.99889E+06	1.99889E+06	1.99889E+06	1.99889E+06	4.66588E-02	2.57386E-02	1.3539	0.48217	0.48217
0.300000	1.97803E+06	1.97588E+06	1.96501E+06	1.96501E+06	1.93955E+06	1.93955E+06	1.93955E+06	1.93955E+06	1.93955E+06	1.93955E+06	4.98339E-02	2.77759E-02	1.3231	0.65219	0.65219
0.310000	1.93495E+06	1.93150E+06	1.91683E+06	1.91683E+06	1.88863E+06	1.88863E+06	1.88863E+06	1.88863E+06	1.88863E+06	1.88863E+06	5.27627E-02	2.98041E-02	1.3268	1.0511	1.0511
0.320000	1.87558E+06	1.87523E+06	1.87078E+06	1.87078E+06	1.84232E+06	1.84232E+06	1.84232E+06	1.84232E+06	1.84232E+06	1.84232E+06	5.53508E-02	3.19419E-02	1.2971	1.2496	1.2496
0.330000	1.82688E+06	1.82559E+06	1.81483E+06	1.81483E+06	1.79327E+06	1.79327E+06	1.79327E+06	1.79327E+06	1.79327E+06	1.79327E+06	5.80595E-02	3.62839E-02	1.2369	1.2114	1.2114

Figure 8.3-6 Example of Minor Edit

There are two added printouts for this failure case (HELP = -1) that are an aid in tracing the code failure. Just preceding the diagnostic edit, information concerning the reason why the code failed is printed out. This information begins with eight asterisks (*****). An example of this printout for the case of a thermodynamic property failure at the minimum time step is shown in the middle of **Figure 8.3-7**. Following this, the old time STATE diagnostic printout is forced out. The other message often printed out for this case (HELP = -1) can usually be buried somewhere within the diagnostic edit. For the example of a thermodynamic property error, information from the STATEP subroutine concerning the faulty volume is printed out (see middle of **Figure 8.3-8**). The information is the label THERMODYNAMIC PROPERTY FAILURE, the volume number (VOLNO), pressure (P), vapor specific energy (UG), liquid specific energy (UF), noncondensable quality (QUALA), liquid void fraction (VOIDF), and vapor void fraction (VOIDG). Further information on the specifics of the thermodynamic property failure such as in which phase the failure occurred is usually printed. This particular printout (using the semi-implicit hydrodynamic scheme) is located between the EQFINL and STATE diagnostic printouts. (No MASS ERROR diagnostic occurs for this failure.)

Failures that result in a diagnostic edit with HELP = -1 can be grouped into two cases. The first case occurs when the user is responsible. The thermodynamic property error mentioned above and shown in **Figure 8.3-7** and **Figure 8.3-8** can occur as a result of this. This can occur when the user inputs state properties that are undetected in input processing and thus get into the transient calculation. Thermodynamic property errors are the same as when either the REDUCE-PROPTY or REDUCE-EXTRAP flags are set in the major edit hydrodynamic volume time step control information block (see Section 8.3.2.8). Another example of a user-caused failure is when material property data are out of range. Two more user-caused failures can occur in the case of valves. If both motor valve trips become true at the same time, a failure will result. In addition, if the control system is set up incorrectly and this results in the servo valve stem position not being between 0 and 1, a failure will result. Another example is when a divide by 0 occurs in a control variable. The second case occurs as the result of a coding failure, which can be caused by a programming error or a model deficiency. Such a failure should be reported to the development staff through the RELAP5 User Services. Such errors often result in negative densities, bad viscosities, bad thermal conductivities, or thermodynamic property errors.

8.4 Plotted Output

The two methods normally used to obtain time plots of computed information are described below.

8.4.1 External Plots

The STRIP option (on Card 100) may be used to obtain ASCII data from the RSTPLT file. **Figure 8.4-1** shows an example of a strip input file. Data for all the parameters listed in the input file will appear on an ASCII file called STRIPF. These data must then be processed to put them into a format that is acceptable to the users' plotting software. XMGR could be used to plot data from the STRIPF file. The users now usually use XMGR5,^{8.1-2} an extension to XMGR^{8.1-3} that adds features to conveniently plot information from restart-plot files or STRIPF files.

```

At time= 0.000000E+00 sec
61-001 502.48 502.48 502.48 502.48 502.48
61-002 569.54 569.54 569.54 569.54 569.54
61-003 656.32 656.32 656.32 656.32 656.32
61-004 732.58 732.58 732.58 732.58 732.58
61-005 835.17 835.17 835.17 835.17 835.17
61-006 931.68 931.68 931.68 931.68 931.68
61-007 961.37 961.37 961.37 961.37 961.37
61-008 1007.7 1007.7 1007.7 1007.7 1007.7
61-009 1057.3 1057.3 1057.3 1057.3 1057.3
61-010 1107.0 1107.0 1107.0 1107.0 1107.0
61-011 1108.4 1108.4 1108.4 1108.4 1108.4
61-012 1084.9 1084.9 1084.9 1084.9 1084.9
61-013 1073.1 1073.1 1073.1 1073.1 1073.1
61-014 991.89 991.89 991.89 991.89 991.89
61-015 911.68 911.68 911.68 911.68 911.68
61-016 829.82 829.82 829.82 829.82 829.82
61-017 729.82 729.82 729.82 729.82 729.82
61-018 663.62 663.62 663.62 663.62 663.62
61-019 610.82 610.82 610.82 610.82 610.82
61-020 591.15 591.15 591.15 591.15 591.15

---Restart no. 0 written, block no. 0---

***** Thermodynamic property error with minimum time step, transient being terminated.

***** Trouble, last advancement being repeated with debug printout.

#####
state Diagnostic printout, timehy = 6.5567970E-02, dt = 1.9073486E-07, ncount = 1339, help = -1, lsuces = 2, fail = F
#####

Volume mixture properties
=====
volno v vo p pps voidf voidg quals quale dotm dotmo quala sigma boron borono soude dsnddp rho satt
=====
005010000 0.00000E+00 2.75790E+05 1.0000 0.00000E+00 0.00000E+00 0.00000E+00 0.00000E+00 0.00000E+00 0.00000E+00 0.00000E+00 1537.8 987.89
0.00000E+00 2.75790E+05 0.00000E+00 -0.15544 0.00000E+00 0.00000E+00 5.26935E-02 2.72067E+06 403.84
006010000 2.83055E-03 1.54960E+06 9.46341E-03 0.20025 146.12 0.00000E+00 0.00000E+00 0.00000E+00 0.00000E+00 0.00000E+00 24.186 11.839
2.83055E-03 1.54960E+06 0.99054 9.54995E-03 146.12 3.78434E-02 0.00000E+00 4.97445E+06 472.99
006020000 2.83055E-03 1.34989E+06 7.20929E-05 0.98079 -963.41 0.00000E+00 0.00000E+00 0.00000E+00 0.00000E+00 0.00000E+00 707.12 3.4851
2.83055E-03 1.34989E+06 0.99993 1.0000 -963.41 3.93030E-02 0.00000E+00 3.67281E+06 466.50
006030000 2.83055E-03 1.26800E+06 2.67725E-04 0.93243 8.6735 0.00000E+00 0.00000E+00 0.00000E+00 0.00000E+00 0.00000E+00 704.85 3.4673
2.83055E-03 1.26800E+06 0.99973 1.0000 8.6735 3.99465E-02 0.00000E+00 3.65922E+06 463.62
006040000 2.83055E-03 1.32421E+06 3.87655E-04 0.91149 14.999 0.00000E+00 0.00000E+00 0.00000E+00 0.00000E+00 0.00000E+00 694.77 3.8216
2.83055E-03 1.32421E+06 0.99961 1.0000 14.999 3.95017E-02 0.00000E+00 3.60120E+06 465.61
006050000 2.83055E-03 1.39871E+06 4.81529E-04 0.90059 18.214 0.00000E+00 0.00000E+00 0.00000E+00 0.00000E+00 0.00000E+00 685.13 4.2117
2.83055E-03 1.39871E+06 0.99952 1.0000 18.214 3.89326E-02 0.00000E+00 3.54662E+06 468.15
006060000 2.83055E-03 1.48181E+06 6.08762E-04 0.88749 19.338 0.00000E+00 0.00000E+00 0.00000E+00 0.00000E+00 0.00000E+00 673.86 4.6874
2.83055E-03 1.48181E+06 0.99939 1.0000 19.338 3.83229E-02 0.00000E+00 3.48886E+06 470.86
006070000 2.83055E-03 1.57004E+06 7.46737E-04 0.87685 22.429 0.00000E+00 0.00000E+00 0.00000E+00 0.00000E+00 0.00000E+00 660.78 5.2366
2.83055E-03 1.57004E+06 0.99925 1.0000 22.429 3.77018E-02 0.00000E+00 3.42392E+06 473.61
006080000 2.83055E-03 1.66580E+06 9.06057E-04 0.86712 39.730 0.00000E+00 0.00000E+00 0.00000E+00 0.00000E+00 0.00000E+00 647.34 5.8687
2.83055E-03 1.66580E+06 0.99909 1.0000 39.730 3.70553E-02 0.00000E+00 3.35791E+06 476.47
006090000 2.83055E-03 1.77436E+06 1.10754E-03 0.85613 103.33 0.00000E+00 0.00000E+00 0.00000E+00 0.00000E+00 0.00000E+00 634.93 6.5977

```

Figure 8.3-7 Example of Printout Before the Diagnostic Edit When a Failure Occurs

```

006180000 19 8.23918E-02 2.29640E-07 3.40446E-07 1.82739E+06 2.62915E+06 8.85948E+05 0.99870 0.00000E+00 9.8622
0.74161 0.00000E+00 0.00000E+00 1.82741E+06 2.62915E+06 8.85958E+05 0.99870 0.00000E+00 35.839
006190000 20 4.48173E-02 7.66357E-08 1.73934E-07 1.41060E+06 2.62187E+06 8.31422E+05 0.99907 0.00000E+00 7.6334
0.35177 0.00000E+00 0.00000E+00 1.41061E+06 2.62187E+06 8.31422E+05 0.99907 0.00000E+00 87.442
006200000 21 2.18821E-02 6.21175E-08 1.16083E-07 8.49402E+05 2.60382E+06 7.31203E+05 0.99948 0.00000E+00 4.6877
0.24154 0.00000E+00 0.00000E+00 8.49410E+05 2.60382E+06 7.31204E+05 0.99948 0.00000E+00 143.65
007010000 22 0.82303 9.74341E-06 1.19936E-05 2.75790E+05 2.65715E+06 5.49024E+05 1.0000 0.00000E+00 1.2690
29.009 0.00000E+00 0.00000E+00 2.75790E+05 2.65715E+06 5.49024E+05 1.0000 0.00000E+00 0.00000E+00

***** Thermodynamic property failure, volno= 6010000, p = 1.550289E+06, ug = 4.326621E+06, uf = 3.459355E+01
***** Liquid phase property call had error.

#####
state Diagnostic printout, timehy = 6.556816E-02, dt = 1.9073486E-07, ncount = 1340, help = -1, lsuccess = 2, fail = F
#####

Volume mixture properties
=====
volno v p pps voidf voidg quals quale dotm dotno quala sigma boron borono sounde dsnddp rho satt
=====
005010000 0.00000E+00 2.75790E+05 1.0000 0.00000E+00 0.00000E+00 0.00000E+00 0.00000E+00 0.00000E+00 1537.8 987.89
0.00000E+00 2.75790E+05 0.00000E+00 -0.15544 0.00000E+00 5.26935E-02 0.00000E+00 2.72067E+06 403.84
006010000 2.83055E-03 1.55029E+06 9.46343E-03 0.22472 146.08 0.00000E+00 0.00000E+00 86.615 10.557
2.83055E-03 1.55029E+06 0.99054 7.01770E-02 146.12 3.78386E-02 0.00000E+00 4.97394E+06 473.01
006020000 2.83055E-03 1.35004E+06 7.25842E-05 0.98070 -924.80 0.00000E+00 0.00000E+00 707.13 3.4857
2.83055E-03 1.35004E+06 0.99993 1.0000 -963.41 3.93019E-02 0.00000E+00 3.67286E+06 466.50
006030000 2.83055E-03 1.26804E+06 2.67783E-04 0.93242 8.6740 0.00000E+00 0.00000E+00 704.84 3.4675
2.83055E-03 1.26804E+06 0.99973 1.0000 8.6735 3.99462E-02 0.00000E+00 3.65918E+06 463.62
006040000 2.83055E-03 1.32424E+06 3.87707E-04 0.91148 15.000 0.00000E+00 0.00000E+00 694.77 3.8217
2.83055E-03 1.32424E+06 0.99961 1.0000 14.999 3.95015E-02 0.00000E+00 3.60118E+06 465.61
006050000 2.83055E-03 1.39873E+06 4.81578E-04 0.90059 18.212 0.00000E+00 0.00000E+00 685.12 4.2118
2.83055E-03 1.39873E+06 0.99952 1.0000 18.214 3.89324E-02 0.00000E+00 3.54660E+06 468.15
006060000 2.83055E-03 1.48183E+06 6.08810E-04 0.88749 19.336 0.00000E+00 0.00000E+00 673.86 4.6875
2.83055E-03 1.48183E+06 0.99939 1.0000 19.338 3.83228E-02 0.00000E+00 3.48884E+06 470.86
006070000 2.83055E-03 1.57006E+06 7.46785E-04 0.87685 22.424 0.00000E+00 0.00000E+00 660.78 5.2368

```

Figure 8.3-8 Example of Printout Buried in the Diagnostic Edit When a Failure Occurs

```

=flecht-seaset separate effects reflood calculation: test 31504
*
100 strip fmtout.
103 0
1001 httemp      006100407
1020 tempg       006040000
1030 voidg       006040000
1040 quale       006040000
1041 gammaw      006040000
1042 vapgen      006040000
1043 velg        006040000
1044 velf        006040000
1050 floreg      006040000
1060 htrnr       006100401
1061 hthtc       006100401
1062 htmode      006100401
.end

```

Figure 8.4-1 Strip Input File

If users plan on using external plots, they should make up their strip input file before generating the RSTPLT file because some parameters they may desire are on the RSTPLT file only if they are specifically requested with 2080XXXX cards.

8.4.2 Internal Plots

This code feature, designed at the INEL for the CDC-176, is not available for the CRAY and the workstations. The following discussion of this capability is printed so that it will be available if the capability is restored.

A plot package has been provided in RELAP5 so that the user may produce graphs of calculational results. However, because each user may have a different use for the plots, many options are provided so that the user may design and vary the quality of plots as desired. In addition, since it is often necessary to compare the results to experiments or other calculations, a means to input plot comparison data tables has also been provided.

For convenience to the user, a check plot option is provided that will produce plots of input data, such as for time-dependent volumes and junctions, general tables, plot comparison data tables, valve area and flow coefficients, etc. This option can be used by the input of the *check plot* general plot request cards. The plots are constructed upon completion of the third phase of input data processing so that all information processed by the code will be included. Once the option is activated, it will remain in effect for all subsequent restarts and plot only jobs, including restarts with renodalization, until cancelled by the user with appropriate input.

It is assumed that each plot must be uniquely identified, and, hence, the run time, date, and code version is written in the plot margin oriented to appear on the edge that would be placed in a notebook binder. The plot heading and title are written at the top of the plot, and the axes labels and titles are written parallel to the left-hand and bottom axes. In addition, the curves plotted must lie within the axes' extremities and yet span as much of the axes as possible. The axes labeling subdivisions are also rounded to the first significant digit in order to produce simple labels.

Results can be plotted for any NEW or RESTART run. In addition, a PLOT run can be performed for which plots can be made of any variable stored in the plot record on the restart-plot file.

Plot input is analogous to the component input for NEW and RESTART problems in that once plot requests have been input, the resultant plot records and plot comparison data records are written to the restart-plot file. Hence, only input to delete, replace, insert, or add plot requests is required for successive RESTART or PLOT runs. In addition, undefined results are not plotted for components added or deleted by renodalization.

Some user inconvenience is apparent for input of plot comparison data tables because this input must be in an 80-character card image and must be part of the user input stream. If each data table is reasonably small, the user may manually produce the card images on a terminal. The tables may then be stored for future use with RELAP5 runs or be made part of each problem input stream as desired. To produce plot comparison data tables from other restart-plot files, the RELAP5 STRIP option may be used to retrieve results and build plot comparison data tables. If the data are contained on user tapes or disk files, the user can provide programs to build plot comparison data tables in the format required by RELAP5.

8.5 RELAP5 Control Card Requirements

When run under the Unix operating system, the code includes processing of the command line that initiates RELAP5 execution. This processing permits specification of some options and the names of the files such as input, output, and restart-plot files needed for execution (Unix is an operating system available on Cray computers and most workstations.) The command line for execution under Unix is documented in the last section of the Input Data Requirements, Appendix A.

8.6 Transient Termination

The transient advancement should not abort (terminate by operating system intervention) except for exceeding available disk space. Other program aborts such as floating point errors, illegal address, or segmentation faults are indications of programming errors and should be reported to the RELAP5 development staff.

The user may optionally specify one or two trips to terminate a problem. Normal termination is from one of these trips or the advancement reaching the final time on the last time step control card. Minor and major edits are printed and a restart record is written at termination. Since trips can be redefined and new time step cards can be entered at restart, the problem can be restarted and continued.

Transient termination can also occur based on two tests on the CPU time remaining for the job. One test terminates if the remaining CPU time at the completion of a requested time step is less than an input quantity. The second test is similar, but the comparison is to a second input quantity and is made after every time advancement. The input quantity for the first test is larger than for the second test because the preferred termination is at the completion of a requested time step. In either case, the termination can be restarted.

Failure terminations can occur from several sources, including hydrodynamic solution outside the range of thermodynamic property subroutines, heat structure temperatures outside of thermal property tables or functions, and attempting to access an omitted pump curve. Attempting to restart at the point of failure or at an earlier time without some change in the problem input will only cause another failure. Problem changes at restart may allow the problem to be successfully restarted.

Additional information on terminating calculations is presented in Section 3 of Volume V.

8.7 Problem Changes at Restart

The most common use of the restart option is simply to continue a problem after a normal termination. If the problem terminated because it approached the CPU time limit, the problem can be restarted with no changes to information obtained from the restart file. If the problem stopped because the advancement time reached the end time on the last time step card, new time cards must be entered. If the problem was terminated by a trip, the trip causing the termination must be redefined to allow the problem to continue. Thus, the code must provide for some input changes for even a basic restart capability.

The ability to modify the simulated system at restart is a desirable feature. The primary need for this feature is to provide for a transition from a steady-state condition to a transient condition. In many cases, simple trips can activate valves that initiate the transient. Where trips are not suitable, the capability to redefine the problem at restart can save effort in manually transcribing quantities from the output of one simulation to the input of another. One example of a problem change between steady-state and transient is the use of a liquid-filled, time-dependent volume in place of the vapor region of a pressurizer during steady-state. The time-dependent volume provides the pressurizer pressure and supplies or absorbs water from the primary system as needed. The time-dependent volume is replaced by the vapor volumes at initiation of the transient. This technique avoids modeling the control system that maintains liquid level and temperature during steady-state calculations when they are not needed in the transient.

Another reason for a problem change capability is to reduce the cost of simulating different courses of action at some point in the transient. An example is a need to determine the different system responses when a safety system continues to operate or fails late in the simulation. One solution is to run two complete problems. An alternative is to run one problem normally and restart that problem at the appropriate time with a problem change for the second case.

The problem change capability could also be used to renodalize a problem for a certain phase of a transient. This has not been necessary or desirable for most problems. For this reason, techniques to

automate the redistribution of mass, energy, and momentum when the number of volumes changes have not been provided.

The current status of allowed problem changes at restart in RELAP5 are summarized below. In all instances, the problem definition is that obtained from the restart tape unless input data are entered for deletions, modifications, or additions. The problem defined after input changes must meet the same requirements as a new problem.

Time step control can be changed at restart. If time step cards are entered at restart, all previous time step cards are deleted. New cards need only define time step options from the point of restart to the end of the transient.

Minor edit, internal plot input data, and expanded edit/plot variables (2080XXXX) cards can be changed at restart. If any of the minor edit cards are entered, all previous cards are deleted. New cards must define all desired minor edit quantities. The internal plot request data and expanded edit/plot variables (2080XXXX) cards are handled in the same manner.

Trip cards can be entered at restart. The user can specify that all previous trips be deleted and can then define new trips. The user can also specify that the previously defined trips remain but that specific trips be deleted, be reset to false, be redefined, or that new trips be added.

Existing hydrodynamic components can be deleted or changed, and new components can be added. An especially useful feature is that the tables in time-dependent volumes and junctions can be changed. If a component is changed, all of the cards for the component must be entered.

Control system components can be deleted, changed, or added.

Heat structures, general tables, and material properties can also be deleted, changed, or added. Heat structures can only be changed at the level of heat structure geometry. When the heat structure geometry is changed, all heat structures referencing that heat structure geometry are affected. Individual general tables and material properties can be added, deleted, or changed.

Reactor kinetics can be added or deleted on restart. A complete set of reactor kinetics data must be input, i.e., individual sections of kinetics data may not be specified as replacement data.

In summary, all modeling features in RELAP5 can be added, deleted, or changed at restart.

9 REVISION HISTORY

Manual Revision History

Date	Notes
2008-07-08	Added this section.
2010-04-01	Added description for end valve.
2010-09-22	Updated title page ISL division name and publication date
2016-06-24	Updated manual for the release of RELAP5/Mod3.3Patch05.
2019-02-28	Update manuals to include all changes through RELAP5 3.3kw: User guide FREETAB table added. User guide discussion on conditional trips and control variables added.
2019-06-24	Update manuals to include all changes through RELAP5 3.3ky.
2019-11-13	Update Revision History.

